PROJECT NEXUS

NEXUS

ENGINEERING & RESEARCH ANNEX

Architecture · Physics · Software · Optics · Neural Systems · Safety



ENGINEERING EDITION / v2.0 / 2026

Document control

Document	NEXUS Engineering and Research Annex
Identifier	NXS-ANX-002
Edition	Engineering edition v2.0. Replaces annex v1.0 (30 September 2026) in full.
Date	19 October 2026
Status	Research architecture. No claim of a finished product, certification, clinical use or manufacturing readiness.
Scope	Architecture, physics, software contracts, intelligence, neural interface, compute fabric, optics, physical engineering, safety, security, human factors, validation, experiments.
Out of scope	Commercial strategy, market claims, medical claims, stimulation of neural tissue (sensing and interaction only).
Evidence rule	Every technical claim carries an evidence class E0 to E5 and, where it concerns a result, a validation rung V0 to V9. Section "How to read NEXUS" defines every label once.
Implementation status	The NEXUS repository was not inspected for this annex. Every implementation statement is labelled REPOSITORY-ASSERTED, and no repository path is reported as verified.
External references	Four external papers were checked against publisher or preprint records during preparation of v1.0 and are carried forward with that status. All other references are carried from project reference lists and are not re-verified. See the Reference register.
Numerical values	Numbers are tagged TARGET, DESIGN PARAMETER, OPEN, MEASURED or DERIVED. No value in this annex is a MEASURED NEXUS result unless it is tagged so; none is.
Revision history	Provenance, changes from v1.0 and corrections are in the Evidence and Revision Register (Appendix R).

READING STATUS

This annex is an engineering program document. It defines contracts, budgets and experiments. A branch rejected on strong measurement is a scientific result, and the annex is written so that branches can be rejected visibly.

How to read NEXUS

Five label families are used throughout. Each is explained here once and then appears as a compact badge. Evidence class describes what kind of support a claim has; validation rung describes how far it has travelled; they are separate attributes.

Family	Labels	Meaning
Evidence	E0 E1 E2 E3 E4 E5	**E0** established physics, mathematics or control theory. **E1** documented platform or engineering technology. **E2** externally demonstrated research or product, partial capability only. **E3** implemented in NEXUS (code, simulator, replay, prototype). **E4** NEXUS engineering hypothesis, testable and unvalidated. **E5** unresolved problem.
Validation	V0 to V9	V0 mathematical model, V1 numerical simulation, V2 software prototype, V3 dataset replay, V4 host VM or Cuttlefish, V5 hardware-in-the-loop, V6 subsystem prototype, V7 integrated wearable, V8 controlled and longitudinal human studies, V9 production engineering. A claim cannot skip a rung by wording.
Decision	FROZEN PREFERRED INVESTIGATING OPEN REJECTED TARGET	FROZEN: changing it would redefine NEXUS. PREFERRED: best current direction, subject to validation. INVESTIGATING: competing branches. OPEN: no architecture selected. REJECTED: ruled out unless new evidence appears. TARGET: desired performance, not demonstrated capability.
Implementation	NOT STARTED PROTOTYPE IMPLEMENTED VERIFIED	VERIFIED means a third party can reproduce it from a repository snapshot, a passing test report and a recorded trace.
Product maturity	M0 to M7	M0 concept, M1 mathematical model, M2 simulation, M3 bench prototype, M4 integrated prototype, M5 human validation, M6 production engineering, M7 certified product.
Value type	TARGET DESIGN PARAMETER OPEN MEASURED DERIVED	TARGET: wanted performance. DESIGN PARAMETER: initial test value, to be overwritten by measurement. OPEN: no value yet. MEASURED: observed on NEXUS hardware. DERIVED: computed from stated assumptions and physics.
Provenance	VERIFIED REPOSITORY-ASSERTED EXTERNALLY DEMONSTRATED PROPOSED UNRESOLVED	Used for implementation status and for claims about current state.

How to open a subsystem

Any load-bearing subsystem is documented by a **Subsystem Engineering Contract** of 24 fixed fields (purpose to next implementation artifact). A technically competent engineer should be able to answer, from the contract alone: what it is, why it exists, which requirement it serves, what it consumes and produces, where it runs, its timing, power, thermal, failure, degradation, interface, calibration, security, manufacturing, test and evidence position, what is unknown, what experiment decides it, what would falsify it, and what is built next.

Page type	Use
Chapter opener (dark)	one thesis sentence and one line diagram per Part
Flagship figure (dark canvas)	master architecture, loops, topologies, fabrics, safety; each figure states the question it answers
Engineering page (light)	dense tables, schemas, equations, requirements
Equation card	physical limit, what it constrains, engineering consequence, experiment or falsifier
Subsystem contract	24-field specification block
Registers (appendices)	evidence, decisions, requirements, failures, budgets, implementation map

Executive architecture map

NEXUS is a research and engineering program for a distributed human-compute operating environment whose physical endpoint is everyday eyewear. The thesis E4: human state, world state, task state, memory, intent, capability, computation and optical presentation should be first-class primitives of one operating architecture, coordinated across tiers that differ in timing, thermal and trust properties.



Figure 1. Master architecture. Colour carries function: cyan sensing, teal world state, violet intelligence, amber action and optics, red safety. The safety path at the bottom depends on nothing above it.

ANSWERS Where does each function live?

NEXUS is organized around five engineering contributions, each of which is a hypothesis with a stated falsifier:

- **Layered locality.** Reflex and safety terminate on the visor; cognition and memory live in a reservoir; scale lives remotely. Scale may improve capability and may never be required for safe operation.
- **A semantic ABI.** Observation, Intent IR, Action IR, Capability, Policy and Trust separate evidence, inference, suggestion, authorization and execution, and keep applications independent of sensors, models, compute location and display technology.
- **Quantified physical constraints.** Control delay against eye motion, pixel and etendue arithmetic, multiplicative optical efficiency, thermal steady state and transient, wire and radio energy, base-rate arithmetic for neural commands.
- **An independent optical safety architecture.** Learned components propose; deterministic, independent hardware decides; human exposure is gated.
- **A falsification program.** For each major hypothesis: a variable, a success criterion, a failure criterion and the decision that follows a negative result.

Engineering status dashboard

The highest rung each subsystem has reached on the evidence available. No entry is raised without an artifact. "External" means demonstrated outside NEXUS.

Subsystem	Maturity	Validation	Implementation	Provenance	Next proof
Semantic kernel, Intent IR, Action IR	M2	V2	PROTOTYPE	REPOSITORY-ASSERTED	replay-equivalence across providers (H1)
Capability graph and policy	M2	V2	PROTOTYPE	REPOSITORY-ASSERTED	gating table enforced with a rejection demo
World, context, memory	M2	V2	PROTOTYPE	REPOSITORY-ASSERTED	memory experiment RX4
Compute fabric and placement	M1	V0 V2	PROTOTYPE	PROPOSED	RX3 on a prototype reservoir
Virtual devices and replay	M2	V3	PROTOTYPE	REPOSITORY-ASSERTED	Aria-type and EEG replay with preserved sync
Android and AIDL integration	M2	V2	PROTOTYPE	REPOSITORY-ASSERTED	Cuttlefish validation (V4)
Eye tracking and gaze control	M1	V0 V1	NOT STARTED	EXTERNALLY DEMONSTRATED	O5 on hardware
EMG interaction	M1	V0	NOT STARTED	EXTERNALLY DEMONSTRATED	N1 to N5 on NEXUS data
EEG interaction	M1	V0	NOT STARTED	UNRESOLVED	N5: incremental information gain
Optical engine	M2	V1	PROTOTYPE	REPOSITORY-ASSERTED	O1 to O3 breadboard
Optical safety architecture	M1	V0 V1	NOT STARTED	PROPOSED	fault-injection harness O7
Reservoir hardware	M1	V0	NOT STARTED	PROPOSED	tethered prototype, RX1
Wearable integration	M0	not reached	NOT STARTED	UNRESOLVED	no integrated prototype exists
Neurodata governance	M1	V0	NOT STARTED	PROPOSED	enforced access-control tests

CRITERIA FOR VERIFIED

A repository snapshot with commit hashes, a test report that runs, and a recorded end-to-end trace (Appendix M) reproducible by a third party.

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PART I

SYSTEM THESIS

Identity · loop · requirements · budgets · evidence



The system is the loop. If measurement shows the closed loop does not reduce human effort at acceptable cost, the architecture is wrong regardless of component quality.

1 NEXUS identity

NEXUS is defined by a coupled closed loop, not by any component. Four logical nodes are frozen: **visor**, **neural node**, **reservoir** and **remote fabric**. Their physical packaging is open. A phone is a possible compute node, not the definition of the architecture.

Node	Optimized for	Owns	Must not	Status
Visor	sensing, reflex, optical control, immediate interaction, mass and comfort	eye state, head pose, frame timing, optical HAL, safety monitor, minimum local interaction, secure root	become the sole home of large models or persistent memory; depend on the network for safe optical output	FROZEN
Neural node	signal quality, low-noise acquisition, local feature extraction	AFE, ADC, impedance and contact monitoring, artifact estimates, uncertainty-aware features	decide or authorize actions; transmit raw neural data without explicit policy	FROZEN
Reservoir	persistence, memory, adaptation, large-model inference, sustained compute	world model, typed memory, planner, action compiler, scheduler, policy gateway	sit inside any safety or reflex loop	FROZEN
Remote fabric	scale, training, archive, long-horizon simulation	offline learning, large-model inference, replay at scale	be required for basic operation; receive raw neural data by default	FROZEN

What NEXUS is not

NEXUS is not just smart glasses, a chatbot, a BCI, an AR display, a VR system, a spatial desktop, a cloud client, a compute puck, a laser projector, a world model or an Android fork. These are candidate components. The system is NEXUS only when the semantic and physical loops are coordinated and measurable.

Rejected positions

- Arbitrary private-thought decoding as a near-term requirement. REJECTED
- A cloud round trip inside the reflex or safety loop. REJECTED
- Raw JSON or IPC transport of continuous sensor tensors. REJECTED
- Application-level direct source authority over the optical emitter. REJECTED
- A requirement to render a uniform 4K screen across the whole field. REJECTED
- Treating one benchmark or dataset as proof of generality, or treating simulation as physical validation. REJECTED

Product commitments carried into the architecture

Optical see-through (real-world photons remain the substrate) is the preferred branch PREFERRED; source and combiner architecture remain INVESTIGATING. The product line is glasses plus audio and wrist-sensing companions; the glasses require the highest engineering precision. Near-work eye strain and sedentary work are treated as coupled problems, with near-work eye strain the top-priority unresolved bottleneck E5.

2 Problem and prior art

Computing capability grew by orders of magnitude while the interface stayed organized around explicit device operation: choose an application, navigate to a target, manipulate a cursor, switch attention between device and task. Learned models improve what computation can do but do not remove the interaction architecture between a goal and a capability. The engineering problem is whether a system that estimates human and world state, infers intent with calibrated uncertainty, resolves capabilities through policy, places computation across heterogeneous nodes and presents results through an adaptive optical interface can reduce interaction burden, and at what cost in power, heat, latency, safety margin and privacy.

Domain	Established or externally demonstrated	NEXUS question
Wearable multi-sensor eyewear	Project Aria research device E2	Can the operating layer above the sensors be semantic rather than app-centric?
Wrist sEMG interaction	Kaifosh et al. 2025 E2	Can EMG be one evidence source in a broader intent system?
EEG decoding	public datasets; wearable-artifact reviews E2	Does EEG add information beyond gaze, EMG and context?
Retinal scanning	de Wit 1999 E2	Can retinal presentation be programmable and adaptive at wearable scale?
PIC-based displays	Shi et al. 2025 E2	Can integrated photonics meet wearable power, thermal and safety limits?
Waveguide holography	Choi et al. 2025 E2	Can large-etendue holography reach a manufacturable eyewear engine?
Visible optical phased arrays	Optics Letters 2025 E2	Can beam control reach useful near-eye information density?

Domain	Established or externally demonstrated	NEXUS question
Foveated rendering	established XR technique E1	Can foveation reduce physical optical resource, not only compute?
Predictive representations	I-JEPA, V-JEPA 2 E2	Can latent prediction serve as distributed world-state representation?
Distributed glasses and host compute	Qualcomm AR2, Meta Orion (reported) E2	Can placement be transparent to the semantic layer?

Combining these components is not a scientific contribution. The integration frontier is the coupled loop among sensing, intent, intelligence, distributed compute and adaptive optics, and whether the coupled system shows a quantitatively meaningful capability.

3 Closed-loop human-compute model

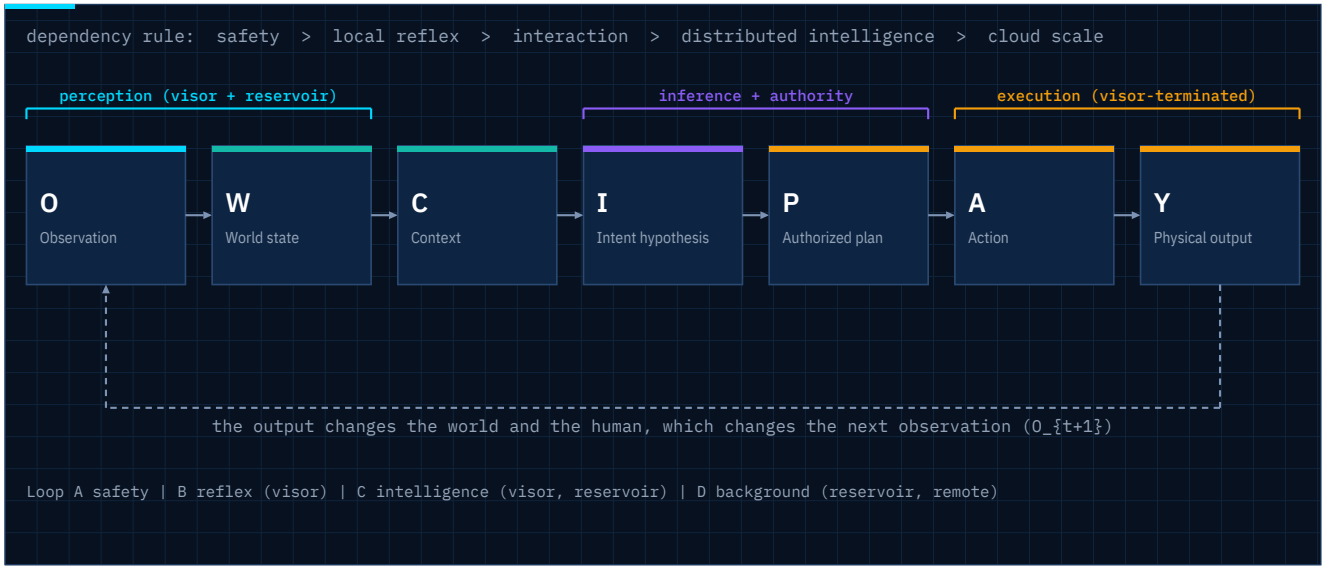


Figure 2. Human-context-compute closed loop. The loop is closed because the output changes the world and the human. It is a partially observed control problem, not a request-response assistant.

ANSWERS What is being controlled, and which part of the loop must never wait on a slower part?

Here O_t is the multimodal observation, W_t the world state, C_t context, I_t the intent representation, P_t the authorized plan, A_t the action and Y_t the physical or digital output. The loop is closed because Y_t changes the world and the human, which changes the next observation. It is a **partially observed control problem**. Formally, with H human, W world, M memory, G gaze and ocular state, O optical output and C compute, energy and network state E4 :

```
H(t+1) = F_H(H, A, W)      W(t+1) = F_W(W, A)      M(t+1) = F_M(M, H, W, A)
I(t)   = F_I(H, W, M, Q)   A(t)   = F_A(I, M, C, T)   O(t)   = F_O(A, G, Theta)   G(t+1) = F_G(G, O)
subject to  A(t) in authorized(T, Trust)   and   O(t) in safe
belief:  b(t+1)(x') proportional to  P(Y(t+1) | x')  *  sum_x P(x' | x, A) b(t)(x)
```

These equations compress the architecture; they are not a solved model. No F is identified. The meaning of a signal depends on context: the same EMG transient can mean different motor goals under different gaze, task and history, so modalities are fused as joint evidence and never as independent voters.

Hypotheses stated so they can fail

Hypothesis	Statement	Class
System	If the system holds an accurate, uncertainty-aware model of intent and environment, computation and optical presentation can be scheduled predictively, reducing explicit interaction, latency and wasted resource.	E4
Optical	A programmable retinal information field can avoid the spatial-energy scaling of a uniform near-eye display by allocating optical degrees of freedom according to gaze, task and perceptual need. A resource-allocation claim, not a thermodynamic loophole.	E4
Neural	Neural and physiological sensing need not decode arbitrary thought; it may add complementary evidence about emerging intent when combined with gaze, voice, context and environment.	E4
Reservoir	Moving persistent intelligence and sustained compute off the head improves the wearable thermal, mass and memory frontier while a local reflex loop preserves latency.	E4
Operating system	A semantic OS lets sensors, models, compute nodes and output devices change independently while human-facing concepts stay stable; testable by backend replacement.	E4

What to prove first. The first proof does not need the final hardware: multimodal observations, coherent state, an intent hypothesis, capability resolution, distributed compute, a semantic action, adaptive visual presentation and a feedback trace, end to end, in simulation and replay. The second proof shows the loop reduces explicit interaction on a matched task against strong baselines. The third shows the optical system can exploit that intelligence to reduce physical resource use without unacceptable degradation. The fourth shows the same loop survives the move to hardware.

4 Canonical end-to-end trace

One trace is the central architectural figure. Every other figure is a zoom into one of its rows. The trace is also the acceptance artifact for the first proof: it must be producible in host simulation, dataset replay, Cuttlefish, hardware-in-the-loop and on the prototype with the same trace schema (Appendix M).

t	STAGE | CARRIED STATE	WHERE IT RUNS	DATA MOVEMENT	LOOP	FAULT
t0	**WORLD** photons, people, objects	ambient	-	-	F0
t1	**CAMERA + EYE + IMU + VOICE + EMG** sensor clocks, hardware timestamps	visor / neural node	handles, not copies	B	F0
t2	**OBSERVATION** id, t, clock domain, quality, pose, calibration_ref	visor	typed descriptor; payload by handle	B	F0
t3	**TIME / CALIBRATION / QUALITY** offset, drift, sigma_t; per-channel quality state	visor	stale data down-weighted, logged	B	F0
t4	**WORLD + HUMAN + CONTEXT** entities, gaze target, task slice, uncertainty	visor, then reservoir	compact features or latents	C	F1 F2
t5	**INTENT HYPOTHESIS** verb, targets, calibrated posterior, evidence	reservoir (visor fallback)	typed Intent IR	C	F1
t6	**CAPABILITY GRAPH** provider resolution; latency, privacy, trust declared	reservoir	typed query	C	F5
t7	**POLICY** gate class; confirmation if irreversible	deterministic, local	policy token	C / A	F4
t8	**COMPUTE PLACEMENT** argmin J subject to tail-latency, thermal, privacy	scheduler	migration logged in trace	C	F6
t9	**ACTION IR** authorized, resource-hinted commitment	reservoir to visor	typed Action IR	C	F4 F5
t10	**GUI / OPTICAL PLAN** what is shown; request {eye, pose, depth, field, brightness}	visor	scene graph, GPU handles	B	F7
t11	**SAFETY ENVELOPE** clips request; independent of software health	visor hardware path	bounded values only	A	F8
t12	**EXECUTION | PHYSICAL OUTPUT** optical field to the eye; audio; haptics	visor	photons	A / B	F7
t13	**HUMAN RESPONSE | NEW OBSERVATION** gaze, EMG, correction, or ignore; O(t+1)	closes the loop	new observation	-	F0

Figure 3. Canonical end-to-end trace. One interaction from photons to photons. Every stage carries a timestamp in a named clock domain, a location, a data-movement form and a fault class (F0 to F8). Stage budgets are in Section 14; energy and thermal state are logged per trace record.

ANSWERS What happens, where, in what form, and what can fail at each stage of one interaction?

Trace field	Content	Why it exists
Timestamps	t_0 to t_{13} in named clock domains with $sigma_t$	stage latency and stale-data accounting
Latency	per-stage T_{sensor} to $T_{optical}$ at p50, p95, p99, jitter and drop count	tail behaviour decides perceived quality
Compute location	visor, reservoir tier, or remote; migration events	placement audit
Data movement	handle, typed message, latent or tensor; bytes moved	bandwidth and energy per task
Energy and thermal state	rail energy per stage; skin-adjacent, junction, source temperatures	energy per completed task; throttle history
Uncertainty	posterior calibration, timestamp and pose covariance, quality state	distinguishes "wrong" from "uncertain"
Policy decision	gate class, confirmation, token, denial reason	authority audit
Failure possibilities	F0 sensor to F8 safety, attributed to a layer	never attributed to "the system"

5 System invariants

Invariants hold in every mode and under every failure. A design change that violates one redefines NEXUS and requires a decision record.

ID	Invariant	Class	Enforced by
INV-1	Dependency order: safety > local reflex > interaction > distributed intelligence > cloud scale. Higher-scale layers may improve capability and may never be required for safe operation.	FROZEN	architecture review; RX2, O7
INV-2	observe, infer, suggest, authorize, execute are distinct. A neural signal is not a thought, an intent or an authorization.	FROZEN	trust chain; policy tests
INV-3	Learned components propose; deterministic components decide. Policy is never learned.	FROZEN	policy envelope; adaptation tests
INV-4	Control plane carries small typed messages; data plane carries payloads by handle. Continuous tensors are never serialized through control IPC.	FROZEN	buffer-ownership audit
INV-5	Class A messages terminate on the visor. Over any link they are advisory telemetry.	FROZEN	transport tests
INV-6	A single software fault must not create uncontrolled optical output. Tracking may improve efficiency but is never the only barrier.	FROZEN	O7 fault injection; Gate 1
INV-7	Dropping data is a decision the system reports, never a silent side effect.	PREFERRED	drop-policy telemetry
INV-8	A missing modality widens the posterior; it does not silently fall back to the prior.	PREFERRED	fusion tests (N5)
INV-9	Memory, world model and policy are separate functions so failures are attributable.	PREFERRED	failure taxonomy F0 to F8
INV-10	Raw neural data does not leave the body-area boundary by default.	FROZEN	access-control tests

6 Requirements hierarchy

Requirements are written as testable statements and trace to a verification artifact. Numeric limits are never invented here: where a limit depends on a standard, a hazard analysis or an experiment, the requirement refers to the budget or experiment that will set it. The full Requirements to Architecture to Verification matrix is Appendix O.

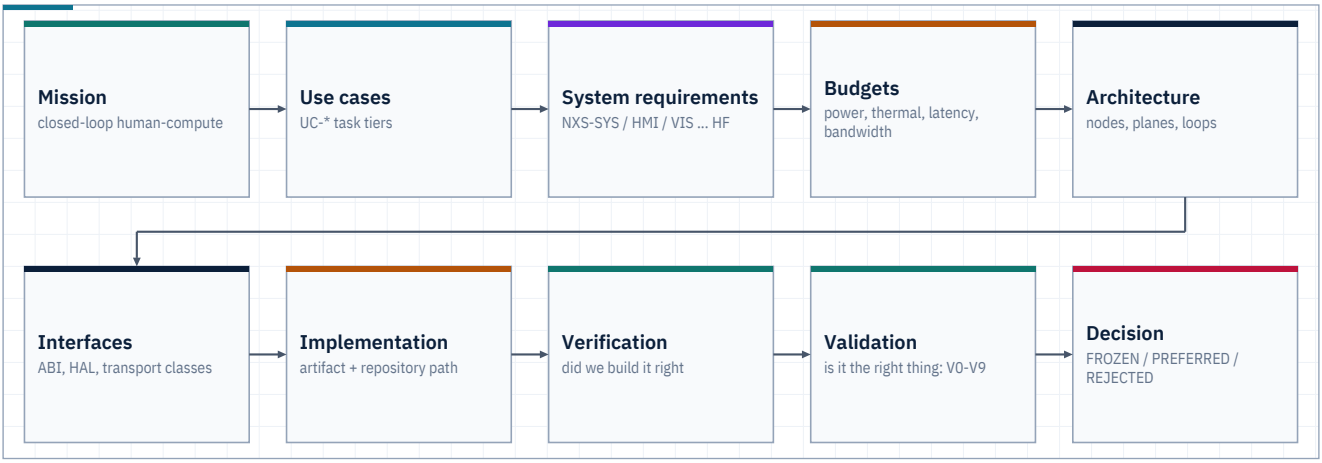


Figure 4. Requirements traceability chain. Every architectural statement terminates somewhere on this chain. A concept that cannot name its requirement, budget, interface, verification and decision is not finished.

ANSWERS Does every major idea terminate in a requirement and an experiment?

NXS-SYS SYSTEM

ID	Requirement	Verification	Rung / gate
SYS-001	The system shall implement the closed loop O to W to C to I to P to A to Y, with every stage reconstructable from a trace record.	trace reconstruction from replay	V3
SYS-002	No function whose failure can create an unsafe optical state shall depend on a network, the reservoir, the GUI or a learned component.	O7 fault injection	Gate 1
SYS-003	The semantic ABI shall be identical on every compute tier and under dataset replay.	backend-swap test H1	V3
SYS-004	Every operating mode (M3 to M0) shall degrade to the next lower mode without loss of safe operation.	RX2 failover	V5

NXS-HMI HUMAN INTERACTION

ID	Requirement	Verification	Rung / gate
HMI-001	Gaze alone shall not commit any action above the lowest gate class.	policy test suite	V2
HMI-002	Every action above gate class 1 shall be reachable through at least two modalities, so users without reliable EMG or voice are not excluded.	modality-removal study	V8
HMI-003	Interaction effort shall be reported as a vector with a paired wrong-action and correction rate.	H1, H2	V8
HMI-004	Proactive behaviour shall expose reason, confidence, reversibility and a suppression control.	H4	V8

NXS-VIS VISOR

ID	Requirement	Verification	Rung / gate
VIS-001	The visor shall emit Observations carrying clock domain and calibration reference.	schema validation	V2
VIS-002	The visor shall run Loops A and B with no reservoir link present.	RX2 disconnect test	V5
VIS-003	Skin-contact temperature shall stay within the standard-defined limit under worst-case sustained load.	thermal test (THM-004)	V7
VIS-004	Mass and centre of mass shall meet limits set by the comfort study (HF-001). Limit: OPEN.	mechanical + wear study	V7

NXS-OPT OPTICAL

ID	Requirement	Verification	Rung / gate
OPT-001	Optical architectures shall be compared by the vector K_{opt}, nominal and worst-case, never by a pixel count.	optical bench O2 to O4	V6
OPT-002	The usable eyebox shall be the set of pupil positions where image quality, luminance and exposure criteria hold simultaneously.	O4	V6
OPT-003	Accessible optical output shall be bounded by hardware independent of source command.	O7; radiometry	Gate 1
OPT-004	State age at any eye-locked element shall not exceed the bound from the registration tolerance, dt <= dtheta_tol / omega.	O5, RX6	V5
OPT-005	Population yield P(Q >= Q_{min}) shall be reported across IPD, pupil, refraction and fit.	population study	V8

NXS-NEU NEURAL

ID	Requirement	Verification	Rung / gate
NEU-001	Raw, feature and hypothesis streams shall have separate access policies.	access tests	V2
NEU-002	Every decoder output shall carry channel quality, artifact state and calibration version.	schema validation	V3
NEU-003	Benchmarks shall separate users, sessions and devices; shuffled within-session accuracy is not a result.	N1 to N4	V3
NEU-004	Neural evidence shall be removable without loss of function (graceful demotion).	N5	V3

NXS-CMP COMPUTE

ID	Requirement	Verification	Rung / gate
CMP-001	Every ComputeNode shall implement admission control that may refuse work; refusal is a logged, expected outcome.	RX3	V4
CMP-002	Placement shall bound a stated percentile of completion time, not the mean.	RX3	V5
CMP-003	A task shall keep its semantic identity across migration, and every migration shall be logged.	RX3, RX7	V4
CMP-004	Reflex-class work shall not be migratable off the visor.	scheduler test	V2

NXS-RF CONNECTIVITY

ID	Requirement	Verification	Rung / gate
RF-001	The link shall report a latency distribution (p50, p95, p99), loss, RSSI and throughput as link-quality state.	RX1	V5
RF-002	Link loss shall have no effect on Loop A.	RX2	V5
RF-003	Wired and wireless transports shall implement the same logical API.	RX7	V5
RF-004	Head-worn RF exposure and electromagnetic compatibility shall be evaluated against applicable standards.	compliance test	V7

NXS-PWR POWER

ID	Requirement	Verification	Rung / gate
PWR-001	Optical-source power shall have an independently enforceable upper bound.	hardware limiter test	Gate 1
PWR-002	Rails shall be instrumented per subsystem so energy per completed task is reportable.	power trace	V5
PWR-003	Battery protection shall be independent of software.	abuse and fault test	V6
PWR-004	Brownout shall drive the visor to a defined safe optical state.	rail fault injection	V5

NXS-THM THERMAL

ID	Requirement	Verification	Rung / gate
THM-001	Skin-adjacent, junction, battery, optical-source and enclosure temperatures shall be instrumented, with redundant sensing for skin and source.	instrumented prototype	V6
THM-002	A throttle ladder with hysteresis shall act before any thermal limit is reached.	O7	V6
THM-003	The human-worn thermal envelope shall be specified separately from the reservoir envelope.	budget review	V0
THM-004	Worst-case compute plus optical plus RF load shall be tested for skin-contact temperature.	O7, wear test	V7

NXS-SAFE SAFETY (INITIAL REGISTRY, CARRIED FROM V1.0)

ID	Requirement	Verification	Rung / gate
SAFE-OPT-001	Accessible optical output shall remain within the verified exposure envelope.	radiometric classification, fault tests	Gate 1
SAFE-OPT-002	Loss of valid eye-state confidence shall force a verified safe optical state.	tracking-loss injection	Gate 1
SAFE-OPT-003	No single software fault shall create an uncontrolled optical output state.	watchdog and fault injection	Gate 1
SAFE-VIS-001	Visual-system testing shall include long-duration exposure, not only short sessions.	H5	Gate 3
SAFE-THM-001	Skin-contact thermal conditions shall be measured under worst-case compute and optical load.	O7	Gate 2
SAFE-PWR-001	Optical-source power shall have an independently enforceable upper bound.	hardware test	Gate 1
SAFE-HF-001	Safety-critical user state shall be communicated predictably and consistently.	usability study	Gate 2
SAFE-MFG-001	Production units shall receive objective safety verification before shipment.	end-of-line fixture	Gate 4

NXS-SEC SECURITY

ID	Requirement	Verification	Rung / gate
SEC-001	Visor, neural node and reservoir shall have a hardware root of trust with secure and measured boot.	attestation test	V6
SEC-002	Every link shall use authenticated encryption with freshness (nonces, counters, sequence numbers).	protocol test	V5
SEC-003	Models and firmware shall be signed with rollback protection; adapters shall deploy by canary with rollback.	update test	V4
SEC-004	Every authorized action shall be recorded in an append-only audit log.	audit test	V2

NXS-SW SOFTWARE

ID	Requirement	Verification	Rung / gate
SW-001	Camera tensors, raw neural streams, high-rate IMU, GPU buffers and optical field data shall never be serialized through the control plane.	buffer audit	V2
SW-002	Every shared buffer shall declare producer, consumers, reference count, fence, timestamp, drop and stale policy.	static check	V2
SW-003	Every service shall declare process, privilege, interface, lifecycle, failure, restart behaviour, version and compatibility.	manifest validation	V2
SW-004	Every Intent shall carry a calibrated confidence, with ECE per verb published.	N5, replay	V3
SW-005	Dataset replay shall be indistinguishable from hardware at the Observation interface.	adapter conformance	V3

NXS-MFG MANUFACTURING

ID	Requirement	Verification	Rung / gate
MFG-001	Every physical subsystem shall have a tolerance budget and a calibration plan.	design review	V6
MFG-002	Optical yield shall be assessed by Monte Carlo of $P(Q < Q_{min})$ and $P(E_{acc} > E_{limit})$ under the fault model.	twin + bench	V6
MFG-003	Calibration data shall be versioned and traceable to each unit.	traceability audit	V6

NXS-HF HUMAN FACTORS

ID	Requirement	Verification	Rung / gate
HF-001	Visual comfort shall be studied longitudinally before any comfort claim.	H5	V8
HF-002	Bystander capture shall be indicated unambiguously.	H6	V8
HF-003	The cognitive-burden construct shall be calibrated against an established instrument before use in any decision.	H3	V8

7 Engineering budgets

The budget sheet fixes what each subsystem is allowed to consume before any subsystem is designed. No budget below is a final product specification. Value types are visually distinct: TARGET wanted, DESIGN PARAMETER initial test value, OPEN not yet set, MEASURED observed, DERIVED computed from stated assumptions. No budget is yet MEASURED.

Budget	Target	Allocation	Current evidence	Verification
Visor mass	OPEN set by comfort study HF-001	frame, lenses, optics, battery, SoC, sensors, haptics	none	mechanical test; wear study
Visor average power	OPEN	$P_{sense} + P_{comp} + P_{opt} + P_{RF} + P_{mem} + P_{conv}$	model only V0	instrumented power trace
Peak power	OPEN	per-rail ceiling; burst scheduling	model only	instrumented load step
Skin temperature	OPEN limit from applicable standard and test condition	SoC, optical source, RF, battery, AFE	lumped model V0	thermal test, worst-case load
Optical latency (eye-to-photon)	DERIVED state age <= dtheta_tol / omega; 1.0 ms at 0.5 deg, 200 us at 0.1 deg (omega = 500 deg/s)	tracking, prediction, control, render, scan	arithmetic	high-speed measurement O5
End-to-end interaction latency	DESIGN PARAMETER p95 <= 200 ms, p99 <= 500 ms for confirmation-level feedback (Loop C)	sensor, sync, fusion, intent, plan, transport, render, optical	none	trace, p50/p95/p99
Wireless bandwidth	DERIVED raw cameras 2.88 Gb/s; encoded features 10 to 100 Mb/s class; EEG raw 1.54 Mb/s	camera features, gaze, control, latents	arithmetic on assumed sensors	RF test RX1
Wireless energy per bit	OPEN	transport	none	power and RF test
Optical efficiency	OPEN (six stages at 0.5 give 1.6 %)	source, modulation, steering, coupling, waveguide, pupil	external papers only E2	optical bench O1 to O3
FOV	OPEN	optical architecture	unresolved	optical bench
Eyebox	OPEN	optical architecture	unresolved	optical bench
MTF	OPEN	optical chain	unresolved	optical bench
Neural SNR	OPEN	electrode, AFE, environment	unresolved	neural bench
False activation rate	DESIGN PARAMETER <= 1 per hour in the lowest gate class, consequence-weighted above	decoder + policy	arithmetic only	N6 human and neural study
Calibration time	OPEN reported separately from steady state	gaze, neural, optics	unresolved	calibration study N2, N4
Reservoir sustained power	TARGET R0 3 to 8 W; R1 10 to 45 W	compute, memory, RF, sensors, conversion	design hypothesis	enclosure thermal and task-energy runs
Battery energy	OPEN ; example: 20 Wh, 90 % usable, 5 W gives 3.6 h	cell, PMIC, system	arithmetic	battery test
Compute memory	TARGET R0 16 to 64 GB; visor OPEN	model, runtime, cache	design	workload test
Reliability	OPEN	per subsystem; serial dependencies avoided on safety path	unresolved	reliability program

Availability arithmetic motivates the safety rule: three serial dependencies of availability 0.99 give 0.970, while two redundant paths of availability 0.9 give $1 - (0.1)^2 = 0.99$. The safety-critical path therefore avoids serial dependence on non-safety components.

8 Evidence model and decision model

A statement is assigned the highest class its support justifies and no higher. Four rules follow: an external demonstration of one component never becomes evidence for the integrated system; a simulation measures the model, not the world; physical limits constrain what is possible and do not show a product is feasible; design-target numbers remain targets until measured on prototype hardware.

Decisions carry one of six statuses and live in the Decision register (Appendix P), each with alternatives, current choice, rationale, evidence, cost, risk and a **revisit condition**, so that bad branches die quickly and visibly. **Evidence** for every important claim is tracked in the Evidence register (Appendix R): claim, evidence, status, artifact, verification, next proof.

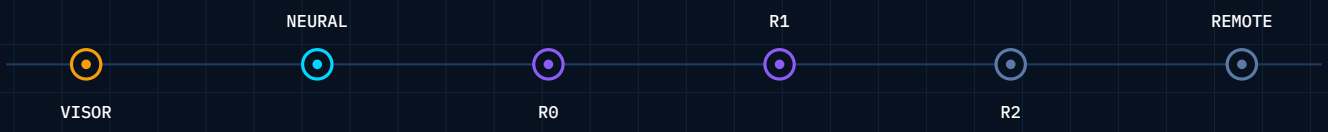
METRIC DEFINITION AND CORRECTION

Three headline metrics from earlier design variants are not usable as written. Interactions-per-success is maximized by a system that acts without asking, which is the over-action the trust model forbids. "Useful completion per joule per attention cost" has no unit for attention cost. The eta-human construct shares that unit problem. Before any is reported as a result, define (a) a measured attention proxy (gaze switches between task and interface, a standard workload instrument, or secondary-task cost) and (b) a paired wrong-action and correction rate that penalizes over-acting. Until then they are hypotheses about metrics.

PART II

SYSTEM ARCHITECTURE

Topology · locality · time · coordinates



Locality is a physical variable. What must be fast, safe or thermally sensitive lives near the eye; what must persist or scale lives elsewhere.

9 Physical topology

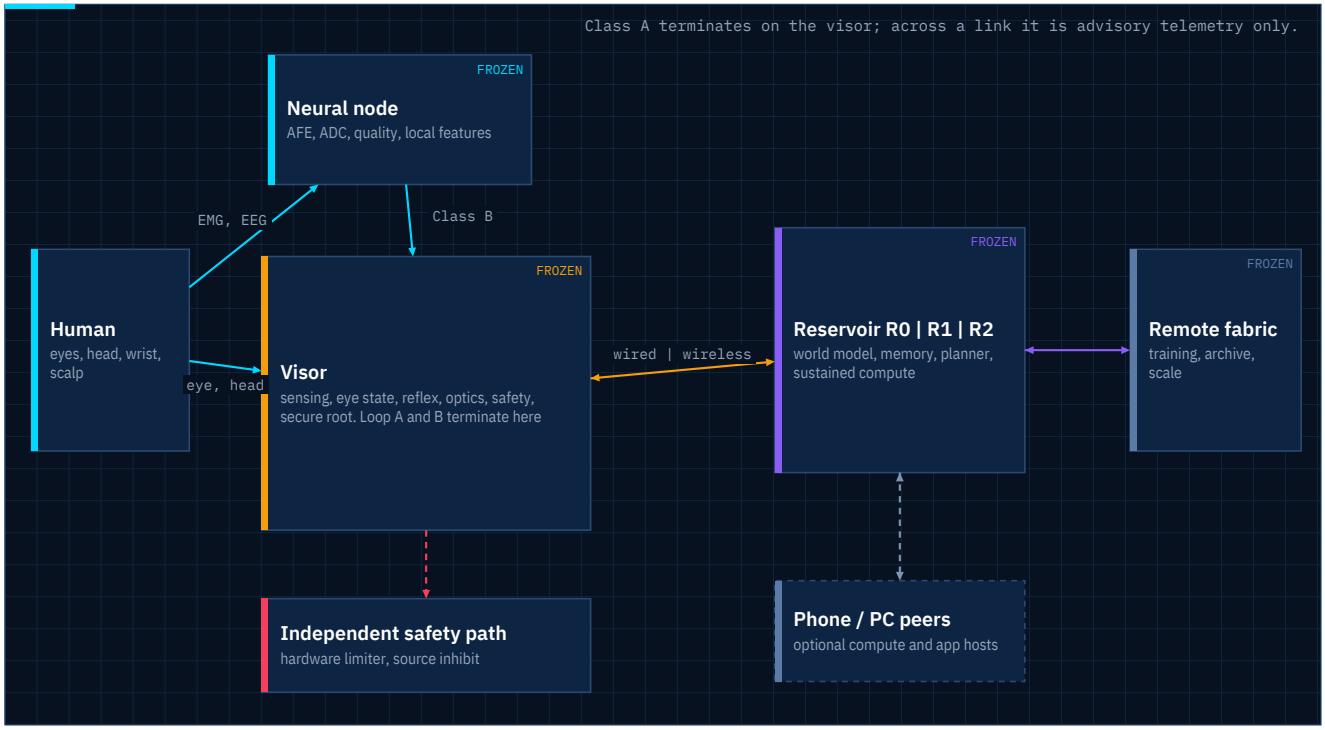


Figure 5. Four-node physical topology. Four logical nodes are frozen; their packaging is open. The neural node may be a wrist band, frame, temple or ear module.
ANSWERS Which node owns which function, and across which links does authority travel?

The architecture is distributed by design, not because the frame cannot host a computer. Sensing, reflex control and optics benefit from proximity to the eye and head, while sustained compute is bounded by skin-contact temperature, mass, battery volume and radio heat at the face. A hardware and software ownership boundary keeps each side testable:

Layer	Hardware concern	Software contract	Test
Neural	electrodes, reference, AFE, gain and noise, ADC, clock	NeuralDevice HAL: raw, feature and hypothesis streams distinct	SNR, impedance, artifact injection
Sensing	cameras, eye cameras, IMU, microphones, time base	Observation with clock domain and calibration reference	timestamp jitter, drift
Optics	source, steering, waveguide, thermal path	Optical HAL with health and safe state	metrology, fault injection
Compute	SoC, reservoir silicon, memory bandwidth	ComputeNode ABI with admission control	task energy and thermal runs
Power	cells, PMIC, rails, tether	power and thermal state exposed to the scheduler	rail and temperature logs
RF	antennas, body blockage	link quality with latency distribution	p99 latency versus body pose

Operating modes

Four execution modes differ only in which nodes are present. The semantic ABI is identical across them, so a mode change is a placement change and never a semantic change.

Mode	Name	Behaviour	Role
M0	glasses-first	reflex, UI, basic sensing and small local models on the visor only	the safe floor
M1	phone-assisted	a phone acts as compatibility host and compute node for app access and connectivity	compatibility
M2	reservoir	persistent memory, assistant, scene model and sustained inference in R0 or R1	normal operation
M3	reservoir plus workstation or cloud	heavy inference, simulation, training and replay move to R2 or remote	research and scale

10 Visor

The visor is a functional decomposition, not a SKU. No silicon is named because none is supported by evidence. Blocks: local SoC (CPU, GPU, NPU, ISP, DSP), memory, storage, clock and synchronization, power management, wireless, an optional wired high-bandwidth link, sensor interfaces, optical driver and monitor interfaces, the safety monitor and a secure hardware root. It owns eye state, head pose, frame timing, the Optical HAL and minimum local interaction, and it never becomes the sole home of large models or persistent memory.

NXS-C01 Visor runtime and SoC

01	**Purpose**	Host everything whose latency or safety consequence forbids a network dependency: sensing, eye state, head pose, frame timing, optical HAL, safety monitor, minimum local interaction, secure root.
02	**System requirement**	SYS-002, VIS-001 to VIS-004, OPT-004, PWR-004, THM-001, SW-001.
03	**Inputs**	World, eye, IMU and microphone Observations; neural-node feature stream (Class B); reservoir results (Class B and C); touch and button input.
04	**Outputs**	Observation descriptors with payload handles; intent evidence; optical requests to the safety envelope; advisory Class A telemetry; audio and haptics.
05	**State**	Clock-domain estimates, calibration set, safety state machine, thermal and power state, small local models, local reduced-interaction state. No long-term memory.
06	**Interface**	Stable AIDL control services; HALs Camera, EyeTracker, IMU, NeuralSensor, AudioInput, OpticalEngine, Battery, Transport; data-plane handles.
07	**Timing requirement**	Loop A bounded by hazard analysis; Loop B state age <= dtheta_{tol} / omega `DERIVED`; no blocking IPC inside the frame loop.
08	**Power budget**	`OPEN` allocated from the visor average-power budget. $P_{visor} = P_{sense} + P_{opt} + P_{local}(w_{local}) + P_{RF}(r, d) + P_{base}$.
09	**Thermal budget**	Skin limit set by applicable standard and test condition `OPEN`. Skin-adjacent, junction and source temperatures instrumented; throttle ladder with hysteresis.
10	**Bandwidth requirement**	`DERIVED` from assumed sensors: raw cameras 2.88 Gb/s and eye cameras 230 Mb/s stay on the visor; outbound encoded features 10 to 100 Mb/s class.
11	**Memory requirement**	`OPEN`. Holds small models only (wake word, eye-state, quality and artifact estimators); sized by workload test, not by benchmark.
12	**Accuracy / quality requirement**	Gaze error budget $\text{sigma}_{tot}^2 = \text{sigma}_{cal}^2 + \text{sigma}_{slip}^2 + \text{sigma}_{pupil}^2 + \text{sigma}_{quant}^2 + \text{sigma}_{lat}^2$, each term measured separately; timestamp uncertainty reported on every Observation.
13	**Safety constraint**	Loop A and the optical inhibit are independent of OS, GPU and network. enter_safe_state() is reachable only through the safety service.
14	**Security boundary**	Secure root; SELinux domain per service; safety supervisor is a separate process; raw eye imagery never leaves the visor.
15	**Failure modes**	Gaze loss, camera failure, IMU drift, clock failure, GPU hang, thermal overload, brownout, radio loss.
16	**Degradation behavior**	M0 is the floor: reflex, UI, basic sensing, small local models. Eye-tracking loss gives wider foveal region, then uniform presentation, then static safe output.
17	**Calibration requirements**	Factory (cameras, IMU, optical alignment and source, clocks), per-user (eye model, IPD), session (fit, slip), online drift. Every Observation carries its calibration reference.
18	**Manufacturing implications**	Thermal path to skin, sensor and optical alignment retention, flex routing, sweat and humidity, serviceability. Tolerance budget by propagation of sigma_i.
19	**Verification method**	Timestamp jitter and drift measurement; thermal map; fault injection; hardware-in-the-loop traces.
20	**Validation experiment**	O5, O6, O7, RX2, RX6.
21	**Acceptance criterion**	Loops A and B continue with the link removed; skin temperature within the standard limit at worst-case load; sync jitter within its allocated budget `OPEN`.
22	**Falsification condition**	If no available SoC and sensor combination holds the Loop B state-age bound, eye-locked optics are infeasible and the design falls back to fixation-onset updating or uniform presentation. If the thermal limit forbids even M0 on skin, local function is reduced.
23	**Current maturity**	`M1` `V0` `V1`; wearable integration `M0`. Implementation `NOT STARTED` for hardware; software surfaces `REPOSITORY-ASSERTED`.
24	**Next implementation artifact**	Visor timing harness: development SoC with IMU and an eye camera on hardware timestamps, producing the first Loop B state-age histogram.

11 Neural node

The neural node is logical and separable. It comprises electrodes, reference and bias, an analog front end, amplification, ADC, local DSP, clock, signal-quality monitor, battery or tether power and a transport with a secure feature path. It processes enough locally to protect the signal and lower interaction latency, and it never becomes the main intelligence computer. Analog constraints are never hidden behind software abstractions. Its full contract is in Part V.

Candidate packaging	Advantage	Limit	Natural modality	Status
Wrist band	demonstrated EMG interface; stable skin contact	not a head signal; arm-posture sensitivity	EMG	OPEN
Frame and temple	integrated with visor; no extra device	restricted scalp coverage; hair and fit variability; EOG contamination	limited EEG, EOG	OPEN
Ear or earbud	mechanically stable near temporal regions	few channels and limited montage	limited EEG	OPEN
Band or cap	broader coverage	social acceptability, donning cost	EEG research	OPEN

Selection depends on experiments N1 to N5 and on comfort and acceptability studies. The wrist band is the near-term product-branch choice for EMG; EEG on frame or temple is a research channel.

12 Reservoir and remote fabric

The reservoir carries persistence, memory, adaptation, large-model inference and sustained compute; the remote fabric carries scale, training, archive and long-horizon simulation. R0, R1 and R2 are a resource hierarchy, not a semantic one. Placement, transport, energy and thermal consequences, and the falsifiable reservoir thesis are Part VI.

13 Locality model

Functions are placed by six properties: latency requirement, thermal sensitivity, bandwidth, privacy, energy cost and fault criticality. The table is a design policy PREFERRED, not a measurement.

Function	Placement	Reason
Eye-state estimation	visor	closes the optical control loop; delay maps directly to angular error
Head pose, frame timing	visor	motion-to-photon path
Optical control, safety inhibit	visor	independent of OS, GPU and network
EEG and EMG quality and artifact estimates	neural node or visor	needs the raw signal; produces compact uncertainty
Wake word, simple intent classification	visor	latency and privacy
Scene understanding, world-model update	reservoir	memory- and compute-heavy; tolerates tens to hundreds of ms
Large language or multimodal reasoning	reservoir, then remote	throughput; not reflex-critical
Memory consolidation, policy training	reservoir, R2, remote	batch; privacy filter first
Final optical command generation	visor	cannot depend on a wireless round trip

PHYSICAL LIMIT: REFLEX LOCALITY

$$\Delta\theta = \omega \, \Delta t$$

CONSTRAINS

Maximum state age of any eye- or head-locked optical element, and therefore where Loop B may run.

ENGINEERING CONSEQUENCE

At 500 deg/s, 1 ms of state age is 0.5 deg and one 90 Hz sample interval (11.1 ms) is 5.6 deg. Foveal resolution is about one arcminute. A loop that holds sub-degree registration during fast motion must close on the visor with local state and local prediction.

EXPERIMENT / FALSIFIER

Experiment O5 and RX6: measured registration error not proportional to delay means a missing term in the model. omega = 500 deg/s is a literature working value, not re-verified E0.

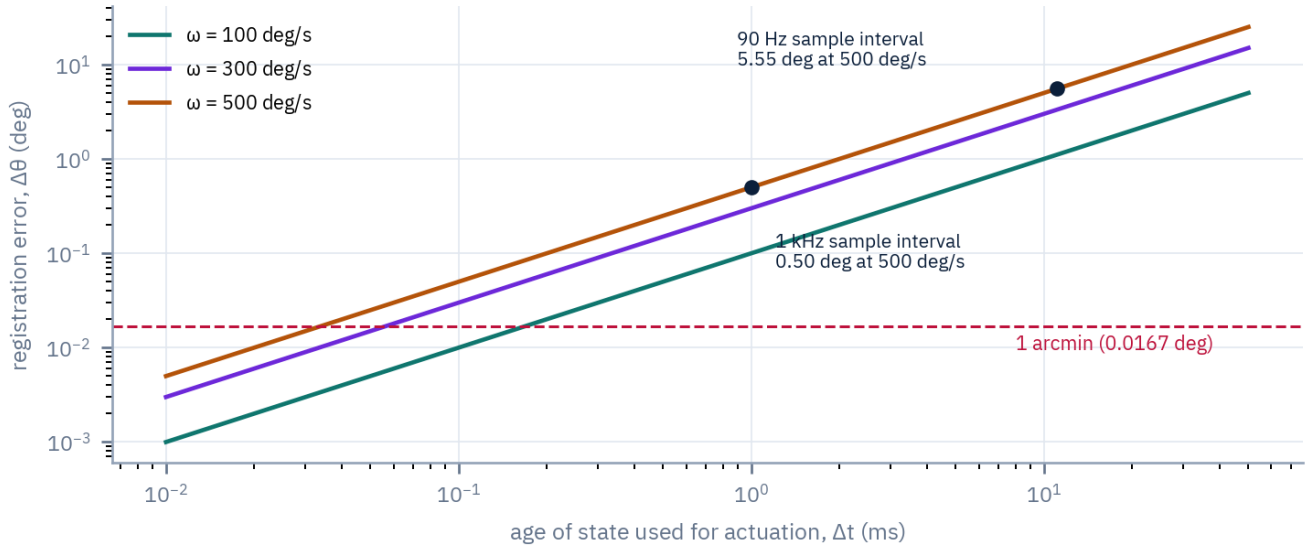


Figure 6. Registration error versus state age. Log axes. Markers show a 1 kHz and a 90 Hz gaze stream. The figure is arithmetic on delta-theta = omega x delta-t; it implies no NEXUS specification.

ANSWERS Why must the reflex loop close on the visor?

State age	Error at 500 deg/s	Multiple of 1 arcmin (0.0167 deg)
20 us	0.01 deg	below one arcminute
100 us	0.05 deg	3x
1 ms	0.5 deg	30x
2 ms (vendor-reported link figure, unverified)	1.0 deg	60x
5 ms	2.5 deg	150x
11.1 ms (one sample at 90 Hz)	5.6 deg	330x
20 ms	10 deg	600x

Two design options follow, neither selected: a gaze sensor much faster than a 90 Hz camera, or a predictor with an uncertainty estimate; or exploiting the documented reduction of visual sensitivity during saccades by updating foveated content at fixation onset. Whether that is perceptually acceptable is experiments O5 and O8. Late-stage reprojection of rendered frames against a fresh head pose is the documented XR model for what the visor keeps local. Even a low-latency wireless link has no worst-case bound, and vendor PHY rates bound throughput, not latency tails.

14 Timing architecture

Time is infrastructure, not a field. Clock domains are sensor, host monotonic, display, neural, reservoir and network. Each domain is tracked for offset, drift, jitter, uncertainty, staleness and sequence loss, and every observation carries its timestamp, clock domain, synchronization uncertainty, sequence number and staleness.

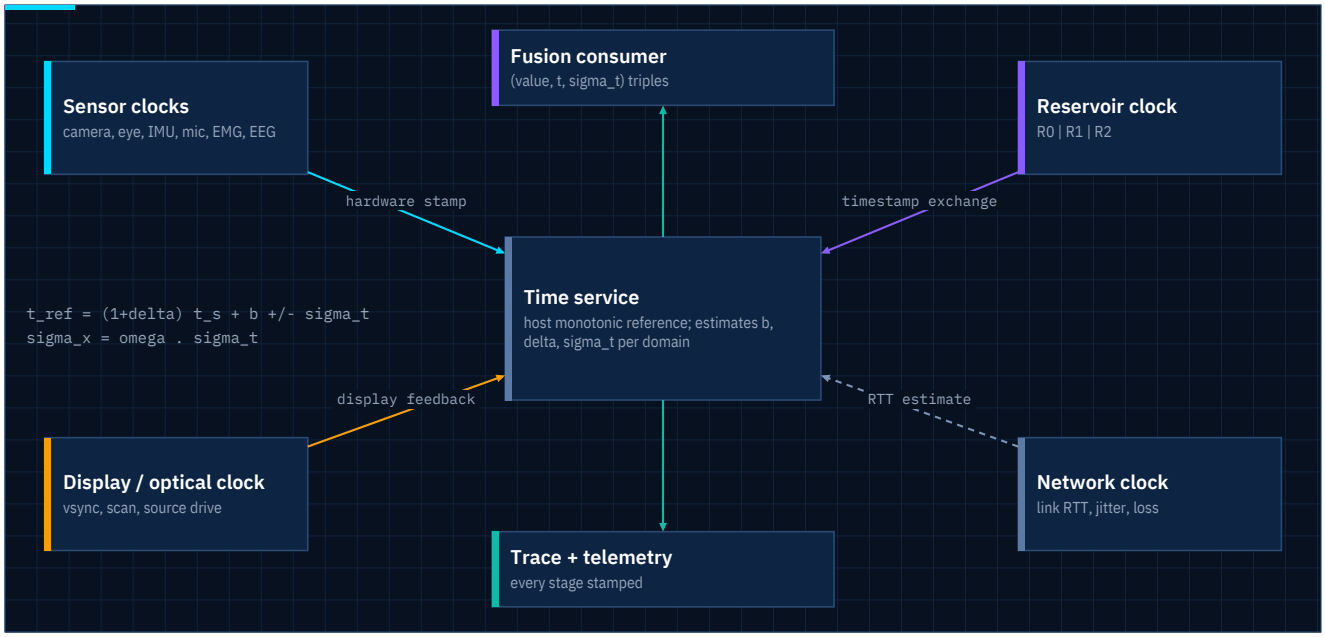


Figure 7. Time synchronization graph. Each clock domain is tracked against a host monotonic reference. Fusion consumes (value, timestamp, uncertainty) triples, never bare values.

ANSWERS How old is each sample, and how uncertain is its time?

LINEAR CLOCK MODEL

$$t_{ref} = (1 + \delta)\, t_s + b \pm \sigma_t$$

CONSTRAINS

Fusion validity: which timestamps may be combined and with what uncertainty.

ENGINEERING CONSEQUENCE

delta is fractional drift and b the offset, estimated by timestamp exchange and re-estimated because drift varies with temperature. Timing uncertainty becomes spatial uncertainty by $sigma_x$ = omega $sigma_t$: at 500 deg/s, 1 ms of timestamp uncertainty is 0.5 deg of gaze uncertainty. For slow quantities the same $sigma_t$ is irrelevant, so timing requirements are stated per fusion pair, not globally.

EXPERIMENT / FALSIFIER

Measured residual of the linear model against a hardware reference exceeding $sigma_t$; temperature-dependent drift not captured E0.

Four loops and their budgets

The latency budget is organized by loop, not by component. Loop A is independent and deterministic; Loop B is the reflex loop; Loop C is intelligence; Loop D is background. Percentile rows state what is reported; values marked OPEN are set by the experiment named in the verification column.

Attribute	Loop A Safety	Loop B Reflex	Loop C Intelligence	Loop D Background
Content	source inhibit, watchdog, hazard monitors	eye state, head pose, optical stabilization, immediate interaction	world model, memory, reasoning, planning, confirmation	learning, sync, archival, model updates
Runs on	visor hardware path	visor only	visor or reservoir	reservoir, remote
Nominal latency	OPEN tau_{tot} from hazard analysis	DERIVED dt <= $dtheta_{tol}$/omega: 1.0 ms at 0.5 deg, 200 us at 0.1 deg	DESIGN PARAMETER p50 target set by H1; start 100 ms	none real-time
p95 / p99	worst-case bound, not a percentile	both must meet the bound; tail counts	DESIGN PARAMETER p95 200 ms, p99 500 ms	n/a
Jitter	bounded and deterministic	counted inside state age	reported	n/a
Maximum stale age	stale monitor reading forces inhibit	dt_{max}; older state discarded	per-modality lifetime; down-weighted, logged	n/a
Allowed packet loss	zero; loss is a fault	sample loss bridged by prediction within horizon	retried; reported	retried
Prediction horizon	none; no prediction in the safety path	bounded where residual sigma stays below tolerance OPEN (RX6)	seconds of context	n/a
Recovery	latch SAFE_OFF, re-run SELFTEST	widen foveal region, uniform presentation, static safe output	local reduced mode on timeout	resync and replay
Safety timeout	independent watchdog	confidence gate drives reduced energy within tau_{tot}	Class B deadline with timeout	none

Which loop may depend on which: a loop may read from itself and from lower-numbered loops only through advisory channels, never through blocking calls, and no loop may wait on a higher-lettered loop.

May depend on	A	B	C	D
Loop A	self	inhibit authority only (A overrides B)	inhibit authority only	inhibit authority only
Loop B	telemetry in, advisory	self	gaze state read, non-blocking	no
Loop C	no	no (B never waits for C)	self	read-only artifacts (models, memory)
Loop D	no	no	no	self; may read anything

LATENCY DECOMPOSITION

$$T_{tot} = T_{sen} + T_{syn} + T_{fus} + T_{int} + T_{pln} + T_{tx} + T_{rnd} + T_{opt}$$

CONSTRAINS

End-to-end interaction latency of Loop C and the eye-to-photon path of Loop B.

ENGINEERING CONSEQUENCE

Each term is reported at p50, p95 and p99 with jitter and drop count, because mean latency hides the tail users perceive. Terms map to owners: sensor and sync (visor), fusion and intent (visor or reservoir), plan and transport (reservoir and link), render and optical (visor).

EXPERIMENT / FALSIFIER

A trace in which the sum of stage medians does not reproduce the measured end-to-end median points to unaccounted queueing or an unsynchronized clock pair.

Contract: time and clock service

NXS-C02 Time and clock service

01	**Purpose**	Maintain a host monotonic reference and per-domain offset, drift and uncertainty so every Observation can be fused with an explicit timing error.
02	**System requirement**	SYS-001, VIS-001, SW-005; supports OPT-004 and every fusion pair.
03	**Inputs**	Hardware timestamps from sensors, display vsync, reservoir and link timestamp exchanges, temperature.
04	**Outputs**	(b, delta, $sigma_t$) per domain; clock health; staleness helper for consumers; trace timestamps.
05	**State**	Linear clock model parameters per domain with covariance; last-exchange age; temperature coefficient estimate.
06	**Interface**	AIDL query for (t_{ref}, $sigma_t$) conversion; hardware sync lines where available; Class A timestamp channel.
07	**Timing requirement**	Estimation updates fast enough that $sigma_t$ stays inside each fusion pair requirement; stale estimates flagged `OPEN`.
08	**Power budget**	`OPEN`; negligible against sensing, but its wake-ups count in duty-cycle budgets.
09	**Thermal budget**	Negligible on the visor `OPEN`.
10	**Bandwidth requirement**	Small control traffic; Class A reserved capacity.
11	**Memory requirement**	Small; kilobytes of model state per domain.
12	**Accuracy / quality requirement**	$sigma_t$ reported with each estimate; residual of linear model against a reference measured per temperature.
13	**Safety constraint**	A clock fault must not create unbounded error silently: it raises a fault and widens every dependent posterior.
14	**Security boundary**	Timestamps are signed by the source domain; replay protection by sequence numbers.
15	**Failure modes**	Clock failure, drift step, exchange loss, sync-line fault, temperature excursion.
16	**Degradation behavior**	Hold the last estimate with growing $sigma_t$; fall back to local timestamps; fusion down-weights, logs and reports.
17	**Calibration requirements**	Offset and drift per sensor pair at factory; online re-estimation; re-estimation trigger on temperature change.
18	**Manufacturing implications**	Hardware sync routing and oscillator selection constrain sensor placement.
19	**Verification method**	Hardware-timestamp loopback; injected drift and step tests; jitter histograms.
20	**Validation experiment**	RX1, RX2, O5.
21	**Acceptance criterion**	Residual after correction within $sigma_t$ budget for every fusion pair `OPEN`.
22	**Falsification condition**	If residual timing error across fusion pairs exceeds the pair tolerance after correction, multimodal fusion over those pairs is invalid and must be dropped.
23	**Current maturity**	`M1` `V0`; clock abstraction `REPOSITORY-ASSERTED`.
24	**Next implementation artifact**	Timestamp-exchange estimator with a recorded-trace conformance test against a reference clock.

15 Coordinate architecture

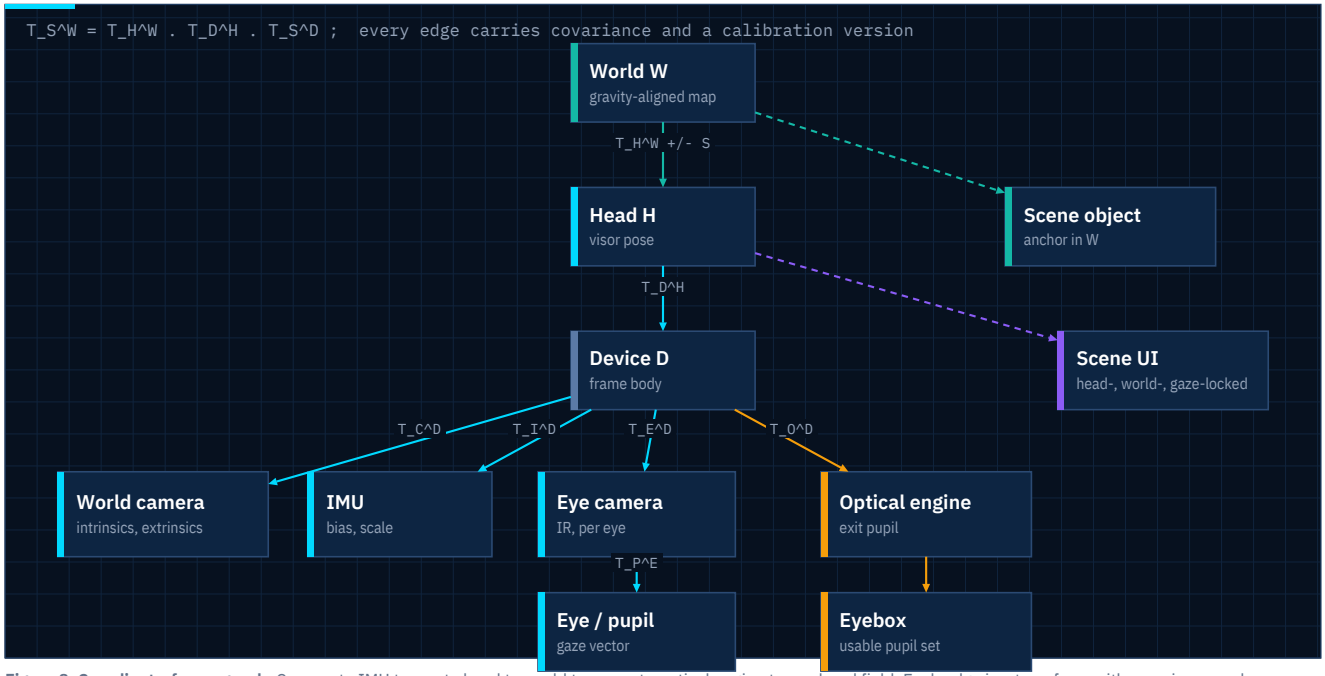


Figure 8. Coordinate-frame graph. Camera to IMU to eye to head to world to scene to optical engine to rendered field. Each edge is a transform with covariance and a calibration version.

ANSWERS How does pose uncertainty propagate to the rendered optical field?

Transforms among world, head, device, camera, eye, pupil, screen, optical engine and scene objects compose as $T_S^W = T_H^W T_D^H T_S^D$. Every spatial quantity carries its frame, the transform, the transform uncertainty and the calibration version, and pose uncertainty propagates to rendering and optical targeting. Content placement mode (head-locked, world-locked, body-relative, workspace-relative, gaze-relative, task-relative) is a system variable chosen by the compositor, not a property of an application.

Transform	Source of estimate	Uncertainty source	Recalibration trigger
T_H^W **head in world**	visual-inertial estimate	tracking quality, motion blur, drift	tracking loss, loop closure
T_D^H **device on head**	fit and slip estimate	frame flex, fit variation	don/doff, slip detector
T_C^D, T_I^D **camera, IMU**	factory calibration	assembly tolerance, thermal expansion	drift against factory reference
T_E^D **eye camera**	factory + per-user	slip, occlusion	confidence drop
T_P^E **pupil and gaze**	eye model	pupil dilation, refractive state	per-user recalibration
T_O^D **optical engine**	factory alignment	thermal drift, shock, ageing	self-test against reference

PART III

SOFTWARE SYSTEM

Substrate · semantic ABI · planes · intent to action



The semantic architecture must be implementable, not merely diagrammable: every object has an owner, a lifetime and a failure behaviour.

16 Android and Linux substrate

NEXUS is a semantic operating layer on an Android/AOSP and Linux substrate PREFERRED. It reuses documented platform mechanisms (E1 , to be re-verified per target release) and adds meaning above them. It does not replace Linux at ring 0, and no claim is made that Android has been replaced. Deeper integration proceeds only where measurement justifies it.

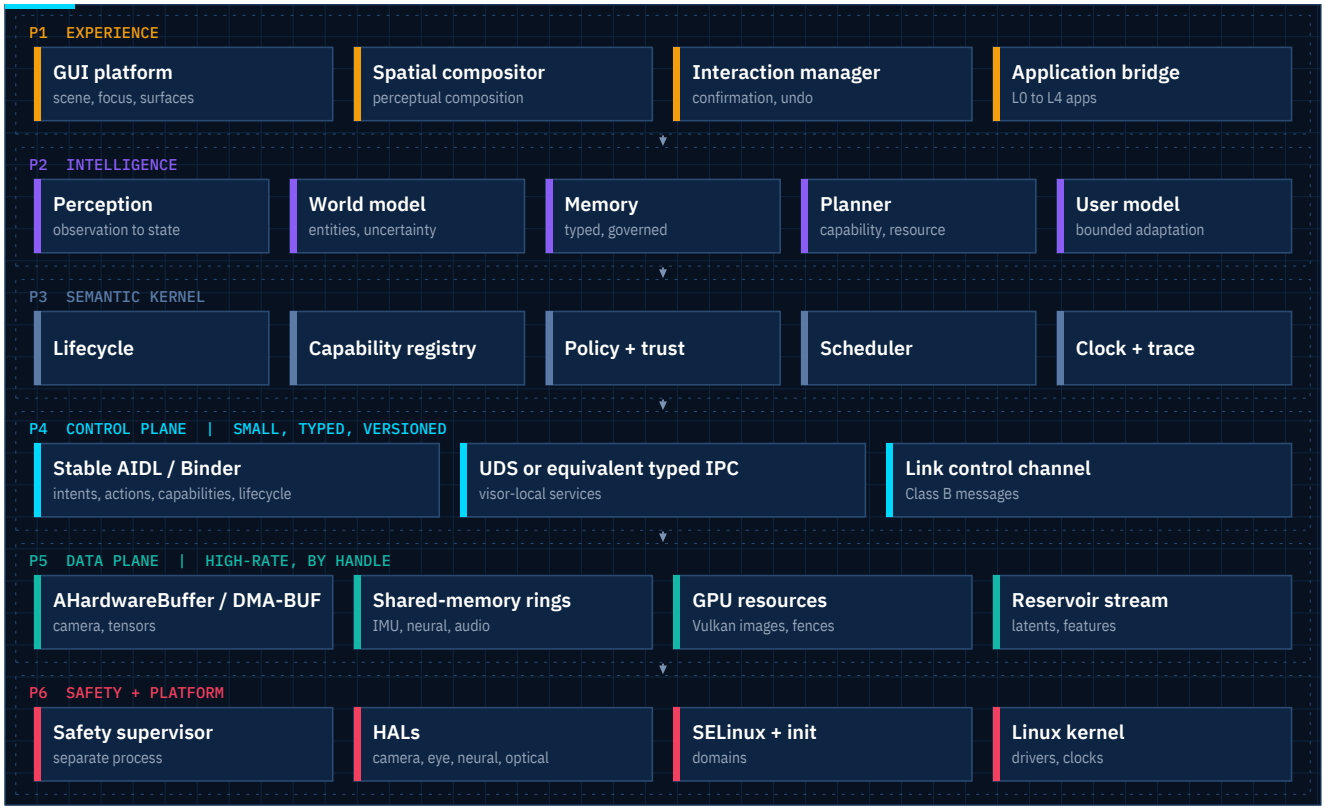


Figure 9. Six-plane software architecture. NEXUS is a semantic layer on an Android and Linux substrate. Control and data planes are separate planes, and the safety supervisor is a separate process.

ANSWERS What runs in which plane, and what may never cross between planes?

Mechanism	Established role	NEXUS use
Binder, stable AIDL	IPC between apps, framework services and HAL processes; Java, NDK, C++ and Rust backends with different stability properties	control plane for intent, action, capability, policy, lifecycle
HALs	hardware abstraction below the framework	Optical, Neural, sensor and compute-node HALs
init, SELinux	process lifecycle and mandatory access control	per-service domains; separation of privileged services (proposal)
SurfaceFlinger, Hardware Composer, BufferQueue, Gralloc	composition of visible layers; buffer allocation	compositor path to display or optical HAL where the configuration exposes it
AHardwareBuffer, DMA-BUF	zero-copy buffer sharing	camera, tensor and render-target data plane
Vulkan	explicit GPU API	NEXUS compositor rendering and compute
OpenXR 1.1	spatial-computing interoperability	one spatial backend, not the semantic definition of NEXUS

Partition placement of NEXUS services (system, system_ext, vendor, product) depends on the Android product configuration and is not asserted. Direct compositor control is claimed only where the selected configuration exposes it. Every semantic abstraction should move through interface, host implementation, virtual device, Android implementation and hardware implementation without the layers above noticing. Integration path E4 : host C++ semantic runtime, Android native runtime, AIDL contracts, Binder services, framework integration, vendor and HAL integration, AOSP device target, Cuttlefish validation, hardware.

17 Semantic kernel and normative ABI

The semantic kernel owns lifecycle, resource discovery, capability registration, policy, trust, scheduling, eventing, timing, fault containment and telemetry. Its objects are typed because they differ in lifetime, authority and failure semantics. A generic event bus erases those differences and makes safety review impossible. The statements below are normative for the ABI.

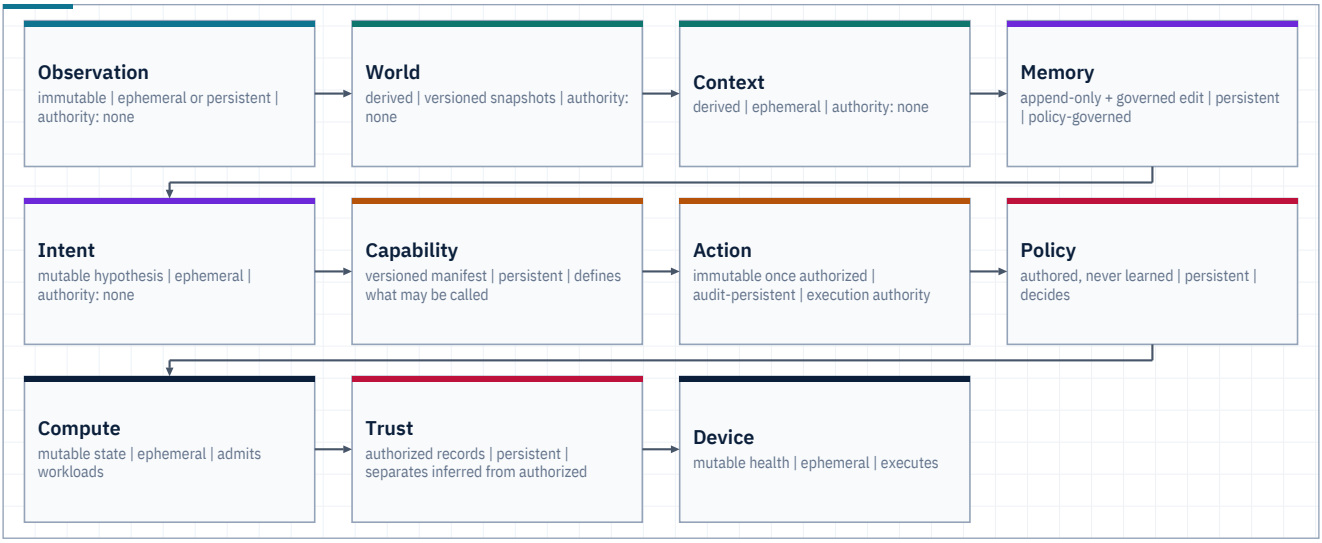


Figure 10. Semantic ABI object classes. Mutability, lifetime and authority differ by object, which is why a generic event bus is rejected.

ANSWERS Which objects are immutable, derived, authorized or ephemeral?

Object	Mutability	Lifecycle and persistence	Authority	Privacy class
Observation	immutable	created by source; ephemeral unless a memory write policy admits it	none	inherits source: camera and audio bystander-sensitive, eye biometric-adjacent, neural neural-sensitive
World	derived; snapshots immutable, graph mutable by the estimator only	continuously updated; checkpointed	none	derived from sensitive inputs
Context	derived	per task episode; ephemeral	none	task-scoped
Memory	append-only with governed edit and delete	retention policy; persistent; encrypted	none; governed by policy	user-controlled, inspectable, exportable
Intent	mutable hypothesis until committed; immutable after commit record	per episode; ephemeral unless written to memory	none	capability-scoped
Capability	versioned manifest	install to revoke; persistent	defines what may be called	declared in the manifest
Action	immutable once authorized	authorization to verification; audit-persistent	execution authority	audit class
Policy	authored and signed, never learned	versioned releases; persistent	decides	system
Compute	mutable node state	node lifetime; ephemeral	admits workloads	low
Trust	append-only authorization records	persistent	separates inferred from authorized	audit class
Device	immutable identity, mutable health	pairing lifetime; identity persistent, health ephemeral	executes	low

Object	Required schema fields	Provenance	Uncertainty	Failure semantics
Observation	id, source, modality, timestamp, clock_domain, payload_handle, pose, coordinate_frame, calibration_ref, confidence, quality, sampling_rate, sequence, compression_state, staleness	device, preprocessing, model versions	quality state, sigma_t, pose covariance	stale, missing, corrupted, out of order
World	entity set, relations, geometry, poses, temporal state, snapshot_id	estimator version, contributing observation ids	per-entity and per-relation covariance	incoherent or stale state
Context	world slice, memory refs, task, gaze and human state, relevance weights	selection model version	relevance confidence	wrong relevance
Memory	type, content, source, confidence, created, updated, scope, retention, sensitivity, encryption, deletion path	writer, policy decision, score	confidence	contamination, leakage
Intent	intent_id, verb, targets, parameters, context_refs, confidence, urgency, reversibility, commitment, evidence, timestamp	fusion model, calibration_ref	calibrated posterior; ECE per verb	misestimated, late

Object	Required schema fields	Provenance	Uncertainty	Failure semantics
Capability	name, version, provider, inputs, outputs, required_context, modalities, latency_class, privacy_class, compute and trust requirements, side_effects, reversibility, cost_model	manifest signer	declared cost model	unavailable, misconfigured
Action	action_id, capability, arguments, authorization, risk, reversibility, resource_hint, presentation	intent id, policy token	risk tier	partial execution
Policy	rule set, gate classes, bounds, version, signature	author, review record	none (deterministic)	misconfigured
Compute	id, tier, accelerators, memory, thermal, power, network, locality, security_domain, admission(w), health	attestation	state age	overload, thermal
Trust	subject, grant, scope, expiry, modality of authentication	issuer	none	bypassed
Device	id, type, capabilities, calibration, health, firmware version	attestation	health confidence	fault, drift

Serialization and compatibility rules

- Control-plane objects use typed, versioned interface definitions (stable AIDL or an equivalent typed IDL). Payloads are referenced by handle, never inlined.
- Within a major version a schema changes additively: new optional fields only. Unknown fields are preserved and ignored. Removing or changing the meaning of a field is a major version.
- Every object carries schema_version. Consumers declare a supported range; producers publish a minimum consumer version. Deprecation windows are stated in the manifest.
- Time is encoded as (int64 nanoseconds, clock-domain id, σ_t). Spatial quantities are encoded with frame id, transform reference and covariance.
- Audit and replay use a canonical deterministic encoding; dataset replays pin the schema version.
- confidence is a calibrated posterior and evidence.weight is attribution. Neither is a probability unless its calibration record says so.

NXS-C03 Semantic kernel

01	**Purpose**	Own lifecycle, capability registration, policy, trust, scheduling, clock, fault containment and telemetry for the typed semantic objects.
02	**System requirement**	SYS-001, SYS-003, SW-001 to SW-005, SEC-004.
03	**Inputs**	Service manifests, capability manifests, policy releases, Observation descriptors, Intent and Action objects, health reports.
04	**Outputs**	Lifecycle events, admission decisions, policy tokens, trace records, audit log entries.
05	**State**	Registry of services and capabilities, policy version, trust records, trace buffer, fault state per service.
06	**Interface**	Stable AIDL services (Runtime, Capability, Compute, Telemetry); manifest schema; registry queries.
07	**Timing requirement**	Control-plane calls are not in the frame loop; registry lookups bounded; policy evaluation on the Loop C budget.
08	**Power budget**	`OPEN`; kernel services must be idle-cheap on the visor.
09	**Thermal budget**	`OPEN`; no sustained load.
10	**Bandwidth requirement**	Small typed messages only; kilobytes per interaction `OPEN`.
11	**Memory requirement**	Registry and trace buffers sized by workload; `OPEN`.
12	**Accuracy / quality requirement**	Deterministic policy evaluation; trace completeness measured as the fraction of actions that can be fully reconstructed.
13	**Safety constraint**	Policy and safety supervisor are outside the learned core; policy cannot be altered by any adaptive component.
14	**Security boundary**	Capability manifests signed; least privilege; SELinux domain per service (proposal); audit log append-only.
15	**Failure modes**	Registry corruption, policy misconfiguration, service crash loop, trace overflow, capability impersonation.
16	**Degradation behavior**	Safe defaults deny; stateless services restart; registry restored from signed snapshot; trace overflow drops by declared policy and logs the drop.
17	**Calibration requirements**	Policy and manifest signing keys; clock service calibration reference.
18	**Manufacturing implications**	Process boundaries and privilege separation affect packaging of services on the visor.
19	**Verification method**	Manifest validation, schema conformance, replay equivalence, rejection demos.
20	**Validation experiment**	H1 backend-swap, safety rejection demonstration.
21	**Acceptance criterion**	A scripted scenario passes unchanged across providers; every action has a complete trace; every denied action cites a rule.
22	**Falsification condition**	If an unmodified semantic scenario cannot run across two providers without application changes, the ABI is not an ABI and the platform thesis weakens.
23	**Current maturity**	`M2` `V2` `REPOSITORY-ASSERTED`.
24	**Next implementation artifact**	Replay-equivalence test suite and a rejection-demo trace committed with the schemas.

18 Observation and data plane

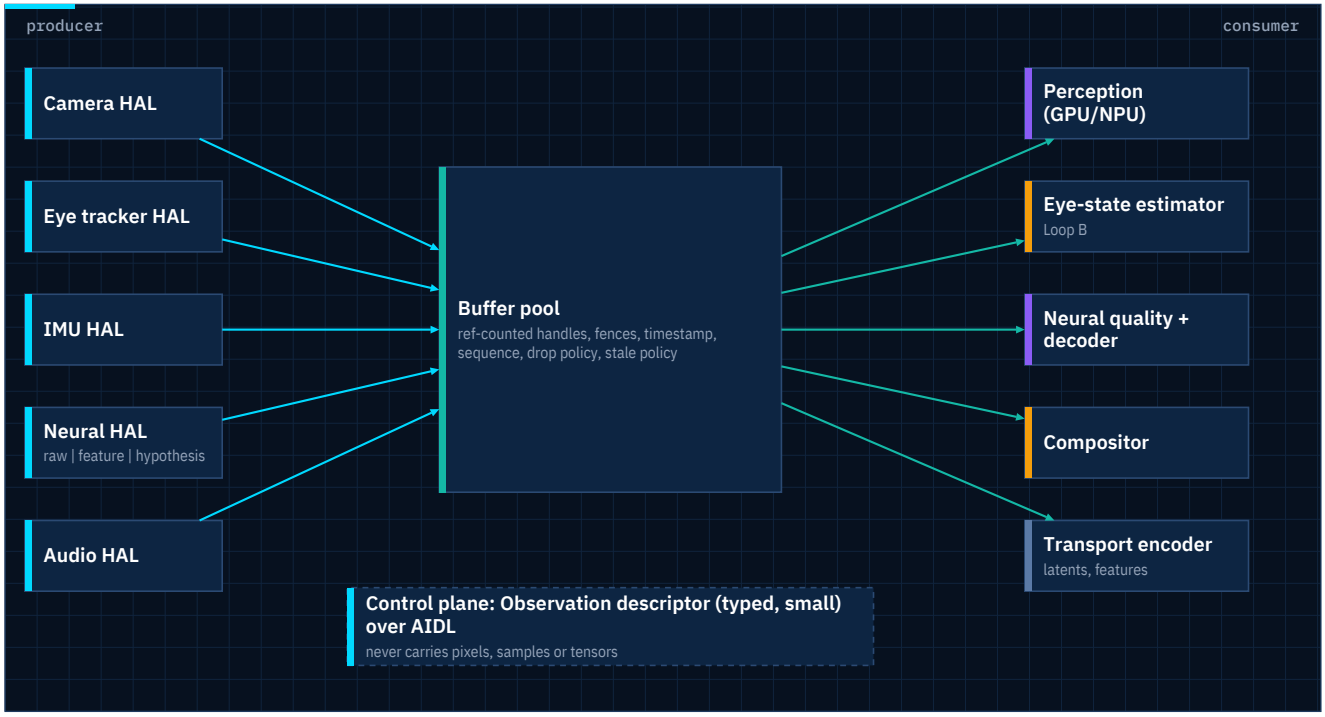


Figure 11. Observation and data plane. High-rate payloads move by handle with a producer, consumers, reference count, fence and timestamp. Only a small typed descriptor crosses the control plane.

ANSWERS How does information physically move, and who owns each buffer?

Continuous camera tensors, raw neural streams, high-rate IMU, GPU buffers and optical field data are **never serialized** through the control plane. They move by handle. Every shared buffer has an owner and a lifetime:

- The producer owns a pool slot until it publishes. After publication the slot is immutable.
- Consumers acquire a reference (reference count increments), wait on the buffer fence, read, and release. The pool reclaims a slot only when its count is zero and its fences have signalled.
- A producer may not overwrite a slot with outstanding references. If no slot is free it applies the stream drop policy and reports the drop.
- Every published buffer carries a timestamp in a named clock domain, a sequence number and the calibration reference of its producer.
- Backpressure is declared per stream; stale data is handled per stream, never silently.

Stream	Producer to consumers	Handle and ownership	Synchronization	Drop and backpressure	Stale policy
World camera frames	camera HAL to perception, compositor, transport encoder	AHardwareBuffer pool; producer until publish, then shared read-only	acquire fence, hardware timestamp	drop oldest unconsumed; report count	frame older than pose lifetime is down-weighted
Eye camera frames	eye HAL to eye-state estimator only	DMA-BUF ring; visor-local	hardware timestamp, sequence	drop oldest; never blocks Loop B	older than state-age bound discarded
IMU samples	IMU HAL to pose estimator, artifact estimators	shared-memory ring, single producer	monotonic sequence, hardware time	overwrite oldest with gap marker	gap marker widens pose covariance
Neural raw samples	neural HAL to quality and artifact estimators	shared-memory ring in a restricted domain	sample clock, sync pulse	no drop on raw; quality estimator reports loss	stale window excluded from fusion
Neural features and hypotheses	quality estimator and decoder to fusion	typed messages; small	window start, end, $sigma_t$	drop with report	down-weighted by age
Audio	audio HAL to wake word, speech	shared-memory ring	sample clock	overwrite oldest	utterance boundary rules
GPU tensors and render targets	perception, compositor	Vulkan resources; ownership by queue family with explicit transfer	timeline semaphore or fence	frame skip by policy	frame older than display deadline dropped
Optical command and field data	optical policy to safety envelope to driver	bounded structures; visor hardware path	frame timing, signed sequence	never queued beyond one frame	stale command discarded and reported

NXS-C04 Observation and data plane

01	**Purpose**	Move high-rate data by handle with explicit ownership, timing and drop semantics, so every consumer knows what a sample is and how old it is.
02	**System requirement**	SW-001, SW-002, SW-005, VIS-001, INV-4, INV-7.
03	**Inputs**	Sensor HAL outputs with hardware timestamps; calibration references; producer policies.
04	**Outputs**	Published buffers, Observation descriptors, drop and staleness reports, per-stream telemetry.
05	**State**	Pool occupancy, reference counts, fence state, sequence counters, drop counters.
06	**Interface**	HAL buffer interfaces; AHardwareBuffer, DMA-BUF, shared-memory rings, Vulkan resources; typed descriptor over stable AIDL.
07	**Timing requirement**	Zero blocking IPC in the frame loop; publication latency counted inside T_{sensor} and T_{sync}.
08	**Power budget**	`OPEN` ; zero-copy by construction, DMA and memory bandwidth dominate.
09	**Thermal budget**	`OPEN` ; memory traffic contributes to SoC hotspot.
10	**Bandwidth requirement**	Camera 2.88 Gb/s and eye 230 Mb/s (assumed sensors) stay local; descriptors are bytes per frame.
11	**Memory requirement**	Pool depth times frame size per stream `OPEN` ; sized from workload tests.
12	**Accuracy / quality requirement**	Zero silent drops; sequence continuity verifiable at consumers.
13	**Safety constraint**	Raw neural and eye streams live in restricted domains; the safety path does not share pools with perception.
14	**Security boundary**	Handles are capabilities; a consumer holds only the handles it is granted; raw streams need explicit enable.
15	**Failure modes**	Pool exhaustion, fence timeout, handle leak, descriptor and payload mismatch, sequence gap.
16	**Degradation behavior**	Declared drop policy per stream; reduce resolution or rate before dropping; report every drop.
17	**Calibration requirements**	Producer clock calibration reference carried on every buffer.
18	**Manufacturing implications**	Buffer alignment and memory bandwidth constrain SoC choice.
19	**Verification method**	Buffer-ownership audit; leak and fence tests; sequence-gap injection.
20	**Validation experiment**	RX1, RX2, S0 to S2.
21	**Acceptance criterion**	No buffer leak over a stated duration; every drop reported; no serialization of payloads over control IPC in a static audit.
22	**Falsification condition**	If handle-based transport cannot meet the Loop B timing on the target SoC, camera and eye processing must be fused into one process on the visor.
23	**Current maturity**	`M2` `V2` `REPOSITORY-ASSERTED` for the host abstraction; Android handles `PROPOSED` .
24	**Next implementation artifact**	Static buffer-ownership checker plus a sequence-gap injection test on the host runtime.

19 Control plane and service topology

The NEXUS service family is not a set of independent microservices. Its members share versioned semantic contracts, one state model, telemetry, policy, lifecycle, permissions and resource ownership. The Safety service and its hardware monitor are separated from all others. The table gives the contract each service must declare; processes and privileges are proposals pending the target configuration.

Service	Process and privilege (proposed)	Interfaces	Owns	On failure	Compatibility
Runtime	system; lifecycle, registry	AIDL control	service registry, manifests	restart from signed snapshot	major.minor, additive
Device	system; HAL clients	AIDL; HAL interfaces	device graph, health	mark device degraded; restart	HAL versions pinned
Sensor	system; sensor HAL access	AIDL + buffer pools	stream pools, Observation emitters	restart; drop policy applies	schema additive
Neural	restricted domain; raw neural access	AIDL + restricted ring	quality and artifact state	drop neural evidence; widen posterior	stream classes distinct
World	system	AIDL + shared snapshots	world graph, estimator	rebuild from observations; mark stale	snapshot version
Memory	isolated; encrypted store	AIDL	typed memory, write policy	read-only mode; restore from snapshot	record schema versioned
Intent	system	AIDL	intent hypotheses, commitment states	fall back to explicit input	verb set versioned
Capability	system; manifest verification	AIDL	capability registry	deny by default	manifest signed, versioned
Compute	system	AIDL	admission, placement, migration log	run local; refuse remote	ComputeNode ABI versioned
Transport	system; link keys	AIDL + link channels	sessions, link quality	M0; resync on return	protocol version
Scene	system; compositor	AIDL + Vulkan	scene graph, surfaces	minimal indicators	scene schema versioned
Optical	system; no source authority	AIDL to safety envelope	optical policy	static safe presentation	Optical Action IR versioned

Service	Process and privilege (proposed)	Interfaces	Owns	On failure	Compatibility
Safety	separate process; hardware path	hardware lines + minimal AIDL	safety state machine, interlocks	latch SAFE_OFF	frozen interface
Telemetry	system	AIDL + trace sink	trace buffer, audit log	drop by policy; report	trace schema versioned

20 Intent IR

Intent IR is the canonical representation of what the human intends (`REPOSITORY-ASSERTED` schema; usefulness as a cross-modal ABI is `E4`). The same Intent can be produced by gaze plus voice, gaze plus EMG, gaze plus a button, a gesture, touch or keyboard. The capability layer is blind to the modality that produced it.

```
{ "intent_id": "uuid", "verb": "select", "targets": ["entity:42"], "parameters": {},
  "context_refs": ["task:current"], "confidence": 0.91, "urgency": 0.20,
  "reversibility": "high", "commitment": "supported",
  "evidence": [ {"source":"gaze","weight":0.54}, {"source":"voice","weight":0.31},
                {"source":"neural","weight":0.15} ],
  "timestamp": "...", "provenance": "fusion-model-vX", "calibration_ref": "..." }
```

Field	Meaning	Constraint
verb	LOOK, SELECT, OPEN, MOVE, PLACE, COMPARE, EXPLAIN, REMEMBER, SEARCH, ACT, CONFIRM, CANCEL	the verb set is a hypothesis; coverage is measured
confidence	posterior probability that this verb and target match the intent	calibrated; ECE per verb published
evidence	list of (source, weight)	attribution of the posterior; weights need not sum to 1
reversibility, urgency	cost of undoing; time pressure	reversibility gates commitment
commitment	emerging, speculative, supported, confirmed, committed, executing, verified; rejected or cancelled exits	state machine with hysteresis
provenance, calibration_ref	model and calibration versions	required for replay

CALIBRATION OF CONFIDENCE

$$\mathrm{ECE} = \sum_b \frac{n_b}{n} \left| \mathrm{acc}(b) - \mathrm{conf}(b) \right|$$

CONSTRAINS

Whether the confidence field means anything.

ENGINEERING CONSEQUENCE

A confidence of 0.91 is meaningful only if about 91 percent of intents reported at 0.91 are correct. Without a reliability diagram per verb the field is decoration. Commitment uses hysteresis: enter SUPPORTED above τ_{enter}, leave below $\tau_{exit} < \tau_{enter}$; thresholds are tuned per verb and reversibility class and no value is asserted here.

EXPERIMENT / FALSIFIER

ECE per verb on held-out users and sessions (N5); too few windows per bin makes ECE itself unreliable.

Prediction is not preparation is not authorized execution. A high-confidence prediction can still be wrong. An uncertain prediction may justify prefetching a result, which is reversible and has no side effects. It never justifies commitment.

NXS-C05 Intent runtime

01	**Purpose**	Turn calibrated evidence into revisable intent hypotheses with explicit commitment state.
02	**System requirement**	HMI-001 to HMI-004, SW-004, SYS-001.
03	**Inputs**	Fused evidence (gaze, voice, EMG, EEG, touch, context, memory), modality quality and timing, task state.
04	**Outputs**	Intent IR objects with calibrated confidence, evidence attribution and commitment state.
05	**State**	Active hypotheses, commitment state per hypothesis, hysteresis thresholds, calibration records.
06	**Interface**	Stable AIDL Intent service; Intent IR schema; commitment query and cancel.
07	**Timing requirement**	Loop C: p95 and p99 per budget; commitment decisions respect minimum dwell per verb.
08	**Power budget**	`OPEN` ; small models run on the visor, larger on the reservoir.
09	**Thermal budget**	`OPEN` ; visor-resident part is small.
10	**Bandwidth requirement**	Evidence rates from kb/s features, not raw samples; tens of kb/s `OPEN` .
11	**Memory requirement**	`OPEN` ; model and memory sized by workload.
12	**Accuracy / quality requirement**	ECE per verb; consequence-weighted false-action rate; correction rate; commitment timing.
13	**Safety constraint**	Never authorizes; cannot raise a gate class; prediction cannot commit.

14	**Security boundary**	Capability-scoped output; no raw sensitive evidence stored by default.
15	**Failure modes**	Miscalibration, late intent, over-commitment, flicker, modality dropout, drift.
16	**Degradation behavior**	Missing modality widens the posterior; fall back to explicit input; speculative preparation only while reversible.
17	**Calibration requirements**	Per-user adapter calibration; recalibration trigger when ECE drifts.
18	**Manufacturing implications**	Not applicable.
19	**Verification method**	Replay evaluation, calibration reports, hysteresis tests, ablations.
20	**Validation experiment**	N5, N6, N7, H1, H2.
21	**Acceptance criterion**	Lower consequence-weighted false-action rate than strong baselines at matched sensitivity on held-out users; ECE within a threshold set before the experiment.
22	**Falsification condition**	If fused intent does not beat a strong gaze-plus-explicit-confirmation baseline on held-out users, intent fusion complexity is removed.
23	**Current maturity**	`M2` `V2` `REPOSITORY-ASSERTED` ; usefulness `E4` .
24	**Next implementation artifact**	Calibration report generator (reliability diagrams per verb) wired to the replay harness.

21 Action IR and capability graph

Intent resolves through the capability graph, policy and authorization into Action IR, which lowers to resources and device-specific operations. `SELECT(entity:42)` lowers to: acquire gaze lock, verify target, prepare selection, request confirmation if reversibility or risk demands it, commit. For optics the same Action IR lowers to `{eye, pose, depth, field, brightness, foveation, temporal profile}` requests submitted to the optical policy, which the independent safety envelope may clip. Action IR is independent of sensor, accelerator, reservoir and display technology.

```
{ "action_id": "uuid", "capability": "document.compare",
  "arguments": {"left": "paper:A", "right": "paper:B"},
  "authorization": "policy-token", "risk": "low", "reversibility": "high",
  "resource_hint": "reservoir", "presentation": {"mode": "spatial", "priority": "normal"} }
```

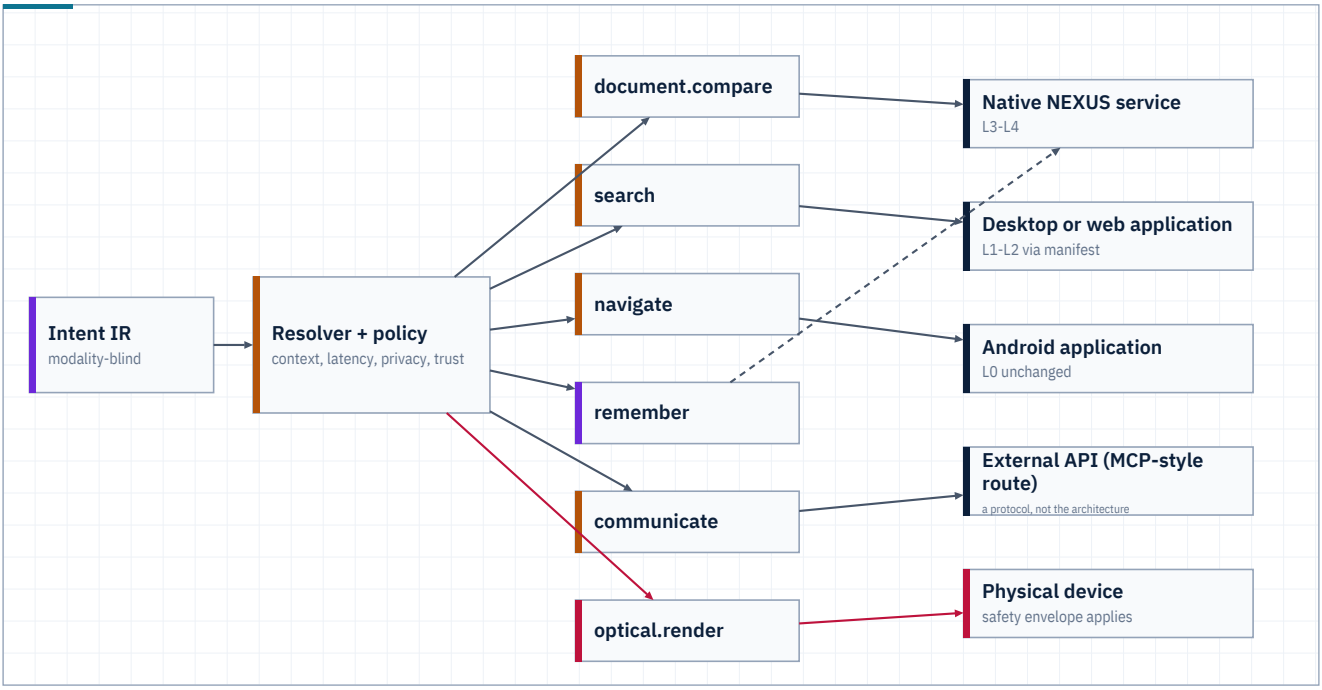


Figure 12. Capability graph. Capabilities are separate from devices. A capability is blind to the modality that produced the intent and to the provider that executes it.
`ANSWERS` How does an intent find a provider without knowing sensors, models or hardware?

A capability is an authorized operation a provider can perform. The graph is separate from physical devices, so NEXUS inherits the existing software ecosystem instead of rebuilding it. Each capability declares required context, input and output modalities, latency class, privacy requirements, compute requirements and trust requirements. External tool protocols (MCP-style) are an integration route; a protocol carries semantics and does not define them. The CUDA comparison is an analogy about a stable abstraction above volatile hardware and nothing more.

Level	Description	What changes
L0	existing Android application runs unchanged	nothing

Level	Description	What changes
L1	NEXUS can launch and route to it	launch and window management
L2	selected capabilities become addressable	capability manifest
L3	application adopts NEXUS semantic APIs	intent and context APIs
L4	application is intent-native	no screen-centric flow required

22 Trust, policy and the learned-policy envelope



Figure 13. Trust architecture. Learned components propose; deterministic components decide. Action classes carry minimum gates.

ANSWERS Where can authority be gained, and where can it only be removed?

Action class	Examples	Minimum gate
1 Reversible, local, no external effect	scroll, highlight, preview, prefetch	policy check; no confirmation
2 Reversible with external visibility	create draft, add to calendar	policy plus a visible undo window
3 Irreversible or external	send message, purchase, delete, share	explicit confirmation through an authenticated modality
4 Physical or safety-relevant	optical source control, machinery, robot, vehicle	deterministic safety layers decide; learned layers only propose

A learned policy chooses within a permitted capability set. It cannot redefine permissions, bypass interlocks, install sensors or rewrite safety constraints. Online adaptation is a subset of the policy envelope: $pi_u(a|s) = pi_0(a|s) + Delta\ pi_{phi}(a|s, M)$, with Delta pi bounded and Delta pi = 0 for class 3 and 4. Neural sensing adds a sixth term to the chain, stimulate; NEXUS neural architecture is sensing and interaction only, and stimulation would sit in a higher risk tier with its own review.

SPECULATIVE EXECUTION RULE

$$EU_{spec} = P\,B_{saved} - (1 - P)\,C_{waste}$$

CONSTRAINS

Whether work may start before an intent is confirmed.

ENGINEERING CONSEQUENCE

Speculation is permitted only if the work is reversible and the expected value is positive: P is the posterior, B_{saved} the latency or effort saved if confirmed, C_{waste} the energy, bandwidth and interruption cost if not. A decision rule, not an algorithm. Parallel active goals form a task graph G = (V, E) with concurrent, speculative and confirmation-gated nodes.

EXPERIMENT / FALSIFIER

H4: correction rate and interruption rate under speculation exceeding a baseline without speculation means parallel speculative work over-acts E5 .

23 Action state machine

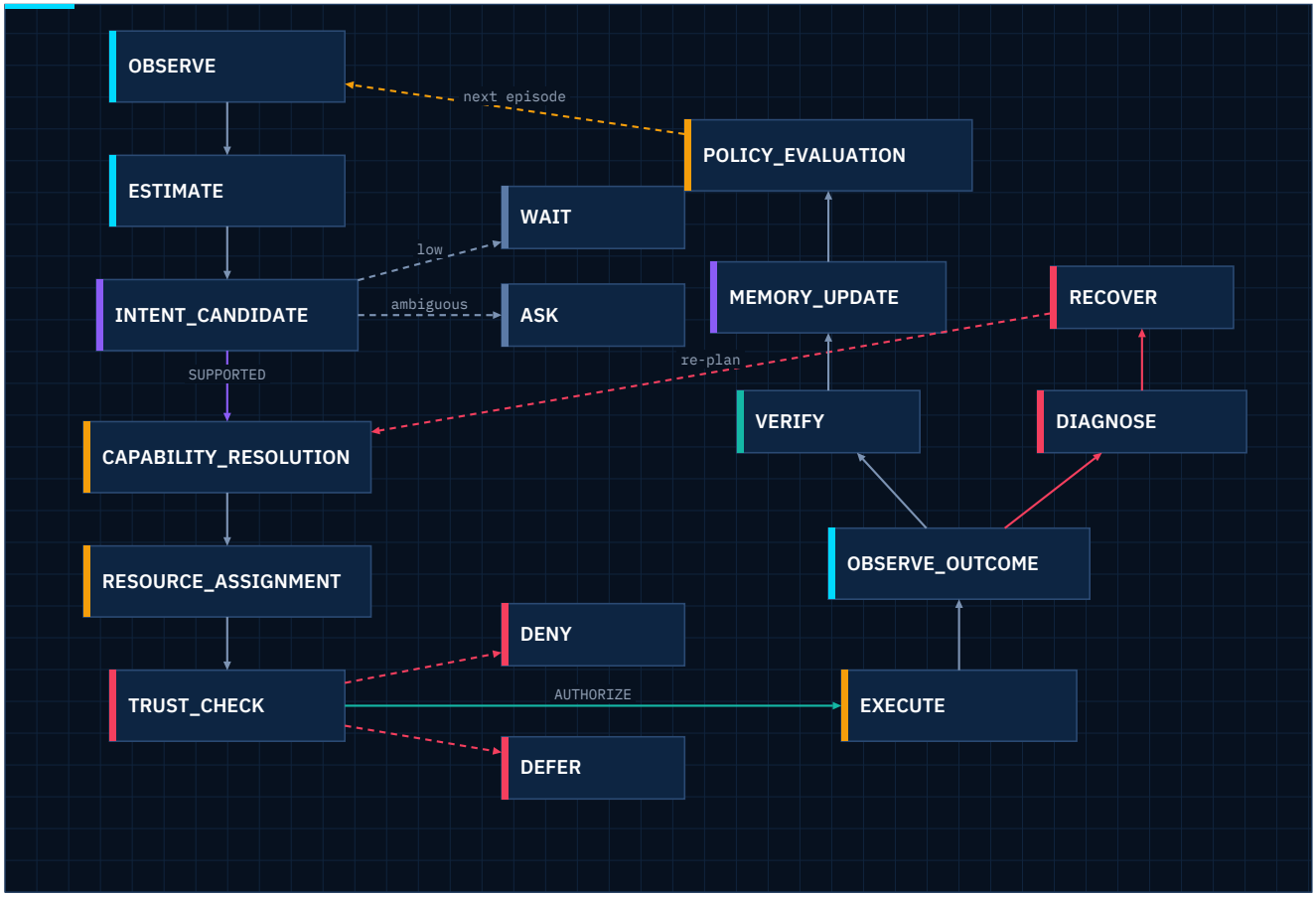


Figure 14. Intent-to-action state machine. Every transition has an input, a condition, an authority, a timeout, a failure mode and telemetry (see the transition table). Dashed edges are exits and recoveries.

ANSWERS What states may an action occupy, and who can move it between them?

Transition	Input	Condition	Authority	Timeout	Failure mode	Telemetry
OBSERVE to ESTIMATE	Observation set	quality above floor	estimator	per window	missing data widens posterior	staleness, count
ESTIMATE to INTENT_CANDIDATE	state estimate	posterior computed over verbs	fusion model	Loop C deadline	F1 miscalibration	posterior, ECE bin
CANDIDATE to WAIT	posterior	below τ_{enter}	commitment policy	window expiry	missed intent	wait reason
CANDIDATE to ASK	posterior	ambiguous top verbs, or low reversibility	commitment policy	user response	interruption cost	ask rate
CANDIDATE to CAPABILITY_RESOLUTION	posterior	SUPPORTED with hysteresis	commitment policy	none	premature commit	commitment state
CAPABILITY_RESOLUTION to RESOURCE_ASSIGNMENT	intent, context	provider satisfies context, latency, privacy, trust	capability registry	resolution timeout	F5 unavailable, then ASK or DEFER	resolver trace
RESOURCE_ASSIGNMENT to TRUST_CHECK	workload descriptor	placement meets constraints	scheduler, admission	admission timeout	F6 refusal, re-place or defer	J, refusals
TRUST_CHECK to DENY	Action draft	gate violated	policy (deterministic)	none	false denial	denial reason
TRUST_CHECK to DEFER	Action draft	confirmation or resource needed	policy	user timeout then CANCEL	interruption	defer reason
TRUST_CHECK to EXECUTE	Action draft	gate satisfied; confirmation if class 3 or 4	policy and user authority	token validity window	token replay (nonce)	token id
EXECUTE to OBSERVE_OUTCOME	command	actuator acknowledges	capability provider	execution deadline	F7 partial execution	execution result
OBSERVE_OUTCOME to VERIFY	outcome	matches expected effect	verifier	verification timeout	false verification	verify result
OBSERVE_OUTCOME to DIAGNOSE	outcome	mismatch or timeout	diagnoser	none	wrong attribution	failure code F0 to F8

Transition	Input	Condition	Authority	Timeout	Failure mode	Telemetry
DIAGNOSE to RECOVER	failure code	attributed to a layer	recovery policy (inside envelope)	retry budget	recovery loop	attempts
RECOVER to CAPABILITY_RESOLUTION or CANCEL	recovery plan	budget remains	recovery policy	retry budget	budget exhaustion	re-plan, cancel
VERIFY to MEMORY_UPDATE	verified outcome	S_{write} and policy permit	memory service	none	contamination, leakage	write score
MEMORY_UPDATE to POLICY_EVALUATION	updated state	correction and interruption rates computed	policy owner (offline, bounded)	none	reward hacking	rates

Learned models participate in ESTIMATE, CANDIDATE, RESOURCE and POLICY_EVALUATION. Permission, safety and physical control sit outside the adaptive core.

24 Failure and degradation

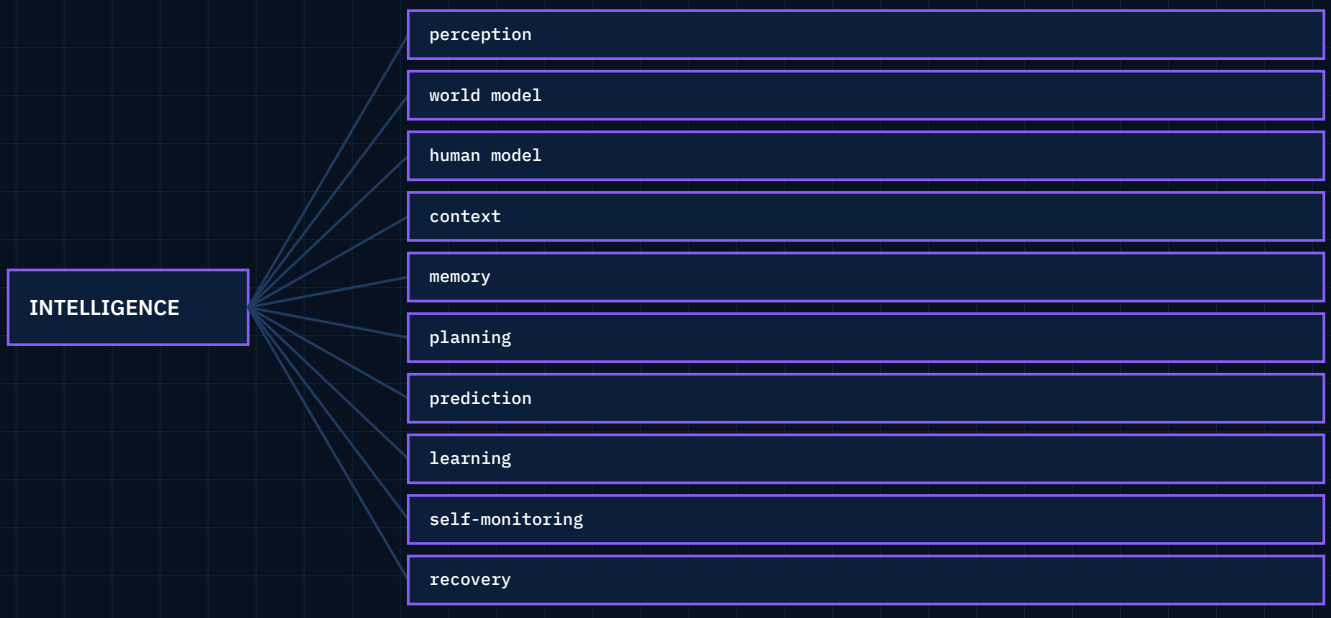
Code	Failure	Meaning
F0	sensor	signal missing, corrupted, saturated or stale
F1	estimation	signal valid but inferred state wrong
F2	context	world or task context incomplete or stale
F3	model	predictive model fails under an unmodelled condition
F4	policy	action selected poorly although state estimation was reasonable
F5	capability	requested operation unavailable or misconfigured
F6	transport	logical operation correct, physical link late or absent
F7	actuation	physical device does not realize commanded state
F8	safety	system permits an unsafe state to persist

Failures are attributed to a layer, never to "the system". Each subsystem defines normal, reduced, degraded, safe, failed and recovery states, and the transitions are fault-injected. The full Failure and Degradation Matrix is Appendix T.

PART IV

INTELLIGENCE

World · human · context · memory · plan · predict



Intelligence is an engineering subsystem with inputs, outputs, uncertainty, cost and a fallback, not a generic layer.

25 Intelligence runtime

The intelligence layer runs the loop observe, represent, predict, reason, act, measure, update. It is a stateful inference system over a persistent task episode, not prompt, model, answer. The assistant is a controlled system process with permitted access to context, memory and capabilities. It does not own hardware; the operating system and the safety supervisor do. The task episode is a unit alongside the application process: the user state and explicit goal, relevant world state, history, memory references, candidate capabilities, compute resources, actions, outcomes, corrections and learned consequences.

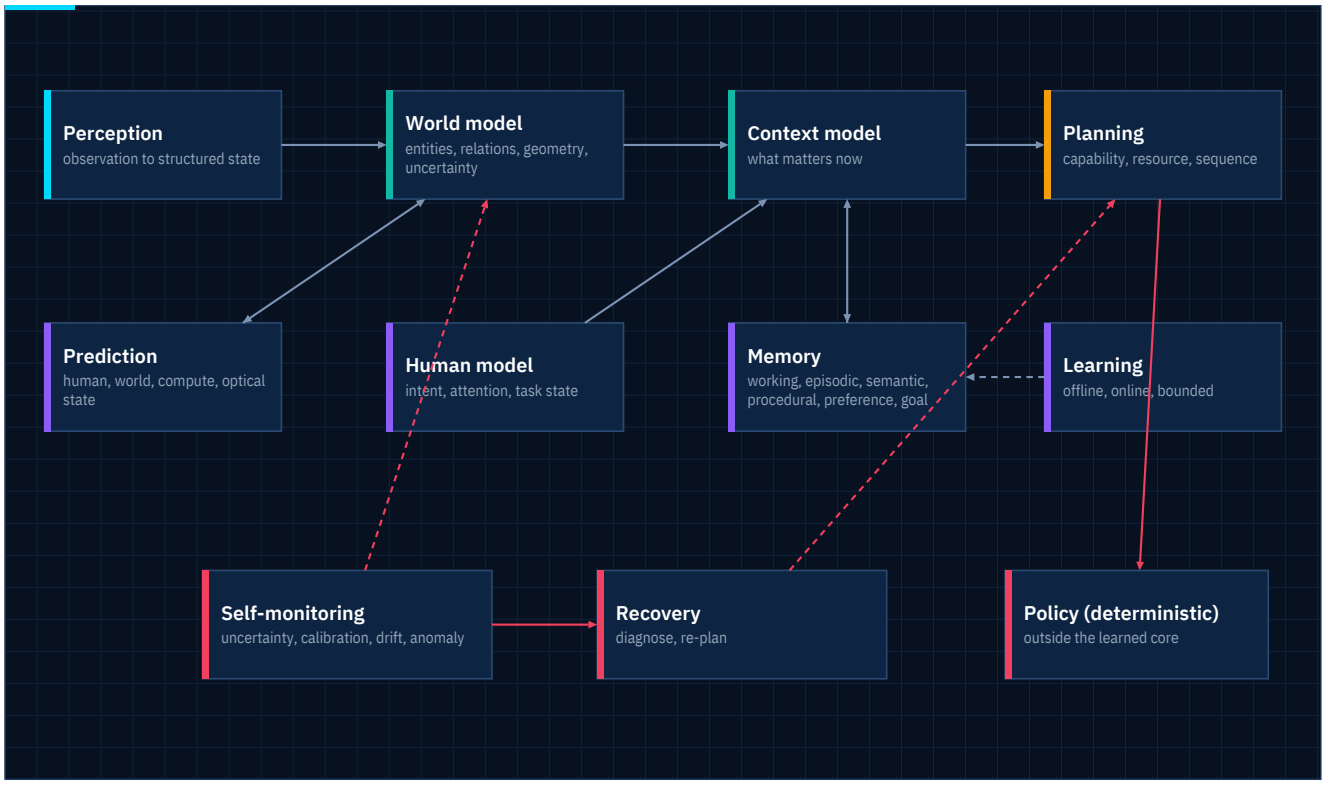


Figure 15. Intelligence runtime. World model, human model, context and memory are separate functions so failures are attributable. Self-monitoring and recovery sit beside the core, and policy sits outside it.

ANSWERS Which intelligence function failed, and what is the fallback?

World model $W_t = (V_t, E_t, S_t)$: entities, relations, geometry, poses, temporal state and uncertainty. **Context** $C_t = f(W_t, M_t, Q_t, G_t, H_t)$ is a relevance-weighted slice conditioned on memory, task, gaze and human state. **Human model** holds intent, attention, preference, physiological evidence and task state as hypotheses with uncertainty. **Memory**, **world model** and **policy** are three separate functions, kept apart so that failures are attributable.

Every learned component states its engineering envelope

Component	Input to output	Uncertainty	Compute, memory, latency	Failure mode	Fallback
Perception encoder	camera handles to entities, features	per-entity confidence	OPEN; GPU or NPU; Loop C tens to hundreds of ms	misdetection (F1, F3)	reduced scene complexity; last-known state with growing uncertainty
Eye-state estimator	IR eye frames to gaze, pupil, confidence	$sigma_{gaze}$; confidence gate	OPEN; visor; Loop B	slip, blink loss, glare	uniform presentation; reduced foveation
Neural quality and artifact estimator	samples, IMU, EOG reference to quality and contamination scores	score calibration	OPEN; neural node or visor	missed artifact	drop neural evidence
Neural decoder	representation to NeuralHypothesis	calibrated probability, uncertainty	OPEN; visor or reservoir	misclassification, drift	drop; widen posterior
World-model updater	observations to World graph	entity and relation covariance	OPEN; reservoir	incoherent or stale state (F2)	rebuild from observations; mark stale
Intent fusion	evidence to Intent IR	calibrated posterior	OPEN; visor small, reservoir large	miscalibration (F1)	explicit input
Human-state model	gaze, task, physiology to attention and workload estimate	hypothesis only, no clinical claim	OPEN	over- or under-interruption	neutral prior; reduce proactivity
Memory writer and retriever	events to typed records; query to ranked records	retrieval relevance	OPEN; reservoir	contamination, leakage	read-only mode
Planner and scheduler	capability and resource selection	cost-model error	OPEN; rules first	poor selection (F4)	rule-based locality table
Predictive world model	z_t, a_t to predicted z_{t+1}	ensemble dispersion	OPEN; reservoir or R2	unmodelled condition (F3)	reactive operation without prediction
Personalization adapter	memory and feedback to bounded adapter	adapter confidence	OPEN; reservoir	poisoning, drift	global model; rollback
Self-monitor	residuals, calibration, drift statistics to flags	detector false-alarm rate	OPEN	late detection	conservative thresholds

26 Memory, prediction and learning

Memory type	Content	Governance
Working	current task state	volatile; cleared at episode end
Episodic	events with time and place	retention policy; inspect and delete
Semantic	facts and relations	provenance and confidence required
Procedural	learned workflows	bounded; versioned
Preference	preferences and thresholds	user-editable; exportable
Goal and project	long-horizon state	user-visible

Every item carries source, confidence, created and updated time, scope, retention, sensitivity, user control and encryption status. The default is not recording. A write occurs only when score and policy permit: S_{write} = alpha N + beta U + gamma F + delta I - epsilon R_p, with N novelty, U expected utility, F predictive value, I explicit importance and R_p privacy risk. A system that remembers everything can be less useful and less safe than one that remembers selectively.

Prediction

Latent prediction is one research direction for world state: z_t = Phi($Y_{0:t}$, $A_{0:t-1}$); predicted z_{t+1} = f_{theta}(z_t, a_t); the ensemble mean over K models with dispersion as one component of epistemic uncertainty. JEPA-family methods (I-JEPA, V-JEPA 2) are externally demonstrated as predictive representation methods E2; their use as persistent user-context state is a NEXUS hypothesis E4. Generative, predictive-latent, world-model and hybrid designs are compared empirically. The Isaac Lab and Twin environment belongs to simulation level S3: it tests representation, dynamics and uncertainty, and does not show that NEXUS intelligence works.

Layer	Scope	Method family	Boundary
L1	representation	self-supervised	offline or slow update
L2	predictive and world modelling	latent prediction, dynamics, ensembles	validated in simulator first
L3	task and policy	RL, MPC or deterministic optimization where each fits	inside the policy envelope
L4	user personalization	memory, bounded adapters	rollback, per user
L5	preference and alignment	preference learning for assistant behaviour	does not train the hardware loop

Reinforcement learning is reserved for sequential decision problems with action, utility and feedback: compute placement, interaction policy, optical allocation, active sensing, adaptive UI. Supervised learning, model-predictive control or deterministic optimization are used where they fit better.

Safe adaptation

Pipeline: experience log, privacy filter, episode construction, quality and label pass, counterfactual replay, simulator evaluation, candidate adapter, offline evaluation, canary deployment, monitoring, promotion or rollback. The previous policy is always retained. Level 1 adapts memory, Level 2 adapts a user policy with bounded Delta pi, Level 3 contributes privacy-filtered aggregate experience to slow global training. Changing a user memory carries different risk from changing a production model, which is why the levels are separate.

```
reward (candidate):
  r_t = a R_success - b C_interaction - g C_latency - d C_error - h C_interrupt - k C_trust
proactivity gate:
  U(a | c) = P(benefit | c) V_b - P(interruption | c) V_i
  the adaptive threshold may not drift downward merely because activity was rewarded;
  every proactive action shows its reason, is reversible and has a suppression control
model routing (minimum machinery that reliably solves the task):
  J(m, q) = a L(m,q) + b E(m,q) + g B(m,q) + d R(m,q) - h Q(m,q)
presentation priority (attention is scarce and the system does not own it):
  A_i = w_r R_i + w_u U_i + w_t T_i + w_c C_i - w_d D_i
```

27 Uncertainty, self-monitoring and recovery

Uncertainty has five carriers in NEXUS: timestamp uncertainty sigma_t, pose covariance, signal-quality state, calibrated posterior, and ensemble dispersion. Each is propagated, not discarded. Self-monitoring tracks calibration (ECE drift), distribution drift, anomaly rate and residuals of the predictive model, and raises a flag that tightens thresholds and reduces proactivity before a failure becomes a wrong action. Recovery is diagnosis followed by re-planning inside the policy envelope, with a retry budget, and failures are attributed to F0 to F8.

Subsystem	Metrics	Baseline	Experiment
Perception	task-specific detection quality; calibration of confidence	strong off-the-shelf detector	S1 replay; O-series for optics coupling
World model	prediction error; calibration of uncertainty; coherence under latency	persistence and constant-velocity predictors	S3 Twin; RX6
Intent fusion	ECE per verb; consequence-weighted false-action rate; correction rate	gaze plus explicit confirmation	N5, N6, N7, H2
Memory	repeated instructions, corrections, interaction time, effort	desktop workflow without memory	RX4, H3
Planner and placement	task completion, energy per task, tail latency	rule table (Section 13)	RX3
Personalization	long-horizon gain per data; rollback correctness	fixed model	ablation in H1
Self-monitor	detection delay; false-alarm rate	fixed thresholds	fault and drift injection
Recovery	time to recover; wrong-action rate during recovery	no recovery logic	RX2, H2

28 Research thread: recursive epistemic intelligence

SEPARATION OF PROJECTS

NEXUS is the human-compute operating environment. UWERT-style intelligence research is a separate research program. It can supply evaluation methodology for world-model quality, active exploration, model mismatch, epistemic uncertainty, error detection, hypothesis formation, experiment selection and adaptive behaviour. It is not part of the product-critical path, and no NEXUS claim depends on it.

The thread defines recursive intelligence as the capacity to improve the processes by which a system forms representations, builds and falsifies models and chooses informative encounters, under continuing interaction with external reality. It is organized as a ladder: R1 state or memory adaptation, R2 model or error adaptation, R3 abstraction creation and revision, R4 representation revision, R5 question or goal refinement, R6 experiment selection, R7 epistemic or architectural self-revision. This is an ablation program E4. Epistemic action selection is formalized as $a^* = \text{argmax}_a [I(Q; Y_{t+1} | a) - \text{lambda } C(a)]$: expected information gain about competing hypotheses minus action cost. Experiment EP1 asks whether a system in a controlled environment with hidden parameters forms a model, notices its error, selects a discriminating experiment, updates and transfers the abstraction. A capability that changes its own evaluation structure must remain externally grounded, otherwise an agent can become self-consistent while less coupled to reality.

PART V

NEURAL INTERFACE

Signal · quality · decoder · hypothesis · intent



Signal quality, decoder quality, intent quality and action safety are four different quantities. None of them is "BCI accuracy".

29 Neural hardware



Figure 16. Neural signal chain. Signal quality, decoder quality, intent quality and action safety are four separate quantities. Nothing downstream recovers information lost at the electrode.

ANSWERS Where can neural information be lost?

Software cannot recover information lost at the electrode. Poor contact impedance, motion artifact, saturation and inadequate front-end noise performance set an upper bound on everything downstream E0. The ADS1299 class is named as an example of a low-noise multi-channel biopotential front end for experimental architectures E1; it is an example, not a component decision. Analog constraints are the specification of what must be measured:

INPUT-REFERRED NOISE BUDGET

$$\mathrm{SNR} = \frac{P_{sig}}{e_n^2 B + P_{el} + P_{mot} + P_{int}}$$

CONSTRAINS

Front-end voltage-noise density e_n, analysis bandwidth B, electrode and contact noise, motion artifact power and coupled interference.

ENGINEERING CONSEQUENCE

Digital processing can exploit statistical structure such as spatial filtering across channels, but it cannot remove noise already summed into the digitized samples at the same time and place as the signal. Channel count and placement set the ceiling.

EXPERIMENT / FALSIFIER

Neural bench: measured output SNR not explained by the four noise terms indicates a missing term, such as reference or grounding noise.

Parameter	Why it matters	Requirement form	Status
Electrode impedance	contact noise and common-mode conversion	distribution over donning, sweat, hair, pressure; monitored online	OPEN
Channel count and montage	sets spatial-filtering ceiling	set by N1 to N5 information gain per channel	OPEN
Sampling rate	bandwidth of EEG, EMG, EOG content	EMG above EEG; example 2 kHz and 1 kHz	OPEN
ADC resolution	dynamic range against EOG and motion transients	no clipping at worst artifact; example 24 bit	OPEN
Input range	saturation under blink and motion	recoverable saturation with flag	OPEN
Noise density	microvolt-scale signals	front-end term below electrode thermal term in the noise budget	OPEN
CMRR	mains and body-coupled interference	measured with real electrodes, not on the bench with shorted inputs	OPEN

Parameter	Why it matters	Requirement form	Status
Reference and bias strategy	common-mode level, drift	compared on N1 data	OPEN
Grounding and shielding	RF and mains coupling; EMC victim	evaluated with the radio running	OPEN
Motion artifact	cable, head and electrode motion	IMU-correlated artifact score available per window	OPEN
EOG contamination	blinks and saccades dominate frontal sites	EOG reference channel; contamination score per window	OPEN
EMG contamination	jaw and neck activity	contamination score per window	OPEN
Clock synchronization	fusion with gaze and IMU	sample-clock to host mapping with σ_t	OPEN
Power and temperature	comfort, battery, drift	duty-cycled; temperature recorded with samples	OPEN
Contact monitoring	detect loss before the decoder does	impedance or equivalent per channel, continuous	OPEN

Raw rates under assumed parameters (arithmetic only): EEG 64 channels, 24 bit, 1 kHz is 1.54 Mb/s; EMG 16 channels, 16 bit, 2 kHz is 0.51 Mb/s. Both are far below what a Wi-Fi or BLE link carries, so local encoding at the neural node is justified by privacy (raw neural signal never leaves the body-area boundary by default), latency (decisions close to the signal) and uncertainty representation (quality-weighted features), **not by bandwidth**.

NXS-C06 Neural node

01	**Purpose**	Acquire biopotential signals with enough fidelity that downstream inference is limited by physiology, not by the front end, and emit quality-tagged features.
02	**System requirement**	NEU-001 to NEU-004, SEC-002, INV-10.
03	**Inputs**	Skin potentials at electrodes; reference and bias; IMU and EOG references; sync pulses.
04	**Outputs**	Raw samples (restricted), per-window quality and artifact estimates, features or latents with uncertainty, health and contact state.
05	**State**	Per-channel contact state, gain and range setting, clock map, calibration version.
06	**Interface**	NeuralDevice HAL: configure, start, stop, get_stream (raw, feature, hypothesis distinct), get_quality, get_calibration, get_artifact_state, get_health.
07	**Timing requirement**	Window latency set by decoder window length OPEN ; sample timestamps with σ_t.
08	**Power budget**	OPEN ; duty-cycled acquisition and local DSP.
09	**Thermal budget**	OPEN ; sits against skin, so its dissipation is a comfort limit.
10	**Bandwidth requirement**	Example 1.54 Mb/s raw EEG and 0.51 Mb/s raw EMG stay on the node or tether; features kb/s.
11	**Memory requirement**	OPEN ; buffers and small estimators only.
12	**Accuracy / quality requirement**	Noise budget terms measured individually; quality estimate calibrated against injected artifacts.
13	**Safety constraint**	Sensing only. No stimulation. A decoder output never actuates directly.
14	**Security boundary**	Authenticated encryption on every link; raw stream off by default; separate permissions per stream class.
15	**Failure modes**	Contact loss, saturation, motion artifact, EOG and EMG contamination, mains pickup, clock error, power loss.
16	**Degradation behavior**	Quality collapse removes neural evidence; the posterior widens and interaction continues on gaze, voice, gesture or manual input.
17	**Calibration requirements**	Impedance and gain calibration per session; per-user decoder calibration time reported separately.
18	**Manufacturing implications**	Electrode materials survive motion, sweat, repeated donning and variance; dry electrodes selected by multi-objective score J_{bio}, not chosen yet.
19	**Verification method**	SNR, impedance and artifact-injection tests; EMC test with the radio active.
20	**Validation experiment**	N1 to N7.
21	**Acceptance criterion**	Quality estimates predict decoder failure on held-out sessions; features carry the information a raw-stream decoder uses within a stated loss.
22	**Falsification condition**	If the front end cannot hold usable quality for realistic motion and wear, the neural channel is demoted and NEXUS stays intent-centric.
23	**Current maturity**	M1 V0 ; EMG externally demonstrated E2 ; wearable EEG E5 .
24	**Next implementation artifact**	Neural development board capturing EEG and EMG with hardware timestamps, plus an artifact-injection fixture.

30 Signal chain and channels

Channel	Answers	Strength	Limit	Status
Gaze	where attention is directed	spatial target; narrows the hypothesis space	calibration drift, slip, blink loss, latency; cannot confirm alone	E2
EMG (wrist)	what motor action is initiated	peripheral, close to motor output, low latency	placement and posture sensitivity, fatigue, co-contraction; low command rate	E2
EEG	what neural-state evidence exists	non-invasive access to cortical-origin signals	microvolt scale, large artifacts, session drift, user variability	E5 at glasses scale
EOG	eye-movement and blink potentials	low-power secondary gaze channel	coarse accuracy; couples into EEG	E1
Voice	semantic content	rich meaning	social, acoustic and privacy cost	E1
Touch or button	explicit confirmation	unambiguous, authenticated	needs a hand and attention	E1

EMG

Surface EMG measures electrical activity of muscle, closely related to the motor commands that drive it E0. Kaifosh et al. (Nature 645, 702 to 711, 2025) describe a wrist-worn dry-electrode sEMG interface with generic decoding models evaluated on held-out people without per-person calibration E2. Reported closed-loop performance: a median of 0.66 target acquisitions per second in continuous navigation, 0.88 gesture detections per second in discrete gestures and handwriting at 20.9 words per minute, with personalization improving handwriting by 16 percent. It establishes that a generic sEMG interface can work across people at a usable but modest rate. It does not establish a rate above about one command per second, robustness during arbitrary arm motion or load, operation for users with limited motor function, or that EMG improves a multimodal intent system. EMG should carry a small vocabulary of confirmations and fine controls, with gaze and context doing target selection. Users who cannot produce reliable EMG must reach the same Intent IR through other modalities.

EEG and EOG

EEG costs are the specification of what must be measured: low amplitude and SNR; contamination by EOG (much larger than cortical signal at frontal sites), jaw and neck EMG, head and cable motion, mains interference and electrode drift; contact variability on a frame or temple; nonstationarity and large inter-user variability; and limited command vocabulary in the standard paradigms (motor imagery, SSVEP, P300). Dry-electrode performance varies widely with design and is not evidence of all-day glasses-scale reliability E5. The right first question is incremental information over gaze, EMG, voice, context and memory at acceptable power and comfort (N5). An implicit error-related signal as post-action verification evidence is a speculation to test only after N1 to N3 show a usable within-user decoder. Electrode materials (noble metals, Ag/AgCl structures, PEDOT:PSS, nanostructured metals, graphene-class carbon, MXene-like films, skin-conformal elastomer composites) are evaluated against J_{bio} = alpha SNR - beta $Z_{contact}$ - gamma A_{motion} - delta D_{drift} - eta R_{skin} - zeta C_{mfg}; no material is selected OPEN.

31 Quality, decoder and hypothesis

Stage	Input	Output	Runs where	Failure
Electrode and reference	skin potential	analog signal	neural node	contact loss, motion
Analog front end	analog signal	amplified, filtered analog	neural node	saturation, noise floor
ADC and timestamp	analog	samples with source time	neural node	clipping, clock error
Quality estimate	samples	per-channel and per-window quality	node or visor	misjudged quality
Artifact estimate	samples, IMU, EOG reference	contamination scores	node or visor	missed artifact
Preprocessing	samples	cleaned signal	visor	over-filtering
Representation	cleaned signal	features or latent with uncertainty	visor or reservoir	drift
Decoder	representation	NeuralHypothesis	visor or reservoir	misclassification
Fusion	hypotheses and other evidence	Intent IR	reservoir (visor fallback)	miscalibration

Quality state per channel and window: SNR estimate, contact impedance where measurable, clipping and saturation flags, dropout, line-noise power, IMU-correlated artifact score, EOG and EMG contamination scores, temperature and a session-drift indicator. Every decoder output carries this state forward so fusion can weight it. Decoder families (classical on engineered features, temporal CNN and TCN, sequence models, hybrid EEG plus EMG) sit behind one output contract:

```
NeuralHypothesis { hypothesis, calibrated_probability, uncertainty, time_window,
                   channel_quality, artifact_state, decoder_version, calibration_version }
```

A hypothesis is evidence, not authority. Personalization is layered: $theta_G$ global model, $theta_U$ user adapter, s_t session state. More pooled data does not by itself solve user variability; the test is whether an adapter closes the cross-session gap within a stated calibration time (N2, N4).

Four separate qualities

Quality	Question	Metrics	Measured by	Fallback when poor
Signal quality	is the physics adequate	SNR, impedance, clipping, dropout, artifact scores	neural bench, artifact injection	drop channel or window
Decoder quality	does the model read the signal	accuracy, FPR, FNR, precision, recall, ECE, per user and session	held-out users and sessions (N1 to N4)	drop decoder; widen posterior
Intent quality	does it add information about intent	incremental information gain over gaze, EMG, voice, context; posterior calibration after fusion	N5	demote EEG; stay intent-centric
Action safety	is acting on it acceptable	consequence-weighted false-action rate, false activations per hour, correction rate	N6, N7, H2	raise gate class; require confirmation

32 Multimodal fusion and commitment



Figure 17. Multimodal fusion. Modalities are evidence, not votes. A missing modality widens the posterior. Correlated modality errors are treated explicitly.
ANSWERS How are modality reliability, timing and correlation handled before an intent is committed?

POSTERIOR ODDS

$$O(I \mid Y) = O(I) \prod_m LR_m$$

CONSTRAINS

How modality evidence combines into one intent posterior.

ENGINEERING CONSEQUENCE

With conditionally independent evidence the posterior odds are prior odds times each modality likelihood ratio. Modalities are not voters: reliability depends on context, a modality may be late, wrong or absent, and a missing modality must widen the posterior. The independence assumption inflates the posterior when errors are correlated, which is common: gaze and context both depend on what the user attends to.

EXPERIMENT / FALSIFIER

N5: reliability diagrams on held-out sessions. A learned fusion with calibration and a conservative treatment of correlated evidence is a requirement, not a refinement.

Evidence (assumed values, illustrative)	Likelihood ratio	Odds after	Probability
Prior: 5 percent this object is being selected now	-	0.053	0.050
Gaze on the target	x 8	0.421	0.296
EMG confirm event	x 6	2.526	0.716
Context says the action fits the task	x 2	5.053	0.835

Evidence (assumed values, illustrative)	Likelihood ratio	Odds after	Probability
Same, but gaze and context correlated: joint LR 10, not 16	joint 10, x 6	3.158	0.759

The last row is the central caution: the independence assumption gives 0.835 where the correlated treatment gives 0.759.

Base rates decide usability

Take a decision window every 100 ms for an hour (36,000 windows), 30 true intents per hour, true-positive rate 0.95 and false-positive rate 10^{-3} per window. Expected true activations are 28.5; expected false activations are 0.001 x 35,970 = 36.0; precision is 28.5 / (28.5 + 36.0) = 44 percent. A decoder that looks accurate per window is unusable as a continuous command channel.

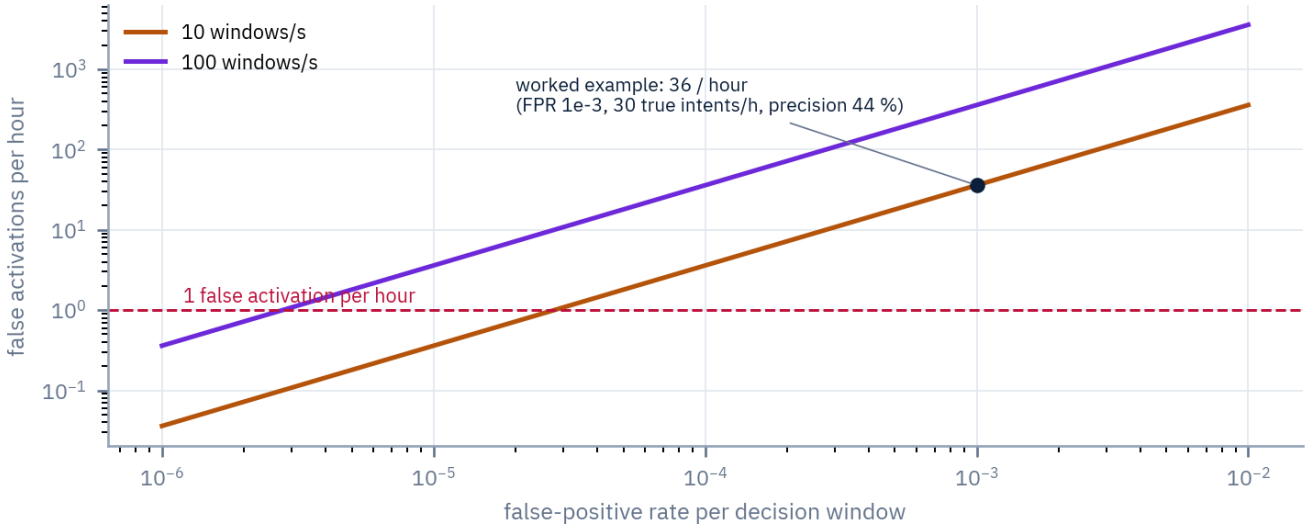


Figure 18. False activations per hour versus per-window false-positive rate. Independent windows assumed. Debouncing and hysteresis reduce the effective rate but do not remove the base-rate problem.

ANSWERS Why is window accuracy the wrong metric for a continuous command channel?

Mitigations are structural: most actions sit in gate class 1 where a false activation costs a dismissal; confirmation across a second modality above that; hysteresis and minimum dwell per verb; and evaluation by consequence-weighted false-action rate alongside accuracy, never accuracy alone.

33 Calibration and neural validation

The central question is whether neural sensing provides incremental information beyond gaze, voice, EMG and context at an acceptable cost. If it does not, the architecture demotes EEG without collapsing. The benchmark separates training users from held-out users, training sessions from test sessions, development devices from evaluation devices, calibration time from steady state, and clean data from realistic motion. Shuffled within-session accuracy is not a result.

Measure	Definition or split	Why
Within-session accuracy	train and test in one session, time-ordered not shuffled	upper bound only
Cross-session performance	train sessions to held-out sessions, same user	drift handling (N2)
Cross-user performance	train users to held-out users	generic-model viability (N3)
Calibration time and adaptation gain	gain per calibration minute	personalization value (N4)
False activations per hour; false-action rate	windows per hour times FPR; weighted by action class	usable operating point (N6)
FPR, FNR, precision, recall	per verb and per window and per event	base-rate honesty
Calibration, ECE	reliability diagram per verb	confidence is meaningful
Consequence-weighted error; correction rate	cost by action class; user corrections per action	agency and trust (H2)
Latency	p50, p95, p99 window to hypothesis	commitment timing (N7)
Robustness	motion, sweat and contact change, fatigue	realistic wear
Energy	joules per decision and per hour	thermal and battery budget

Calibration layer	Covers	Notes
Factory	amplifier gain and offset, clock references	versioned; traceability

Calibration layer	Covers	Notes
Per user	montage fit, decoder adapter, EMG calibration	calibration time reported (N2, N4)
Session	contact quality, fit, impedance	re-estimated every wear
Online	drift correction	never silently violates a hard limit

PART VI

COMPUTE FABRIC

R0 → R1 → R2 → REMOTE



Locality is a physical variable: compute moves when the energy, thermal and latency arithmetic says it should, and never when safety says it must not.

34 Compute hierarchy and the reservoir thesis

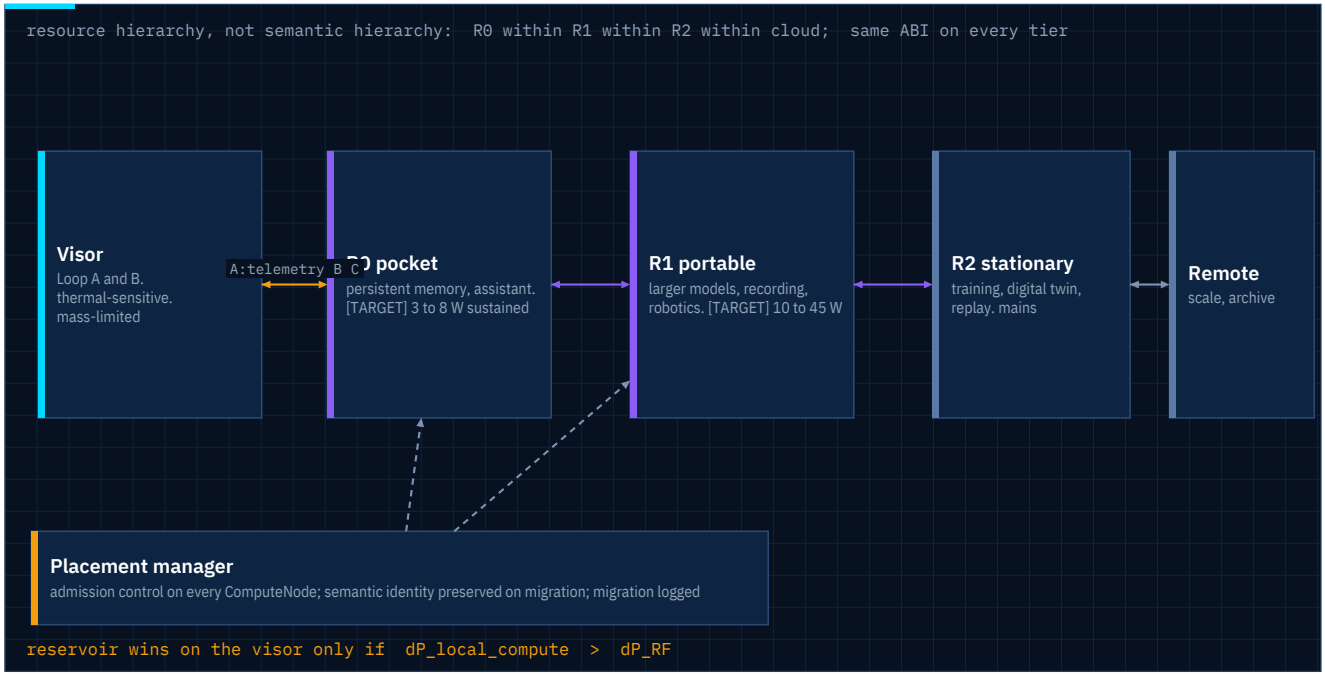


Figure 19. Compute reservoir. R0, R1 and R2 are a resource hierarchy with the same semantic ABI on every tier. Power envelopes are targets, not measurements.

ANSWERS Where does sustained compute live, and why does a workload move?

Tier	Role	Sustained power	Memory	Persistent state	Status
Visor	Loop A and B, small local models, minimum interaction	bounded by skin-contact thermal budget OPEN	OPEN	no long-term memory	FROZEN
R0 pocket	persistent assistant, memory, scene model, small and medium inference, sensor fusion	TARGET 3 to 8 W	TARGET 16 to 64 GB	yes	PREFERRED
R1 portable	larger models, recording, robotics, offline replay	TARGET 10 to 45 W	OPEN	yes	INVESTIGATING
R2 stationary	training, digital twin, replay at scale	mains-powered	large	yes	PREFERRED
Remote	scale, archive, long-horizon simulation	not applicable	not applicable	privacy-filtered only	PREFERRED

The tiers differ in power, memory, thermal envelope and persistence, not in semantics: the same ABI runs on each. R0 is a pocket-class node; no silicon is named because none is supported by evidence. The human-worn thermal envelope is specified separately from the reservoir envelope, which has its own enclosure, ventilation and skin-contact conditions.

RESERVOIR INEQUALITY

$$\Delta P_{local} > \Delta P_{RF}$$

CONSTRAINS

Whether moving a workload off the visor reduces visor power and heat at all.

ENGINEERING CONSEQUENCE

Offloading helps the visor only if the local compute power avoided exceeds the added RF (or tether) power, and the visor-side tail latency still meets the loop it serves. Radio energy depends on rate, distance and body blockage, so the inequality must be measured per workload class. The reservoir thesis is the claim that for the workload mix of Loop C it holds with margin.

EXPERIMENT / FALSIFIER

RX1: instrumented visor power with each workload local versus offloaded. If offloaded power is not lower, or if p99 latency violates the budget, the reservoir thesis is rejected for that workload class and the class stays on the visor.

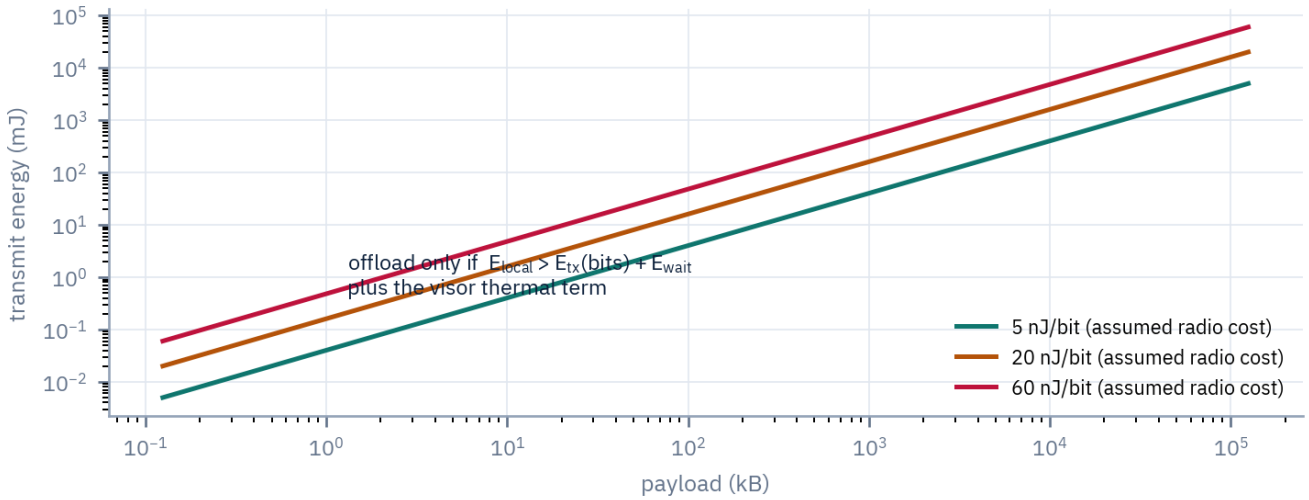


Figure 20. Offload break-even payload. Transmit energy against payload for three assumed radio costs. Radio energy per bit is an assumption to be measured in RX1.
ANSWERS When does sending work away cost more than doing it locally?

The wire is not a free alternative: conductor loss is P^2R/V^2, so tether voltage is a design variable, and a cable adds mass, stiffness, strain-relief and connector reliability to the head-worn side.

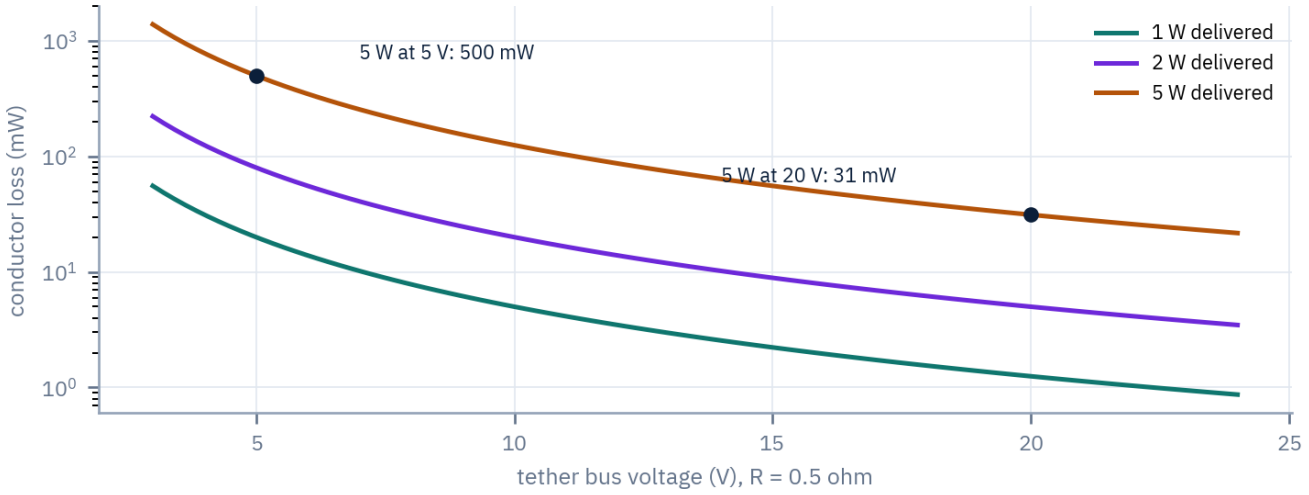


Figure 21. Conductor loss versus tether bus voltage. R = 0.5 ohm. Converter loss, insulation and EMC are not modelled.
ANSWERS Why is tether voltage a design variable?

35 Workload model and placement optimization

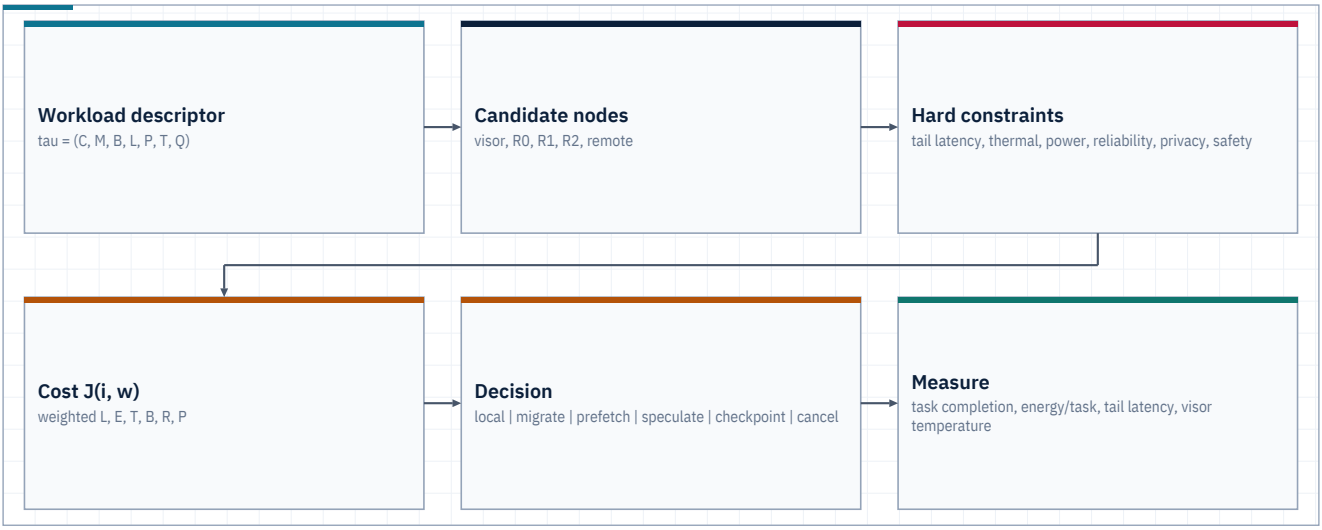


Figure 22. Compute placement optimization. The figure defines the problem; no solver is claimed. Rules come first, and a learned scheduler must beat them on the measured frontier.

ANSWERS Why does a workload move, and what stops it from moving?

A workload descriptor tau = (C, M, B, L, P, T, Q) records compute, memory, bandwidth, latency class, privacy class, thermal sensitivity and quality target. Placement chooses a node i from the candidate set by minimizing a weighted cost under hard constraints:

PLACEMENT COST

$$J(i, w) = \lambda_L L + \lambda_E E + \lambda_T T + \lambda_B B + \lambda_R R + \lambda_P P$$

CONSTRAINS

Which node runs which workload.

ENGINEERING CONSEQUENCE

Hard constraints are applied first, not weighed: L <= L_{max} at the declared percentile, T <= T_{max}, P <= P_{max}, R <= R_{max} (reliability), privacy class, and safety class. Reflex-class work has an empty candidate set apart from the visor. The weights lambda are fitted from measurement, not asserted; a learned scheduler must beat the rule table on the measured frontier before it replaces it.

EXPERIMENT / FALSIFIER

RX3: placement quality measured as task completion, energy per task, tail latency and visor temperature versus a rule table. If a learned placement does not dominate the rules, it is removed.

LATENCY ALONG A PLACED PATH

$$L_{tot} = L_{queue} + L_{tx} + L_{comp} + L_{rx} + L_{jitter}$$

CONSTRAINS

Which term of Loop C latency dominates for an offloaded task.

ENGINEERING CONSEQUENCE

L_{queue} is admission and scheduling wait, L_{tx} and L_{rx} link transit including retries, L_{comp} execution, L_{jitter} the tail contribution. Percentiles do not add; the budget is checked on the measured distribution, not on a sum of means.

EXPERIMENT / FALSIFIER

A sum of stage medians that does not reproduce the measured end-to-end median points to unaccounted queueing.

Execution mode	Trigger	Constraint	Recovery
Local (visor)	reflex class; link absent; privacy; cheap task	thermal and power headroom	throttle ladder
R0	persistent state, medium inference	R0 admission and thermal	fall back to local or defer
R1	sustained models, recording	R1 admission; link class C	migrate down or queue
R2	training, replay, twin	privacy filter first	retry; local reduced
Remote	scale, archive	privacy filter; never raw neural	resync on return
Migration	node thermal rise, link degradation, battery state	semantic identity preserved; checkpoint exists	roll back to checkpoint
Prefetch	high-probability reversible need	no side effects; bounded energy	cancel and discard
Speculative execution	EU_{spec} positive and work reversible	class 1 only	cancel; log waste
Checkpointing	long task or migration risk	state size versus bandwidth	resume from checkpoint
Cancellation	intent revised, link lost, budget exhausted	must complete within a deadline	release resources; report

Execution mode	Trigger	Constraint	Recovery
Failure recovery	node fault, link loss	retry budget	degrade to the next lower mode

Measured outcome	Why it is measured
Task completion	the quantity users care about
Latency (p50, p95, p99)	tail behaviour
Energy per task	battery and thermal
Visor temperature	skin-contact budget
Network utilization	link headroom for Class A and B
Failure rate	availability
Model quality	did offloading change the answer
Recovery time	graceful degradation

36 Transport, wired and wireless

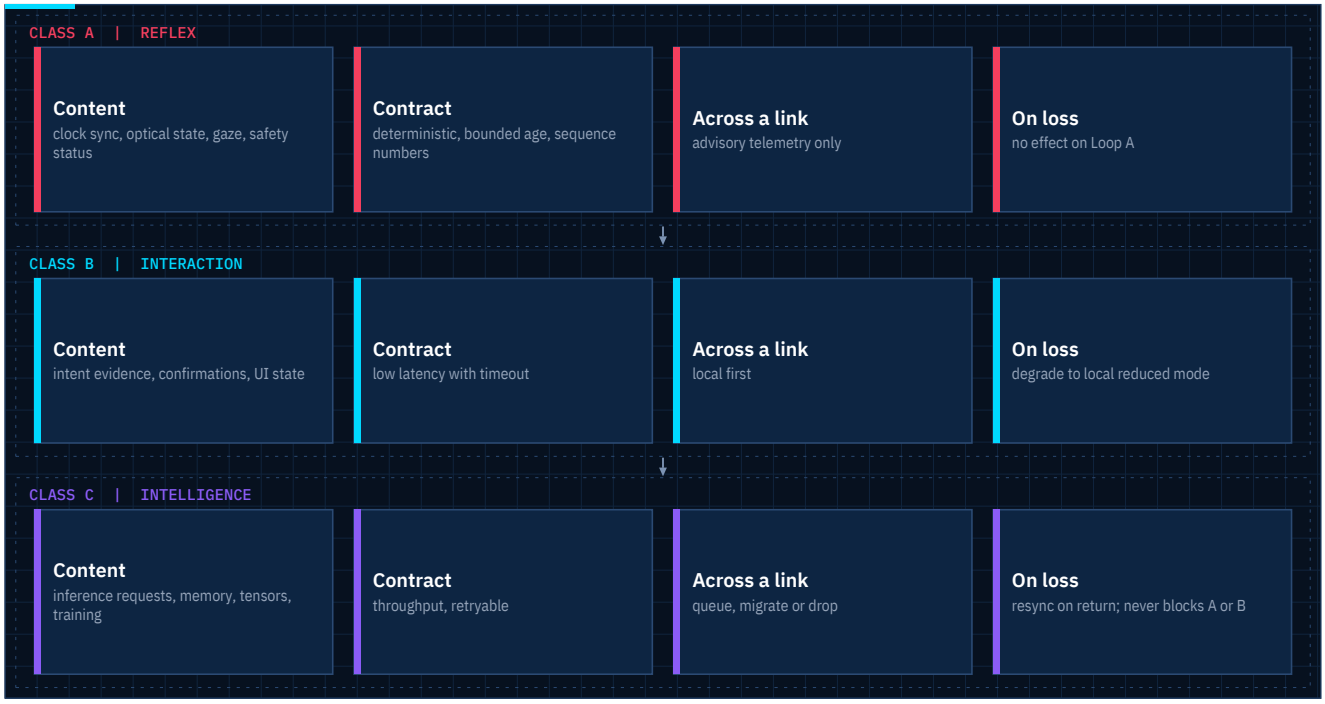


Figure 23. Transport architecture. Classes are timing contracts, not network labels. Large transfers may never block Class A or B.

ANSWERS What must still work when the link is slow, lossy or gone?

Transport classes are timing contracts, not network labels. Class A is deterministic and bounded-age, and across a link it is advisory telemetry only (INV-5). Class B is low latency with a timeout and falls back to a local reduced mode. Class C is throughput-oriented and retryable, may be queued, migrated or dropped, and may never block A or B.

Stream (assumed sensors, arithmetic only)	Raw rate	What crosses the link
Two world cameras, 2 Mpx, 60 Hz, 12 bit	2.88 Gb/s	encoded features or events, 10 to 100 Mb/s class DERIVED
Two eye cameras, 400 x 400, 8 bit, 90 Hz	230.4 Mb/s	nothing raw; gaze state only
EEG 64 ch, 24 bit, 1 kHz	1.536 Mb/s	features by default; raw only on explicit enable
EMG 16 ch, 16 bit, 2 kHz	0.512 Mb/s	features or latents
Microphone, 16 kHz, 16 bit	256 kb/s	wake-word local; speech by policy
IMU, 6 axes, 16 bit, 1 kHz	96 kb/s	pose state

Sending raw camera data wirelessly is a design choice, not a given. The first compute decision is therefore where the first neural network runs and what it emits, which is the compression point of the whole architecture. Session lifecycle: discovery, mutual attestation, key agreement, class negotiation, continuous link-quality reporting (latency distribution, loss, RSSI, throughput), and seamless handoff between wired and wireless transports behind one logical API (RF-003).

NXS-C07 Reservoir and compute fabric

01	**Purpose**	Provide persistent, thermally unconstrained compute for memory, world model and sustained inference, with admission control and placement across tiers.
02	**System requirement**	CMP-001 to CMP-004, SYS-004, THM-003, PWR-002.
03	**Inputs**	Workload descriptors, node state, link quality, battery and thermal state of visor and reservoir.
04	**Outputs**	Placement decisions, results, migration logs, energy and latency records, admission refusals.
05	**State**	Node inventory, admission state, running tasks and checkpoints, migration log, energy ledger.
06	**Interface**	ComputeNode ABI: admit(w), run, checkpoint, restore, cancel, report_health; Compute AIDL service; Class B and C channels.
07	**Timing requirement**	Placement decision on the Loop C budget; migration must finish inside a deadline or roll back; no tasks in Loop A or B.
08	**Power budget**	`TARGET` R0 3 to 8 W sustained; R1 10 to 45 W; visor side measured through RX1.
09	**Thermal budget**	Own enclosure thermal envelope `OPEN`; separate from the human-worn envelope.
10	**Bandwidth requirement**	Class C throughput; sized by the largest tensor stream after the first network.
11	**Memory requirement**	`TARGET` R0 16 to 64 GB; checkpoint size versus bandwidth is a constraint.
12	**Accuracy / quality requirement**	Task completion, energy per task, tail latency, model quality unchanged by migration.
13	**Safety constraint**	No reflex or safety work is placeable here; policy gateway outside the learned scheduler.
14	**Security boundary**	Mutual attestation of nodes; signed models; privacy filter before remote; no raw neural to cloud by default.
15	**Failure modes**	Node thermal overload, link loss, memory exhaustion, migration failure, scheduler error, model corruption.
16	**Degradation behavior**	Fall back to local, reduce model size, defer or cancel; the visor continues in M0.
17	**Calibration requirements**	Energy and latency model coefficients refitted from measured traces.
18	**Manufacturing implications**	Reservoir enclosure ventilation, cable or radio, connector reliability, serviceability.
19	**Verification method**	Instrumented task runs; fault injection on link and node; failover tests.
20	**Validation experiment**	RX1, RX2, RX3, RX7.
21	**Acceptance criterion**	Offloaded workloads show lower visor power and heat and meet the stated percentile; failover keeps M0 working with the link removed.
22	**Falsification condition**	If offloading does not reduce visor power or violates tail latency for the Loop C workload mix, the reservoir thesis is rejected for that mix and the mix stays on the visor.
23	**Current maturity**	`M1` `V0` `V2`; hardware `NOT STARTED`; software placement `PROTOTYPE` `REPOSITORY-ASSERTED`.
24	**Next implementation artifact**	Placement benchmark harness on a tethered development board with a rule table baseline and recorded visor power.

NXS-C08 Transport link

01	**Purpose**	Carry Class A, B and C traffic between visor, neural node and reservoir over wired or wireless links with declared timing contracts.
02	**System requirement**	RF-001 to RF-004, SYS-004, SEC-002.
03	**Inputs**	Typed messages with class tags, payload handles, encrypted sessions, link-quality probes.
04	**Outputs**	Delivered messages, link-quality state (latency distribution, loss, RSSI, throughput), session events.
05	**State**	Session keys, sequence counters, per-class queues, link-quality estimate.
06	**Interface**	Transport AIDL; link channels per class; one logical API for wired and wireless.
07	**Timing requirement**	Class A reserved capacity and bounded age; Class B deadline with timeout; Class C best effort.
08	**Power budget**	`OPEN` radio energy per bit; measured in RX1.
09	**Thermal budget**	RF front end heat is a skin-adjacent source `OPEN`.
10	**Bandwidth requirement**	Class C dominant; Class A and B small and reserved.
11	**Memory requirement**	Per-class queue depth `OPEN`.
12	**Accuracy / quality requirement**	Latency distribution reported at p50, p95, p99 under body motion and blockage.
13	**Safety constraint**	Link loss must not affect Loop A; Class A over a link is advisory.
14	**Security boundary**	Authenticated encryption, replay protection by counters, session attestation.
15	**Failure modes**	Congestion, body blockage, handoff failure, key expiry, interference.
16	**Degradation behavior**	Class C queued or dropped, Class B to local reduced mode, Class A advisory only; wired fallback if present.
17	**Calibration requirements**	Link-quality model calibrated per body pose.
18	**Manufacturing implications**	Antenna placement against body blockage; connector wear for the tether.
19	**Verification method**	RF test chamber, body-pose sweeps, packet-loss and latency injection.
20	**Validation experiment**	RX1, RX2, RX7.
21	**Acceptance criterion**	Stated p99 latency under defined body-motion conditions; no effect on Loop A under link removal.
22	**Falsification condition**	If p99 transit latency exceeds Class B deadlines under ordinary body motion, the wireless branch is demoted to Class C only and wired becomes mandatory for interaction-class offload.
23	**Current maturity**	`M1` `V0`; `NOT STARTED` in hardware.

24	**Next implementation artifact**	Link characterization rig recording latency distribution versus body pose for two radio classes.

37 Migration, energy per task and thermal impact

Battery arithmetic (example)	Value
Energy	20 Wh at 90 % usable = 18 Wh
At 5 W average	3.6 h
At 2 W average	9 h
At 1 W average	18 h

Energy per completed task E_{task} = integral of P dt over the task, measured per rail, is the quantity placement minimizes. Offloading is not free to the visor: it trades compute power for radio power, and may help temperature even when it does not help energy, because RF dissipation can sit in a different thermal position than the SoC hotspot. Migration preserves semantic identity: the task keeps its id, its checkpoint and its trace, and the migration is logged with the reason (thermal, link, battery, privacy, cost).

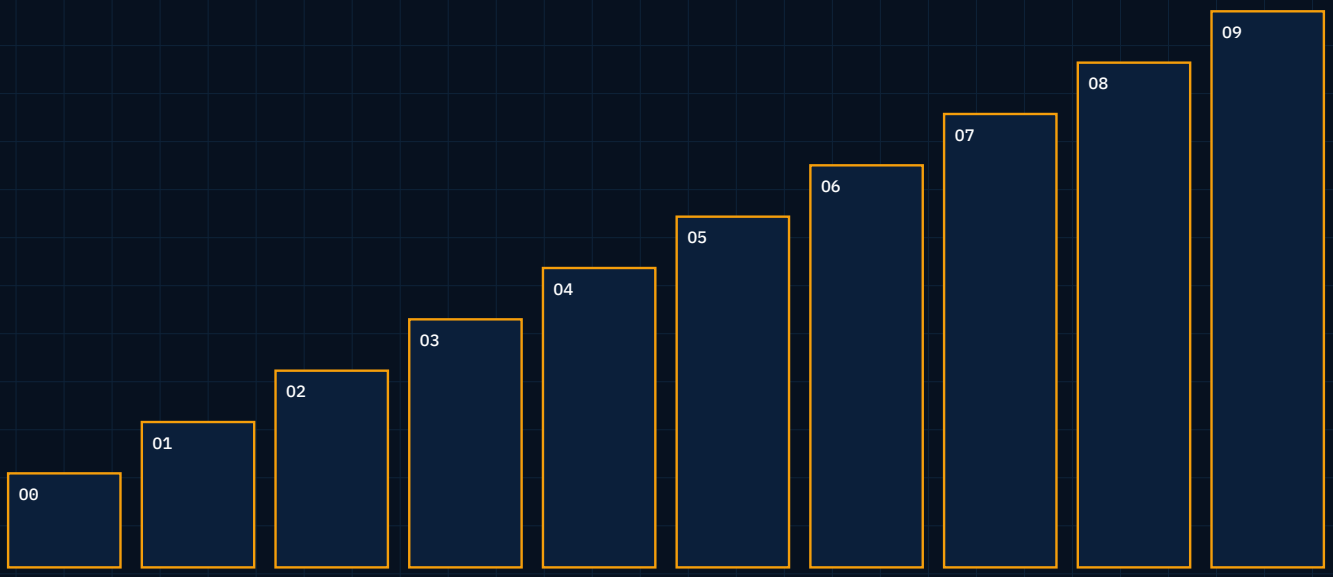
38 Failure and recovery of the fabric

Event	Detection	Response	Result for the user
Link loss	heartbeat and link-quality drop	Class B and C to local reduced mode; Class A unaffected	reduced assistant, UI and safety intact
Reservoir loss	node health timeout	M2 to M0; memory read-only locally if cached	basic operation continues
Reservoir thermal overload	node thermal state	refuse new work; migrate down; reduce model size	slower or lighter answers
Migration failure	checkpoint mismatch or deadline	roll back to the checkpoint; log	task restarts or is reported failed
Scheduler error	trace audit; measured regression	revert to the rule table	none visible
Cloud loss	connection failure	local and reservoir only; queue privacy-filtered uploads	no scale features

PART VII

OPTICAL SYSTEM

Etendue · allocation · control · safety · yield



The optical engine is the part of NEXUS where physics is least forgiving. The task is to deliver useful retinal information at the lowest optical resource and heat the physics permits.

39 Optical requirements and the architecture invariant

The central research question is how little physical optical resource is needed to reproduce the visually relevant information a human needs for a given task. It is not "fewer pixels". It is compression, coordination, adaptation, perceptual relevance and optical controllability together. Two clarifications are required by physics. Removing a display panel does not remove energy cost E0: energy conservation is fully respected, and the claim concerns resource allocation and wasted photons. The measurable objective is useful retinal information per unit of electrical energy and thermal load at matched task performance, not a pixel count.

Optical architectures are compared by the vector K_{opt} = (FOV, eyebox, eye relief, resolution and MTF, luminance, contrast, distortion, chromatic performance, speckle, efficiency, thermal load, electrical power, mass and volume, safety margin, manufacturability, yield), nominal and worst-case, never by a single number. The **invariant** is controllable optical presentation; source, steering and combiner are replaceable until experiments select them. The visible-light, see-through direction is PREFERRED; the technology branch is INVESTIGATING.

Ambient transmission T(x, y, theta, lambda, t) is a separate subsystem from image generation (normal, dimmed, selectively attenuated, blocked), and depth cues (vergence, accommodation, focus) are a separate design question from field of view. Accommodation-consistent depth is E5 for every candidate family.

40 Visual bandwidth

Ashraf, Chapiro and Mantiuk (Nature Communications, 2025, DOI 10.1038/s41467-025-64679-2) measured the resolution limit of the eye in pixels per degree: about 94 ppd for foveal achromatic gratings, about 89 for red-green and about 53 for yellow-violet, falling at 10 and 20 degrees of eccentricity E2 (checked against the publisher abstract; the 18-observer sample is a secondary source). Three cautions: values are for a tested sample under stated conditions; they are for gratings, not arbitrary content; and they are the resolution limit of the eye, not a display requirement, which also depends on contrast, luminance, uniformity, colour, temporal behaviour and comfort.

Field (uniform raster at 94 ppd, arithmetic only)	Pixels	Mpx	Samples/s at 60 Hz
3840 x 2160 raster covers 40.9 x 23.0 deg	3840 x 2160	8.29	498 M
50 x 30 deg	4700 x 2820	13.3	795 M
100 x 60 deg	9400 x 5640	53.0	3.18 G

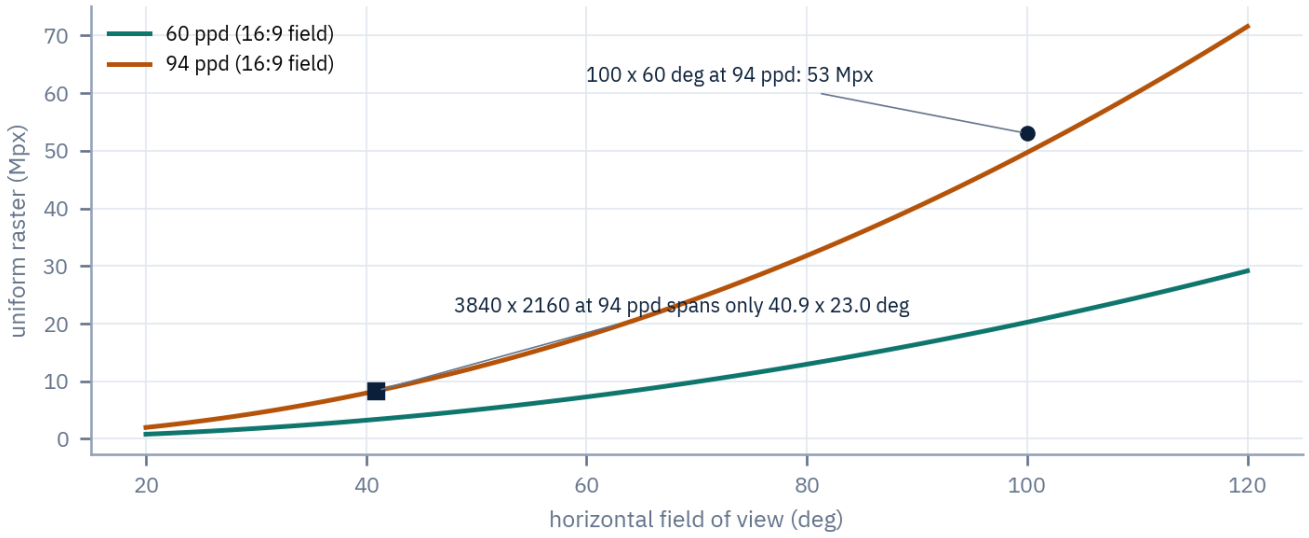


Figure 24. Uniform raster pixel count versus field of view. The 3840 x 2160 raster is one point: at 94 ppd it spans about 41 x 23 degrees.
ANSWERS Why is a pixel count not an optical specification?

41 Field of view, eyebox and etendue

PHYSICAL LIMIT: ETENDUE

$$G = n^2 A \Omega$$

CONSTRAINS

Aperture A, solid angle Omega, refractive index n: the product of field of view, eyebox and radiance a lossless passive system can deliver.

ENGINEERING CONSEQUENCE

FOV, eyebox, radiance and compactness cannot all be increased together without paying somewhere. A candidate must create sufficient optical etendue through aperture, pupil replication or another physical mechanism, and each such mechanism has an efficiency and uniformity cost.

EXPERIMENT / FALSIFIER

Measure the actual FOV-eyebox frontier on the optical bench (O4). An architecture whose measured product exceeds what its stated aperture and replication can supply has an unaccounted source of etendue.

SLM SPACE-BANDWIDTH LIMIT

$$E \cdot FOV_{rad} \approx N\lambda$$

CONSTRAINS

Eyebox E = l arcsin(lambda/d) and FOV = 2 arctan(S/2l) for an SLM of pixel pitch d, side S = N d, at eye relief l.

ENGINEERING CONSEQUENCE

Worked example (assumed): lambda 520 nm, d 8 um, N 1920, l 20 mm gives E = 1.30 mm, FOV = 42.0 deg = 0.733 rad, product 0.954 against N lambda = 0.998 mm.rad. For E = 10 mm at 42 deg, N is about 14,100 pixels per axis (about 199 Mpx) without enlarging effective etendue, which is why pupil replication and synthetic-aperture methods matter.

EXPERIMENT / FALSIFIER

Relation from Choi et al. supplementary material, source checked E0. Bench: eyebox versus pixel pitch at fixed wavelength.

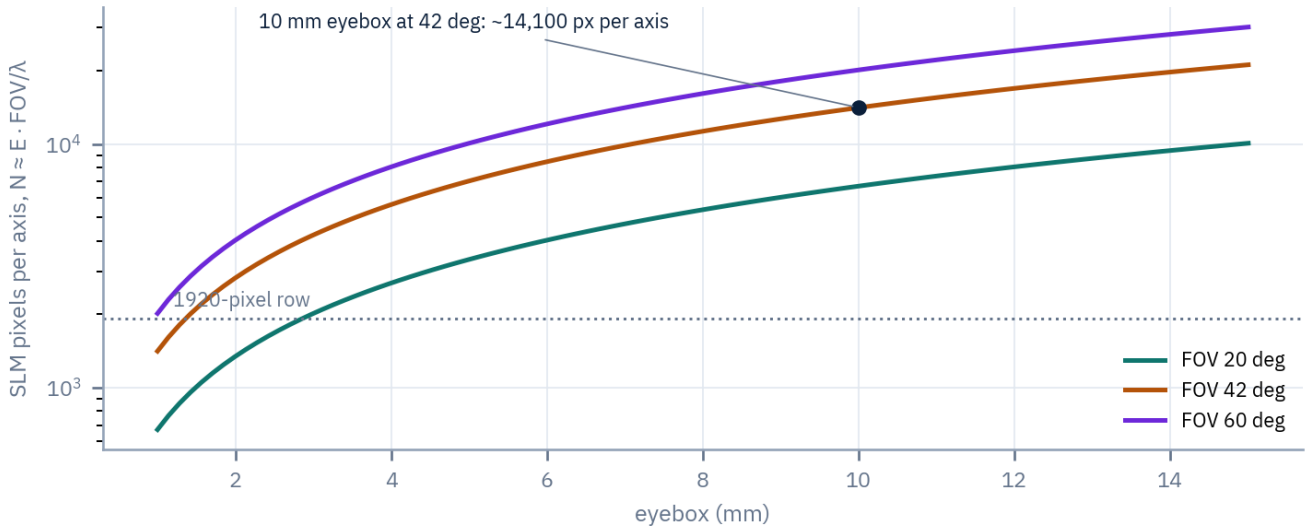


Figure 25. SLM pixels per axis versus eyebox and field of view. E x FOV = N lambda at 520 nm, without synthetic aperture or pupil replication. The dotted line is a 1920-pixel row.

ANSWERS What does etendue force the optical architecture to supply?

Guide	Index	Critical angle against air
n = 1.5	1.5	41.8 deg
n = 1.8	1.8	33.7 deg
n = 2.0	2.0	30.0 deg

Total internal reflection requires $n_{core} > n_{clad}$ with $\sin\theta_c = n_{clad} / n_{core}$. A lower critical angle widens the range of guided angles, which is why higher-index materials bear on achievable field of view. Real devices add polarization dependence, angular response, and efficiency and uniformity trade-offs from pupil replication.

Diffraction sets the angular scale of a finite aperture: theta approximately 1.22 lambda / D, which for D = 1 mm at 520 nm is 0.63 mrad (0.036 deg). Wave-optics propagation for field generation uses the angular-spectrum transfer function H = exp(i 2 pi z sqrt(1/lambda^2 - f_x^2 - f_y^2)) for propagating components, with a grid that resolves the highest propagating spatial frequency, 1/(2

Delta) >= sin theta$_{max}$ / lambda, and band-limiting to avoid aliasing at large z E0. The research question is embedding these standard tools in a real-time closed loop, not inventing them.

42 Candidate architectures and technology ladder

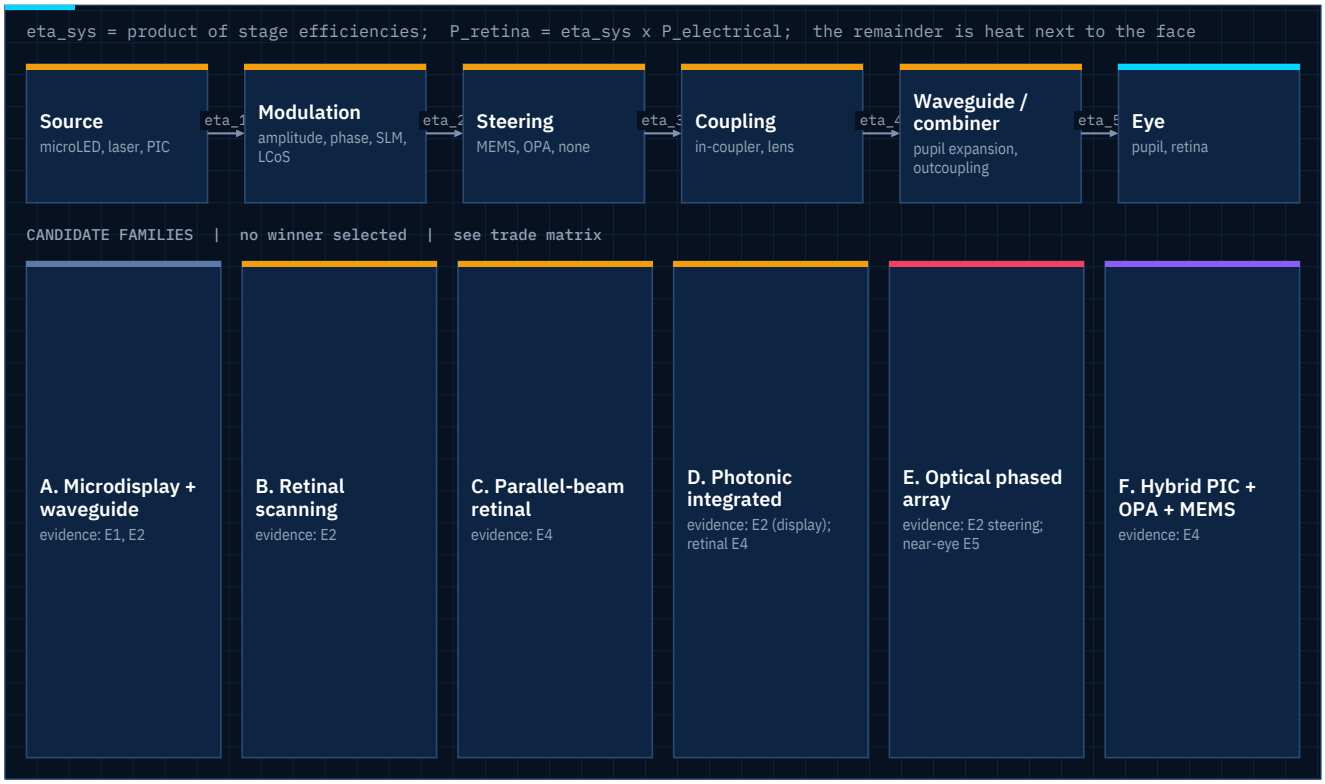


Figure 26. Optical architecture and candidate families. The invariant is controllable optical presentation. Source, steering and combiner are replaceable until experiments select them.

ANSWERS Where does optical power go, and which families compete?

No family is selected. The matrix gives the known basis for each cell; where no measurement or documented property exists, the cell says OPEN rather than carrying an intuition.

Criterion	A Microdisplay + waveguide	B Retinal scanning	C Parallel-beam retinal	D Photonic integrated	E Optical phased array	F Hybrid PIC + OPA + MEMS
Source	microLED, OLED or illuminated LCoS	visible laser	laser or emitter array	on-chip visible laser, coupled	coherent visible laser	laser through PIC
Steering / modulation	pixel array plus projection optics	scanning mirror	multiple steered beams	PIC routing plus LCoS (Shi et al.)	phase control of an emitter array	PIC routing, OPA fine, MEMS coarse
Combiner / aperture	waveguide or reflective combiner	pupil-forming combiner	pupil-forming	thin flat panel (about 2 mm reported)	free-space or waveguide coupling	OPEN
FOV and eyebox	set by waveguide index and replication	scan angle and small exit pupil	beam count supplies etendue	OPEN ; LCoS pixels limit	lobe-free field set by pitch	OPEN
Efficiency	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN
Brightness, MTF, distortion	OPEN ; waveguide uniformity trade-offs documented	OPEN	OPEN	OPEN	OPEN	OPEN
Chromatic performance	metagrating waveguides improve it [C]	OPEN	OPEN	OPEN	per-wavelength pitch	OPEN
Speckle	low (incoherent source)	present; reduced by diversity C = 1/sqrt(N)	present	present	present	present
Thermal load	heat at emitter and driver	OPEN	OPEN	laser and LCoS heat OPEN	OPEN	OPEN
Eye-safety analysis route	photobiological, applicability TBD	laser-class, TBD	laser-class, TBD	laser-class, TBD	laser-class, TBD	laser-class, TBD
Manufacturability, yield	established supply chains	MEMS mature elsewhere	OPEN	thousands of components per chip; yield OPEN	pitch near lithography limit	integration complexity
Calibration complexity	moderate	scan calibration	per-beam alignment	phase and routing	per-element phase	highest

Criterion	A Microdisplay + waveguide	B Retinal scanning	C Parallel-beam retinal	D Photonic integrated	E Optical phased array	F Hybrid PIC + OPA + MEMS
Volume and mass	OPEN	OPEN	OPEN	over 80 % volume reduction vs LCoS stack (reported)	OPEN	OPEN
Evidence class	E1 E2	E2	E4	E2 display; retinal E4	E2 steering; near-eye E5	E4
Principal risk	efficiency, brightness, eye strain	exit-pupil size, speckle	alignment, count	LCoS and thermal retained	visible pitch	integration

VISIBLE OPA PITCH

$$d < \frac{\lambda}{1 + \sin\theta_{max}}$$

CONSTRAINS

Emitter pitch d of an optical phased array that steers to angle theta_{max} without grating lobes.

ENGINEERING CONSEQUENCE

For a +/-90 deg field at 520 nm the pitch must be below 260 nm; for +/-20 deg below 387 nm. Both are far below conventional visible emitter spacing, so steering range and pitch trade directly. A visible OPA is E2 for steering and E5 for near-eye information density.

EXPERIMENT / FALSIFIER

Measure lobe-free angular range versus pitch on a fabricated array; side-lobe level above the model indicates phase-error or coupling terms not modelled.

SPECKLE CONTRAST

$$C = \frac{1}{\sqrt{N}}$$

CONSTRAINS

Number N of independent speckle patterns averaged within the eye integration time, for coherent sources.

ENGINEERING CONSEQUENCE

N = 400 gives C = 0.05. Averaging diversity (wavelength, polarization, angle, time) costs etendue, bandwidth or efficiency, and so appears in the K_{opt} trade rather than as a free fix.

EXPERIMENT / FALSIFIER

Measured contrast on the bench versus the 1/sqrt(N) prediction; excess contrast means correlated patterns.

Technology ladder O0 to O9

Stage	What it proves	Instrument	Exit criterion	Gate
O0 Analytical optical model	K_{opt} sensitivities	twin G, S, W, P, E, H, F	predictions with stated assumptions	none
O1 Source characterization	power, spectrum, divergence, noise, thermal	radiometer, spectrometer, camera	source model fits measurement	Gate 0
O2 Beam and field generation	field fidelity before the combiner	wavefront sensor, camera	field matches model within a stated error	Gate 0
O3 Combiner and waveguide	coupling efficiency, uniformity, ghosts	bench with eye-aperture camera	efficiency and uniformity measured	Gate 0
O4 Eyebox and FOV	the FOV-eyebox frontier, MTF	pupil-scanning camera	frontier measured; compared to etendue bound	Gate 0
O5 Gaze-conditioned allocation	registration error, foveation saving	gaze simulator, high-speed camera	saving versus gaze error and delay	Gate 0
O6 Closed-loop optical control	latency, stability, thermal drift under control	high-speed trace, thermal chamber	loop stable inside bounds	Gate 0
O7 Thermal and safety integration	fault-injection survival; thermal envelope	fault harness, radiometry	nominal-, single-fault- and misalignment-safe	Gate 1
O8 Human visual evaluation	comfort, task performance, artifacts	controlled studies (ethics approved)	comfort and performance at matched task	Gate 2 then 3
O9 Integrated wearable optical subsystem	all of the above on one device	full instrumentation	population yield $P(Q >= Q_{min})$	Gate 2 to 4

43 Optical resource allocation and foveation

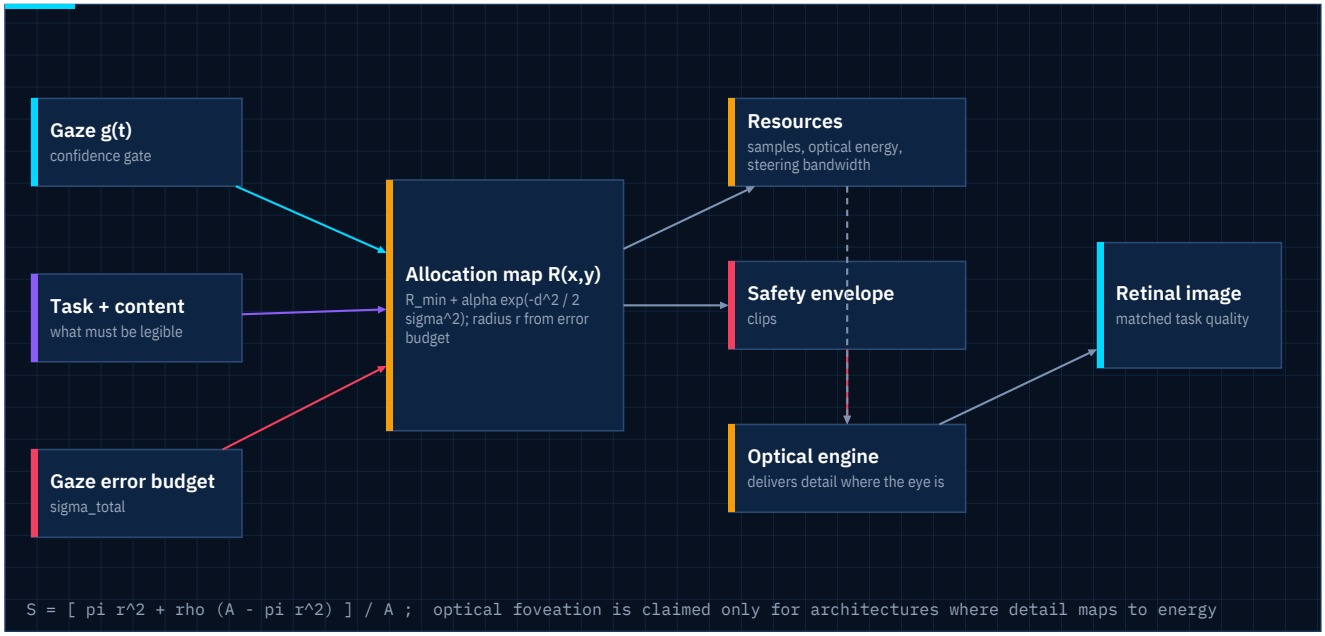


Figure 27. Optical resource allocation. The allocation radius is set by the gaze error budget, so a saving is conditional on measured gaze error and latency.

ANSWERS How much physical optical resource does a task need?

A gaze-contingent allocation $R(x, y) = R_{min} + \alpha \exp(-d(x, y)^2 / 2\sigma^2)$ assigns samples, optical energy and steering bandwidth by distance d from the gaze point. The full-detail radius r is set by the gaze error budget, so the saving is conditional. The claim is about optical resource, not only render cost: it holds only for architectures where detail maps to energy.

FOVEATION SAVING

$$S = \frac{\pi r^2 + \rho (A - \pi r^2)}{A}$$

CONSTRAINS

Fraction S of uniform-resolution samples needed when full detail covers radius r inside field area A and the periphery runs at relative density rho.

ENGINEERING CONSEQUENCE

For a 40 x 23 deg field ($A = 920$ deg^2), $r = 5$ deg and $\rho = 0.1$: $S = 0.177$, a 5.7-fold reduction. Adding a 3 deg gaze-error margin ($r = 8$ deg) gives $S = 0.297$, a 3.4-fold reduction. The saving falls quickly as gaze error and delay grow.

EXPERIMENT / FALSIFIER

O5: measured saving versus gaze error and latency, plus psychophysical thresholds for peripheral degradation (O8). Peripheral motion sensitivity is not modelled here.

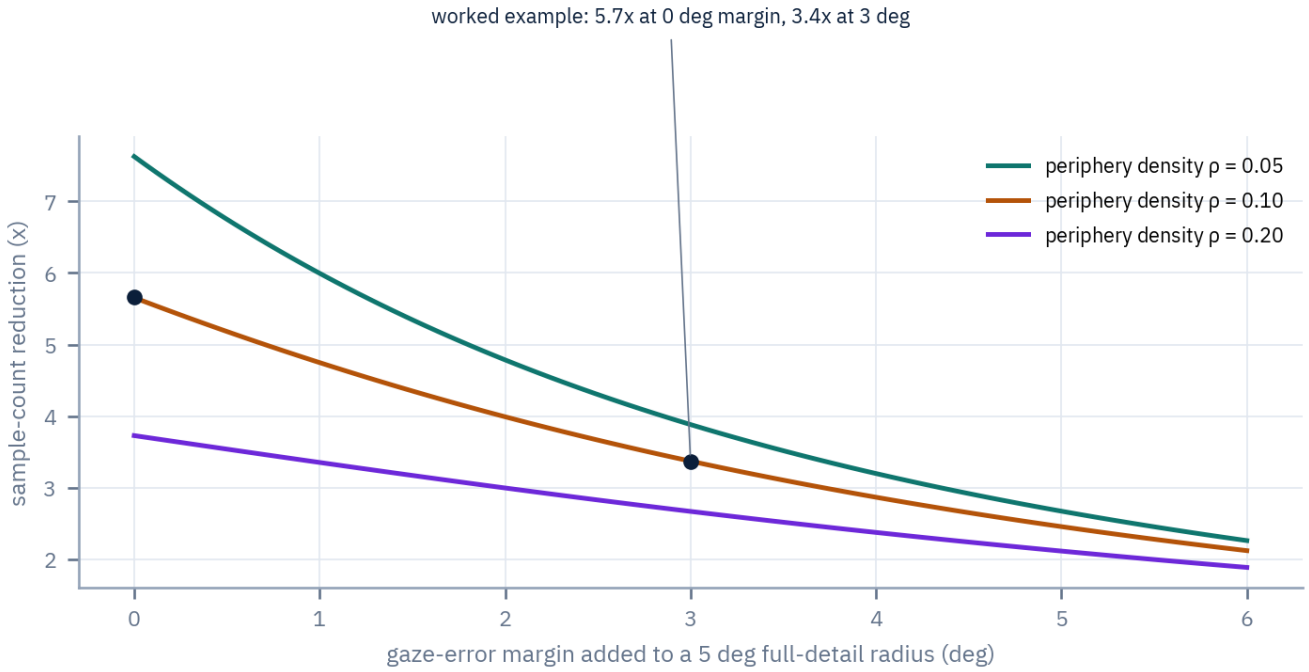


Figure 28. Sample-count reduction versus gaze-error margin. Arithmetic only. Real gains also depend on peripheral motion sensitivity and latency during saccades.
ANSWERS How fragile are foveation savings to gaze error?

Gaze error term (budget)	Source	Measured by
$\text{sigma}_{\text{cal}}$	eye-model calibration residual	per-user calibration test
$\text{sigma}_{\text{slip}}$	frame movement on the head	slip detector, fit study
$\text{sigma}_{\text{pupil}}$	pupil-center shift with dilation	varying luminance protocol
$\text{sigma}_{\text{quant}}$	sensor and estimator quantization	bench
$\text{sigma}_{\text{lat}}$	state age times angular rate: omega dt	timing harness

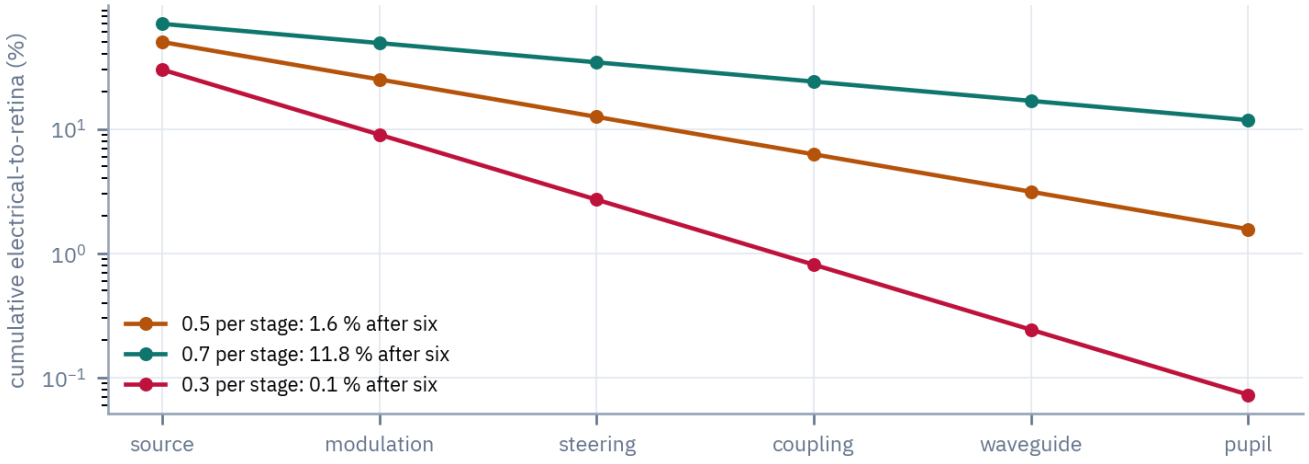


Figure 29. Cumulative electrical-to-retina efficiency. Six multiplicative stages. Illustrative arithmetic, not a NEXUS measurement.
ANSWERS Why are optical and thermal architecture one problem?

MULTIPLICATIVE EFFICIENCY CHAIN

$$\eta_{sys} = \eta_{src}\eta_{mod}\eta_{steer}\eta_{cpl}\eta_{wg}\eta_{pup}$$

$$P_{retina} = \eta_{sys} P_{el}$$

CONSTRAINS

Stage efficiencies from electrical input to retina, and therefore the heat deposited next to the face.

ENGINEERING CONSEQUENCE

Six stages at 0.5 give 1.6 percent (a 64-fold loss); at 0.7, 11.8 percent; at 0.3, 0.07 percent. The remainder is heat. Optical and thermal architecture are one problem, and a stage efficiency measured on a breadboard does not carry to the integrated path.

EXPERIMENT / FALSIFIER

O1 to O3: stage-by-stage efficiency measured in place; product of stages compared with end-to-end measurement.

44 Predictive optical control and calibration

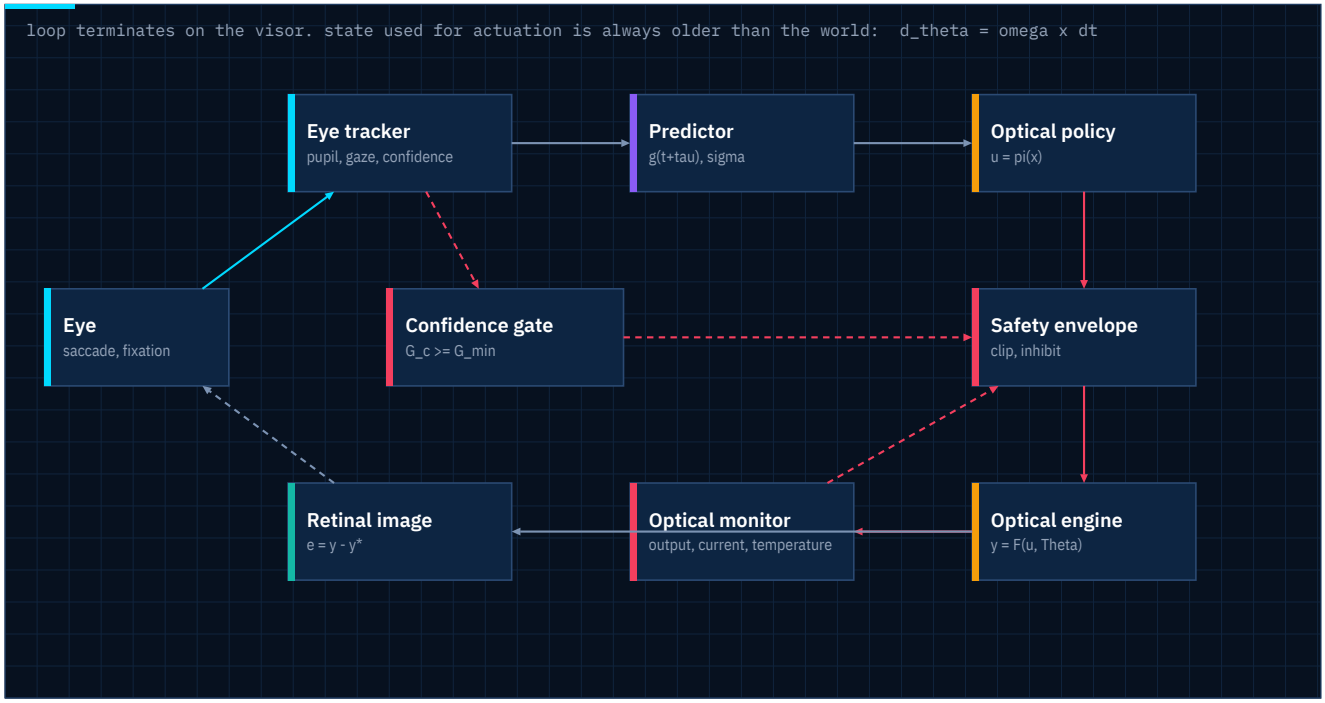


Figure 30. Optical control loop. The loop closes on the visor. The safety envelope sits between policy and engine, and a confidence gate can force a reduced-energy state.
ANSWERS Where is the loop closed, and where is it clipped?

```
state     x = (gaze, head pose, pupil, source state, temperature)
command   u = optical request (eye, pose, depth, field, brightness)
plant     y = F(u, Theta);  Theta = optical parameters, calibration, thermal state
objective   min  sum_t || Q_task(t) - Q_retina(t) ||^2 + lam_E E + lam_T T
constraint  u in safe_envelope(Cs)
prediction  g_hat(t + tau) with covariance
gate        if sigma_gaze > sigma_max: widen the foveal region and reduce energy
```

The loop runs on the visor. A confidence gate monitors eye-state confidence; when it fails, the response is graded: widen the foveal region, then uniform presentation, then reduced-energy static output, then safe-off, each bounded in time. Prediction horizon is limited to where residual uncertainty stays below the registration tolerance (RX6). Calibration is layered: factory (source output versus drive, alignment, pose of each element), per user (IPD, eye relief, pupil, refraction), session (fit and slip) and online (drift, with self-test against factory references).

Thermal coupling

Source temperature changes wavelength, efficiency and alignment, so thermal state is an input to optical control and a source of drift in calibration. Redundant thermal sensors at the source and near-eye structure feed compensation tables and the throttle ladder; thermal-sensor failure is a fault-injection case (O7).

45 Optical safety



Figure 31. Optical safety architecture. The performance path may be wrong; the independent path does not trust it. Tracking may improve efficiency but is never the only barrier.

ANSWERS What remains safe when every adaptive subsystem fails?

Safety is an architectural property, not a warning label. The abstraction is a closed-loop near-eye optical engine whose accessible output is constrained by passive optics, monitored continuously, and attenuated or disabled whenever the verified safe state is lost. The design is never described as the absence of a source: the actual source technology is identified and the applicable standards applied. The principles are fail-safe rather than fail-open, safety case rather than checklist, and no single control failure may create uncontrolled energy at the eye. Tracking may improve efficiency but is never the only barrier preventing hazardous output.

HAZARDOUS-ENERGY REMOVAL TIME

$$\tau_{tot} = \tau_{det} + \tau_{dec} + \tau_{act} + \tau_{opt}$$

CONSTRAINS

Total time from fault onset to bounded output: detect, decide, actuate and the optical system's own response.

ENGINEERING CONSEQUENCE

The maximum allowable tau_{tot} is derived from source class, exposure model, fault mode and governing standard, not selected. The safety confidence $C_s(t) = f(C_{eye}, C_{thermal}, C_{power}, C_{optical}, C_{fault})$; if $C_s < C_{min}$ the eye power goes to P_{safe}. Neither C_{min} nor P_{safe} is invented here; both are outputs of the hazard analysis.

EXPERIMENT / FALSIFIER

O7: measured removal time per injected fault; any case exceeding the bound fails the gate.

Hazard	Mechanism	Primary mitigation	Verification
Excess accessible output	source-control or optical fault	passive energy limits plus independent monitor	radiometric classification, fault tests
Misalignment	tracking or mechanical failure	attenuation outside the valid eyebox	deliberate misalignment test
Tracking loss	eye or pupil not confidently localized	safe-off or reduced-energy state	sensor dropout injection
Stuck-high drive	driver or DAC fault	independent current ceiling	hardware fault injection
Optical element failure	fracture, displacement, coating degradation	mechanical retention, enclosure	shock, thermal cycling, ageing
Software runaway	control logic fault	independent hardware safety path	watchdog and fault injection
Thermal drift	temperature changes source or alignment	thermal sensing, compensation or shutdown	temperature sweep
Calibration drift	long-term parameter change	self-test against factory references	ageing and recalibration study

Hazard	Mechanism	Primary mitigation	Verification
Unexpected reflection	external optical surface	exposure envelope analysis	representative environments
User variability	pupil size, refractive state, eye geometry	eyebox margin and adaptive control	population study

Failures are evaluated in combination (for example tracking failure together with a high source command and thermal drift): the question is the maximum accessible exposure when multiple credible failures occur within the defined fault model. The fault-injection harness injects at least: eye-tracker dropout, false eye detection, stale gaze data, driver saturation, ADC or reference error, monitor-sensor disconnection, thermal-sensor failure, watchdog timeout, clock failure, memory corruption in control variables, alignment offset, rail under- and over-voltage, unexpected reset, and loss of communication between safety-relevant subsystems. Record per injection: fault, detected or not, response, residual exposure, recovery.

Gate	Name	Entry evidence
0	Research safety envelope	source technology selected; hazard model defined; applicable standards mapped; worst-case exposure model; calibrated instrumentation; shutdown path demonstrated
1	Optical subsystem safety	nominal-safe and single-fault-safe and misalignment-safe, before any human wearing
2	Integrated device	adds battery, wireless, thermal load, full software, enclosure and a real eye tracker
3	Long-duration human study	only after previous gates produce evidence
4	Production qualification	frozen alignment tolerances, source calibration, material stack, firmware safety limits, test fixtures, traceability
5	Field monitoring	returned-device analysis, abnormal thermal logs, drift monitoring, battery fault statistics, discomfort reports, firmware safety regression

Candidate standard (applicability to be determined)	Relevance
IEC TS 60825-20:2025	laser products intentionally exposing face or eyes; headset and glasses classes
IEC 60825-1:2014; ANSI Z136.1-2022; FDA laser guidance	laser classification and use; US pathway
IEC 62471:2006; ISO 15004-2:2024	photobiological safety; devices directing radiation into or at the eye
IEC 62368-1:2023; IEC 62133-2:2017+A1:2021	ICT equipment safety; portable lithium cells
IEC 62311:2019; IEEE C95.1-2019	human exposure to electromagnetic fields
IEC 61000-4-2:2025; CISPR 32:2015+A1:2019	ESD immunity; multimedia equipment emissions
ISO 9241-392:2015; ISO 9241-210:2019; ISO 26800:2011	stereoscopic visual fatigue; human-centred design; ergonomics
ISO 10993-1:2025	biological evaluation, depending on classification and claims

Standards are carried from the project reference list and are not re-verified; edition, applicability and claims must be confirmed for the selected architecture. Certification is not established by a software simulation, a mathematical barrier function or an internal test. Claims the program can aim to support: accessible output is bounded; source control is fail-safe; loss of tracking drives the system to a verified safe state; thermal behaviour stays inside a validated envelope; battery faults are contained; EMC and exposure are evaluated; visual comfort is studied longitudinally; production units are screened. Claims to avoid: zero risk, impossible to harm the eye, approved everywhere, safe for every person and condition, or that a technology name alone makes it intrinsically safe.

46 Optical manufacturing and yield

TOLERANCE STACK

$$\sigma_{sys}^2 = \sum_i \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_i^2$$

CONSTRAINS

How component tolerances propagate to a system parameter f (alignment, eyebox position, exit-pupil placement).

ENGINEERING CONSEQUENCE

Independent tolerances add in quadrature, so a few dominant terms set the budget; thermal expansion and ageing add drift terms. Active alignment and factory calibration trade assembly cost against tolerance.

EXPERIMENT / FALSIFIER

Measured unit-to-unit spread of f exceeding the propagated sigma_{sys} indicates a correlated or unmodelled term.

POPULATION YIELD

$$Q_{pop} = P(Q \geq Q_{min})$$

CONSTRAINS

Fraction of units, across IPD, pupil, refraction, fit and tolerance, that meet minimum image quality.

ENGINEERING CONSEQUENCE

A beautiful laboratory prototype is not enough. Yield is a separate question for image quality, $P(Q < Q_{min})$, and for safety, $P(E_{accessible} > E_{limit})$ under the validated fault and tolerance model. Monte Carlo on the optical twin estimates both; production fixtures verify them.

EXPERIMENT / FALSIFIER

O9 population study versus twin prediction; any safety-yield failure blocks Gate 4.

NXS-C09 Optical engine

01	**Purpose**	Deliver controllable optical presentation to the eye at the lowest resource and heat the physics permits.
02	**System requirement**	OPT-001 to OPT-005, PWR-001, THM-001, SYS-002.
03	**Inputs**	Optical requests {eye, pose, depth, field, brightness, foveation, temporal profile} after the safety envelope; calibration; thermal state.
04	**Outputs**	Retinal image; health, monitor and thermal telemetry; optical state estimate.
05	**State**	Source state, alignment calibration, thermal state, last request, health, safety state.
06	**Interface**	Optical HAL: configure, submit(request), get_state, get_health, enter_safe_state, get_calibration.
07	**Timing requirement**	Request-to-photon latency inside the Loop B budget `DERIVED`; one frame of queue at most.
08	**Power budget**	`OPEN`; $P_{retina} = eta_{sys} P_{el}$.
09	**Thermal budget**	Source and driver heat adjacent to the face; `OPEN`.
10	**Bandwidth requirement**	Request stream small; field data generated locally; reservoir sends scene content, never source commands.
11	**Memory requirement**	`OPEN`; field computation memory is local.
12	**Accuracy / quality requirement**	K_{opt} nominal and worst-case; registration error; MTF; speckle; uniformity; all measured, none asserted.
13	**Safety constraint**	Accessible output bounded by passive optics and independent hardware (OPT-003); never application-controlled.
14	**Security boundary**	Signed optical firmware; no network path to the driver; calibration data integrity-protected.
15	**Failure modes**	Source degradation, misalignment, drive fault, thermal drift, contamination, calibration drift, element failure.
16	**Degradation behavior**	Reduced foveation, then uniform presentation, then reduced-energy static output, then safe-off.
17	**Calibration requirements**	Factory alignment and source calibration, per-user calibration, online self-test.
18	**Manufacturing implications**	Alignment tolerances, adhesives and thermal expansion, contamination, test fixtures; population yield.
19	**Verification method**	Radiometry, wavefront and MTF measurement, thermal chamber, fault injection.
20	**Validation experiment**	O1 to O9.
21	**Acceptance criterion**	K_{opt} measured with stated margins; closed loop stable; yield reported over a population.
22	**Falsification condition**	If no candidate family reaches the target FOV-eyebox frontier at an acceptable efficiency and heat, the see-through programmable-field concept is revised (for example to a narrower field with eye-locked foveation).
23	**Current maturity**	`M2` `V1`; optical engine software `PROTOTYPE` `REPOSITORY-ASSERTED`; hardware `NOT STARTED`.
24	**Next implementation artifact**	Breadboard source and beam characterization rig (O1, O2) with radiometric logging.

NXS-C10 Optical safety path

01	**Purpose**	Bound accessible optical output independently of every software layer, and drive the engine to a verified safe state on loss of confidence or fault.
02	**System requirement**	SAFE-OPT-001 to OPT-003, SAFE-PWR-001, OPT-003, PWR-001, PWR-004, SYS-002, INV-6.
03	**Inputs**	Pupil presence, output monitor, drive-current sensor, thermal sensors, rail state, watchdog, eye-state confidence (advisory).
04	**Outputs**	Attenuate or inhibit command to the driver; latched state; safe-state indication.
05	**State**	Safety state machine (OFF, SELFTEST, READY, ACTIVE, DEGRADED, SAFE_OFF, FAULT); fault latch.
06	**Interface**	Hardware lines and a minimal frozen interface; no dependency on OS, GPU, reservoir or network.
07	**Timing requirement**	$tau_{tot} = tau_{det} + tau_{dec} + tau_{act} + tau_{opt}$, bounded by hazard analysis `OPEN`.
08	**Power budget**	`OPEN`; always on while the source is armed.
09	**Thermal budget**	Negligible `OPEN`.
10	**Bandwidth requirement**	Hard-wired signals; no streaming.
11	**Memory requirement**	Minimal; state machine and thresholds.
12	**Accuracy / quality requirement**	Detection coverage and removal time measured per fault class.
13	**Safety constraint**	Passive-first barrier stack: source limit, beam shaping, aperture, wavelength management, verified exit pupil; independent active barriers on top.

14	**Security boundary**	Isolated from application processes; tamper-evident configuration; signed thresholds.
15	**Failure modes**	Sensor failure, false pupil detection, stuck-high drive, watchdog failure, common-mode software fault, power fault.
16	**Degradation behavior**	Fail-safe: on any uncertainty reduce or remove the source; DEGRADED returns to ACTIVE only on recovered confidence, otherwise SAFE_OFF.
17	**Calibration requirements**	Monitor and sensor calibration against factory references; self-test at start-up and periodically.
18	**Manufacturing implications**	Per-unit verification of limiter thresholds (SAFE-MFG-001).
19	**Verification method**	Fault-injection harness; radiometric classification; combined-failure analysis.
20	**Validation experiment**	O7, Gate 1.
21	**Acceptance criterion**	Nominal-safe, single-fault-safe and misalignment-safe, with measured removal time inside the bound.
22	**Falsification condition**	Any credible fault combination in the defined model that leaves accessible output above the limit falsifies the architecture for that source class.
23	**Current maturity**	`M1` `V0` `V1`; `NOT STARTED` in hardware.
24	**Next implementation artifact**	Fault-injection harness specification and the first hazard-analysis table with measured removal-time slots.

NXS-C11 Eye tracking and gaze control

01	**Purpose**	Estimate gaze, pupil and eye pose with confidence at the rate and latency Loop B requires, and feed the optical controller and safety path.
02	**System requirement**	OPT-004, SAFE-OPT-002, VIS-002, HMI-001.
03	**Inputs**	IR eye images, IMU, calibration, optical state.
04	**Outputs**	Gaze vector with covariance, pupil position and size, blink state, confidence, slip estimate.
05	**State**	Eye model, per-user calibration, recent gaze history, confidence.
06	**Interface**	EyeTracker HAL; gaze state over a bounded-age local channel; Observation descriptors for the semantic layer.
07	**Timing requirement**	State age <= $dtheta_{tol}$ / omega `DERIVED`; estimator latency counted in T_{sensor}.
08	**Power budget**	`OPEN`; always on during use.
09	**Thermal budget**	Illuminators and camera heat near the eye `OPEN`.
10	**Bandwidth requirement**	Eye frames stay local (example 230 Mb/s).
11	**Memory requirement**	`OPEN`.
12	**Accuracy / quality requirement**	$sigma_{tot}^2$ budget term by term; confidence calibrated.
13	**Safety constraint**	Eye-state confidence may improve efficiency; it is never the only safety barrier.
14	**Security boundary**	Raw eye imagery visor-local; gaze state minimized before leaving.
15	**Failure modes**	Slip, blink, glare, occlusion, drift in eye model, latency growth.
16	**Degradation behavior**	Confidence drop widens foveal region, then uniform presentation; safe-off on timeout.
17	**Calibration requirements**	Per-user eye model; session slip estimate; online drift.
18	**Manufacturing implications**	Camera placement and illuminator constraints in the frame.
19	**Verification method**	Gaze simulator, high-speed camera, population study.
20	**Validation experiment**	O5, O6, RX6.
21	**Acceptance criterion**	Measured registration error below tolerance at stated angular rates and delays.
22	**Falsification condition**	If no gaze sensor and predictor combination meets the registration tolerance at 500 deg/s class motion, eye-locked foveation is restricted to fixation-onset updating.
23	**Current maturity**	`M1` `V0` `V1`; `EXTERNALLY DEMONSTRATED` for eye tracking in XR products.
24	**Next implementation artifact**	Registration-error measurement rig on a gaze simulator with a high-speed reference camera.

PART VIII

PHYSICAL ENGINEERING

Mass · power · heat · RF · batteries · manufacturing

SOURCES · SoC, optical, RF, battery, AFE

SPREADING · frame and temple

LIMITS · skin contact, ambient

CONTROL · sensors, throttle ladder, shutdown

Power and heat are architectural constraints, not a section near the end. A laboratory prototype is not a product until it can be built, calibrated and tested repeatedly.

47 Mechanical architecture and mass distribution

Mass, centre of mass, temple loading, nose-bridge pressure, hinge and flex routing, and fit variation across head shapes set comfort before any other property. Mass limits are set by the comfort study (HF-001); none is asserted here `OPEN`. Placement of mass is as important as its total: battery and heavy electronics are candidates for temple tips to move the centre of mass rearwards, at the cost of thermal contact on the ear region and cable routing through hinges. The mechanical contract is a tolerance stack with calibration: alignment of cameras, IMU, optical engine and eye tracker is held by structure or restored by calibration, and shock, vibration, thermal expansion and ageing are tested against the calibration validity interval.

48 Power architecture

SYSTEM POWER

$$P_{sys} = P_{sense} + P_{comp} + P_{opt} + P_{RF} + P_{mem} + P_{conv} + P_{loss}$$

CONSTRAINS

Allocation of the average and peak power budget among sensing, compute, optics, RF, memory, conversion and loss.

ENGINEERING CONSEQUENCE

Every rail is instrumented so energy per completed task is reportable. Conversion and conduction loss are first-class terms: conductor loss is P^2R/V^2, so bus voltage is a design variable, and PMIC efficiency at low load decides standby energy.

EXPERIMENT / FALSIFIER

Per-rail power traces on a prototype versus the model; unexplained residual points to an unmodelled load such as radio retries or memory traffic.

Rail	Loads	Protection	Brownout response	Instrumentation
Compute	SoC cores, GPU, NPU, ISP	current limit, thermal throttle	save state; reduce to M0	rail energy per task
Memory	DRAM, storage	sequenced power	flush then halt	rail energy
Sensor	cameras, IMU, microphones	filtered, low noise	drop non-critical sensors first	rail current
Optical source	driver, source, monitor	independent hardware current and energy ceiling	source to safe-off before any other rail	current, voltage, optical monitor
Neural analog	AFE, ADC, reference	isolated low-noise rail	report loss, discard window	rail noise, current
RF	radio, front end	burst limiter	link down; M0	energy per bit
Haptics and audio	actuators, speakers	current limit	mute	rail current

Burst workloads (a short model inference, a reflow of the scene) and sustained workloads (continuous perception, optical drive) are budgeted separately; duty cycling moves energy in time without changing it, and migration moves it to the reservoir at a radio cost (Section 34).

49 Thermal architecture

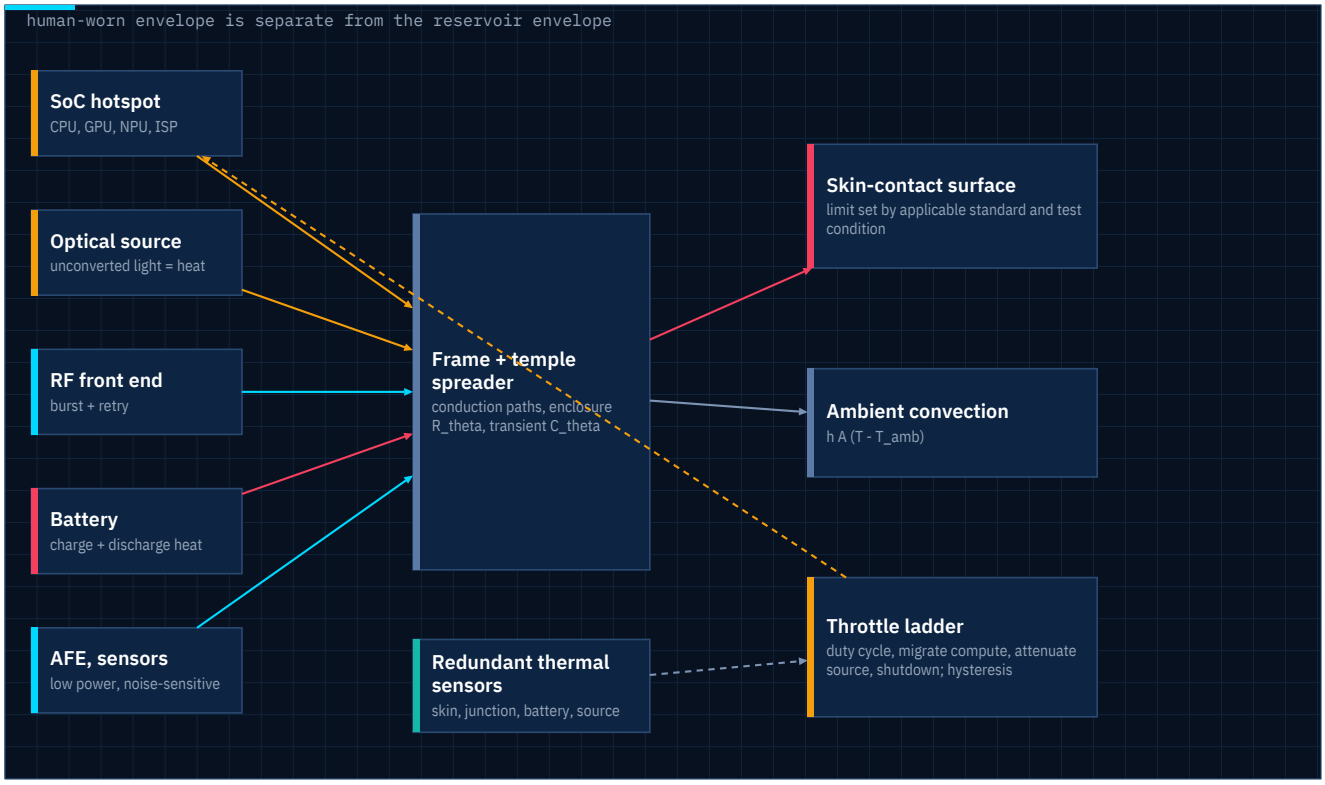


Figure 32. Thermal and power architecture. Heat sources share one small skin-contact budget. A throttle ladder with hysteresis acts on redundant thermal sensing.
ANSWERS Where does heat originate and where does it go?

LUMPED THERMAL MODEL

$$C_\theta \frac{dT}{dt} = P(t) - \frac{T - T_{amb}}{R_\theta}$$

$$\Delta T_{ss} \approx P\,R_\theta$$

CONSTRAINS

Temperature response of a body with thermal capacitance C_{theta} and thermal resistance R_{theta} to ambient, driven by power P(t).

ENGINEERING CONSEQUENCE

Steady state rise is P R_{theta}: 5, 10 and 20 K/W give 5, 10 and 20 K per watt. Transient behaviour matters as much: with an assumed 20 J/K, tau = R_{theta} C_{theta} is 100 to 400 s, so a burst of minutes is far from steady state while a sustained load is not. The lumped form is only a start. Real design needs the skin-contact thermal path, the optical-source path, the SoC hotspot, battery constraints, RF contribution and enclosure resistance as separate nodes.

EXPERIMENT / FALSIFIER

Instrumented prototype: measured step response against the lumped prediction; a multi-time-constant response means more nodes are needed.

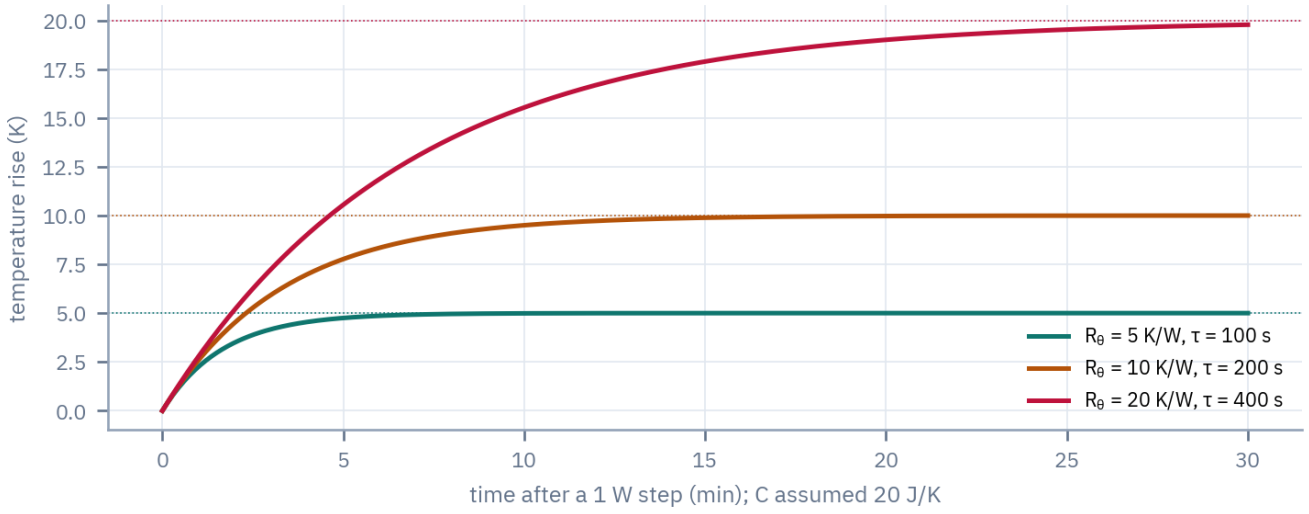


Figure 33. Transient thermal step response. Lumped model, 1 W step. Capacitance 20 J/K is assumed; thermal resistances are assumed, not measured.
ANSWERS How long before a burst workload becomes a sustained one?

Thermal topic	Architecture requirement
Skin-contact path	conduction through frame to skin is the binding limit; limit set by the applicable standard and test condition `OPEN` (touch-temperature limits in ICT equipment standards are candidates, applicability to be determined)
Optical-source path	unconverted light is heat adjacent to the face; thermally isolated from skin; independently monitored
SoC hotspot	spreading and local sensing; throttle before the junction limit
Battery	charge and discharge heat; independent protection; thermal-runaway containment
RF	burst and retry heat is a separate source from the SoC and sits in a different place
Enclosure	R_{theta} and C_{theta} per enclosure; hair, headwear and ambient as variable boundary conditions
Burst and sustained workloads	scheduler knows the thermal state and its derivative, not only temperature
Thermal sensors	redundant for skin and source; disagreement is a fault
Degradation modes	drift of optical alignment and wavelength with temperature; battery ageing
Human-worn versus reservoir	separate envelopes; the reservoir has its own enclosure and ventilation

Throttle level	Trigger	Action	User-visible
T0	normal	none	none
T1	temperature rising faster than a bound	duty-cycle non-critical sensing; migrate compute to the reservoir	none
T2	approach to a skin or junction threshold	reduce optical duty and brightness; reduce model size	dimmer or lighter features
T3	threshold exceeded	attenuate source; suspend non-essential features	notice shown
T4	any sensor disagreement or hard limit	thermal shutdown with hysteresis	device off, safe state

NXS-C12 Power and thermal architecture

01	**Purpose**	Allocate power, bound heat at the skin, and expose power and thermal state to the scheduler and the safety path.
02	**System requirement**	PWR-001 to PWR-004, THM-001 to THM-004, VIS-003, SAFE-THM-001.
03	**Inputs**	Rail currents and voltages, temperatures (skin-adjacent, junction, battery, source, enclosure), workload schedule, ambient.
04	**Outputs**	Throttle commands, power and thermal telemetry, shutdown, energy ledger.
05	**State**	Thermal state and derivative per node, throttle level with hysteresis, rail state, battery state.
06	**Interface**	Power and Thermal HAL; state to Compute service; hard-wired shutdown lines.
07	**Timing requirement**	Throttle response faster than the thermal time constant of the fastest source `OPEN`; independent hardware shutdown.
08	**Power budget**	`OPEN` system budget; allocation by rail.
09	**Thermal budget**	Skin-contact limit from the applicable standard `OPEN`; separate reservoir envelope.
10	**Bandwidth requirement**	Telemetry only; small.
11	**Memory requirement**	Small `OPEN`.
12	**Accuracy / quality requirement**	Temperature accuracy sufficient for the limit margin `OPEN`; energy per task resolution.
13	**Safety constraint**	Optical source rail has an independent ceiling and goes first in brownout; battery protection independent of software.
14	**Security boundary**	Throttle state is not writable by applications; signed thresholds.
15	**Failure modes**	Sensor failure or disagreement, thermal runaway, brownout, regulator fault, battery fault, stale thermal state.
16	**Degradation behavior**	Climb the throttle ladder; sensor disagreement is treated as the worse reading; shutdown with hysteresis.
17	**Calibration requirements**	Thermal sensor calibration; R_{theta}, C_{theta} identification per unit class.
18	**Manufacturing implications**	Thermal interface materials, spreader tolerances, assembly variance affect R_{theta}.
19	**Verification method**	Instrumented thermal mapping; load steps; chamber tests; fault injection on rails.
20	**Validation experiment**	O6, O7, RX1, RX2, wear test.
21	**Acceptance criterion**	Skin temperature within the standard limit at worst-case compute plus optical plus RF load; throttle ladder acts before any limit.
22	**Falsification condition**	If the skin-contact limit cannot hold at a load that supports M0 with optical presentation, optical duty or local function must be reduced or the optical branch changes.
23	**Current maturity**	`M1` `V0`; `NOT STARTED` in hardware.
24	**Next implementation artifact**	Instrumented dummy-head thermal rig with step-load profiles and the lumped model fitted to it.

50 RF, EMI and EMC, battery

Radio sits centimetres from microvolt-scale neural front ends and eye-tracking sensors. RF performance is a body-worn problem: head and hand blockage, antenna placement, and exposure evaluation for devices worn at the head (candidate standards IEC 62311:2019 and IEEE C95.1-2019; applicability to be determined). Electromagnetic compatibility is verified with the radio active, not only on the bench: shielding, filtering, clock-harmonic placement and reference routing are design variables for the neural analog rail. Battery design starts from independent protection and containment (candidate standard IEC 62133-2); placement is a mass and thermal decision; runtime follows the arithmetic in Section 37.

51 Sensors, packaging, calibration architecture

Sensor	Purpose	Locality	Privacy class	Failure modes
World cameras	perception, anchoring	visor; features out	bystander-sensitive	occlusion, blur, thermal drift
Eye cameras and illuminators	gaze, pupil, safety	visor only	biometric-adjacent	slip, glare, blink
IMU	head pose, artifact reference	visor	low	bias drift
Microphones	speech, context	wake word local; speech by policy	bystander-sensitive	noise, wind
EMG, EEG, EOG electrodes	intent evidence	neural node	neural-sensitive	contact loss, motion
Don/doff and proximity	wear state	visor	low	false state
Thermal sensors	throttle and safety	visor	low	drift, disagreement
Ambient light	transmission and brightness control	visor	low	saturation

Packaging decisions are volume allocation under thermal, RF and optical constraints: flex routing through hinges, connector reliability, sweat and humidity ingress, serviceability of the optical engine, and mechanical retention of optics for shock.

Calibration architecture

NEXUS depends on many calibrated systems. Every calibration is a record with the same fields:

```
CalibrationRecord { id, subject, version, timestamp, reference, validity_interval,
                    confidence, drift_estimate, recalibration_trigger, fallback }
```

Calibration	Reference	Validity and trigger	Fallback
Gaze and eye geometry	per-user fixation targets	confidence drop, slip, don/doff	wider foveal region, uniform presentation
Pupil model	varying luminance protocol	dilation outside calibrated range	conservative centre offset
Camera intrinsics and extrinsics	factory targets	drift against factory reference	down-weight visual pose
IMU	factory rotation table	temperature change, bias drift	increase pose covariance
Coordinate transforms	factory and online residuals	residual growth	inflate transform covariance
Optical alignment and source	factory metrology	self-test failure, thermal drift	reduced energy; safe-off
Neural electrodes, EMG, EEG	impedance and per-user task	each wear; ECE drift	drop neural evidence
Thermal sensors	reference thermometer	disagreement	use the worse reading
Timing	hardware reference	temperature change; exchange loss	widen sigma_t
Reservoir link	link probes	latency distribution shift	local reduced mode

NXS-C13 Calibration architecture

01	**Purpose**	Hold every calibrated quantity as a versioned record with confidence, validity and a fallback, so no consumer uses a stale parameter silently.
02	**System requirement**	MFG-001, MFG-003, SW-005, OPT-004, NEU-002.
03	**Inputs**	Factory data, per-user procedures, online residuals, self-test results, temperature.
04	**Outputs**	CalibrationRecords, validity flags, recalibration requests, drift reports.
05	**State**	Records per subject; validity clocks; drift estimates.
06	**Interface**	Calibration service AIDL; record schema; calibration_ref carried on every Observation.
07	**Timing requirement**	Validity checks are cheap and inline; online drift updates run in Loop C or D.
08	**Power budget**	Negligible `OPEN`.

09	**Thermal budget**	Negligible.
10	**Bandwidth requirement**	Small.
11	**Memory requirement**	Kilobytes per record; `OPEN` total.
12	**Accuracy / quality requirement**	Residuals and drift estimates reported with confidence.
13	**Safety constraint**	Hard limits are never relaxed by calibration; out-of-validity optical calibration forces reduced energy.
14	**Security boundary**	Records signed; factory references integrity-protected.
15	**Failure modes**	Expired validity, drift outside estimate, record corruption, wrong-unit record.
16	**Degradation behavior**	Fallback per record; consumers widen uncertainty or demote the dependent function.
17	**Calibration requirements**	Defines itself; recalibration procedures versioned.
18	**Manufacturing implications**	Factory calibration time and fixtures are cost drivers; calibration time per unit is reported.
19	**Verification method**	Record conformance tests; injected drift; replay with historical calibrations.
20	**Validation experiment**	S1, S4, O5, N2.
21	**Acceptance criterion**	Every Observation resolves to a valid CalibrationRecord or is flagged.
22	**Falsification condition**	If online drift cannot be bounded between recalibrations, the dependent function cannot be offered at its stated accuracy.
23	**Current maturity**	`M1` `V0`; abstraction `REPOSITORY-ASSERTED` for calibration references.
24	**Next implementation artifact**	CalibrationRecord schema, validity-clock library and a drift-injection replay test.

52 Manufacturing, reliability and serviceability

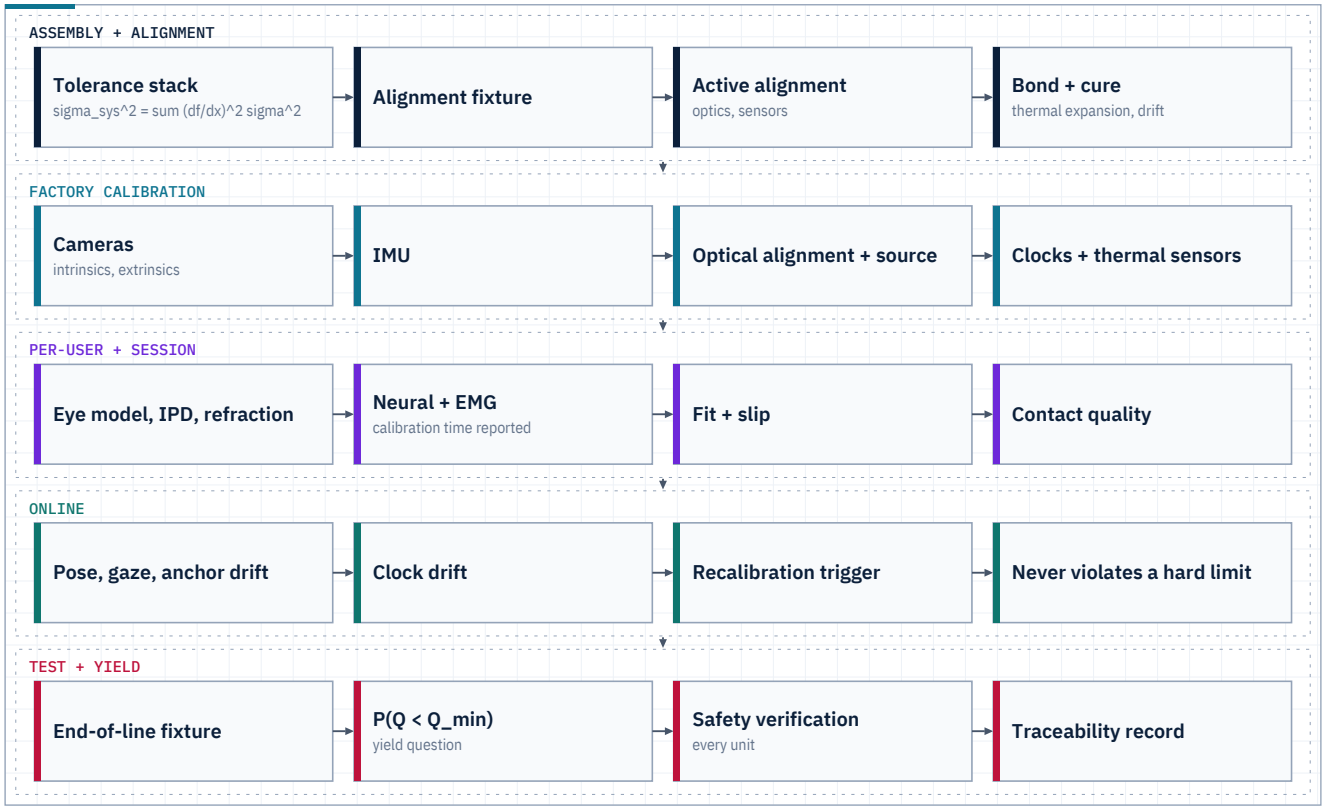


Figure 34. Manufacturing and calibration stack. What must be aligned, calibrated and tested before a unit ships, and where yield is decided.

`ANSWERS` What must be aligned and calibrated, and in which order?

Per-subsystem definition	Why it must exist before design is frozen	Artifact
Tolerance stack	alignment budget; yield	$sigma_{sys}$ analysis per subsystem
Calibration	what is corrected in software versus held by structure	calibration plan
Assembly sequence	access, bonding, cure, test points	process flow
Alignment	active versus passive	fixture specification
Thermal expansion and mechanical drift	calibration validity interval	thermal-cycle results
Connector reliability	flex and hinge cycles	cycle test

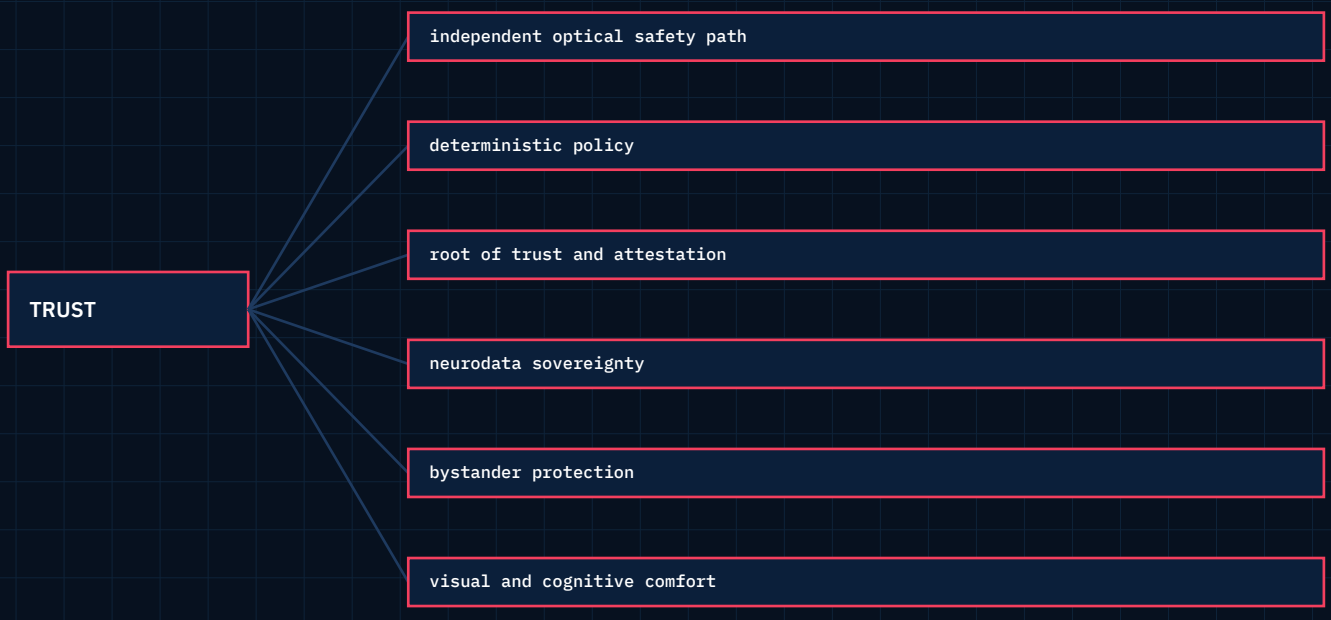
Per-subsystem definition	Why it must exist before design is frozen	Artifact
Optical contamination	sweat, skin oils, dust	cleanability and seal
Ageing	source, coatings, adhesives, battery	accelerated ageing
Shock and vibration	drops; retention of optics	shock and vibration test
Sweat and environment	ingress; corrosion of electrodes	environmental test
Serviceability	what is replaceable, what requires recalibration	service procedure
Yield	cost and safety	Monte Carlo, then production data
Test fixture	end-of-line safety verification	fixture design
Factory calibration time	unit cost	measured per unit

Reliability is `OPEN`. The structural rule is that the safety path avoids serial dependence on non-safety components. Reliability growth is planned from the field-monitoring gate (Gate 5): returned-device analysis, abnormal thermal logs, calibration-drift monitoring and battery fault statistics.

PART IX

SAFETY, SECURITY, HUMAN FACTORS

What remains safe · who holds authority · who is affected



Safety is an architecture, not a label. Security bounds who can gain authority. Human factors decide whether the system is worn at all.

53 Safety architecture and hazard analysis

System-level safety combines the optical safety architecture (Section 45), thermal and battery protection (Sections 49 and 50), software degradation (Section 24) and human-factor safety. Hazard analysis is organized by consequence: optical exposure, skin-contact heat, battery and electrical, RF, mechanical, wrong-action in the physical world, privacy harm, and cognitive or visual harm from long use. Each hazard maps to a mitigation, a safety requirement (NXS-SAFE registry), a verification record and a development gate. Combined failures are analysed explicitly, not only single faults. Bioelectronic safety follows the same separation: observing a signal does not confer the right to act; decoding does not confer permission; a planner may suggest without executing. Neural stimulation, if ever developed, is a higher risk tier and out of current scope, and NEXUS makes no medical-efficacy or clinical claim E5.

Safety case element	Content	Status
Claims	bounded accessible output; fail-safe source control; verified safe state on tracking loss; thermal inside envelope; battery fault containment; EMC and exposure evaluated; longitudinal comfort studied; production screening	PROPOSED
Argument	passive-first barrier stack plus independent active path; combined-failure analysis within the defined fault model	PROPOSED
Evidence	radiometry, fault-injection records, thermal maps, abuse tests, studies	NOT STARTED
Gates	Gate 0 to 5 (Section 45)	PROPOSED

54 Security architecture



Figure 35. Security architecture. Compromise of any higher layer must not be able to produce an unsafe optical state.

ANSWERS Where can an attacker gain authority, and what stops it?

Security protects authority and privacy. Hierarchy: a hardware root of trust anchors secure and measured boot; a key hierarchy derives device identity, per-link session keys and per-stream keys; attestation proves node state to peers; firmware and models are signed with rollback protection; capability authorization and policy tokens bound what any component may do; audit logs record every authorized action; updates are authenticated with canary and rollback; compromise recovery uses revocation, re-provisioning and memory wipe. Cloud exposure is minimized: privacy filter first, no raw neural data by default.

Asset	Attacker	Attack surface	Mitigation	Residual risk	Verification
Neural data	malicious app, rogue node, eavesdropper	HAL, link, storage	restricted domain; per-stream permissions; encryption; raw off by default	side channels via features	access-control tests; leakage audit

Asset	Attacker	Attack surface	Mitigation	Residual risk	Verification
Camera and audio of bystanders	malicious app, insider	capture permissions, uploads	explicit permissions; indicator; local processing; retention limits	misuse by an authorized user	permission tests; H6
Optical safety state	compromised software	driver path, configuration	independent hardware path; no network path to driver	physical tampering	O7 fault injection
Models and firmware	supply chain, update tamperer	update channel	signing; measured boot; rollback protection	compromised signer	update tests; attestation
Memory store	malicious app, stolen device	storage, backup	encryption; policy-governed writes; user control	weak user key handling	write-policy tests
Identity and keys	impersonator	pairing, provisioning	hardware root; mutual attestation	provisioning-time compromise	attestation tests
Reservoir	network attacker, malicious tenant	link, services	attested nodes; authenticated encryption; least privilege	zero-day in a service	penetration test
Capability authorization	confused-deputy app	manifest, tokens	signed manifests; tokens with nonce and expiry	policy misconfiguration	rejection demos
Update channel	downgrade attacker	OTA	signed, versioned, canary	delayed revocation	rollback tests
Audit log	insider erasing traces	log storage	append-only; remote anchoring	loss before anchoring	tamper tests

NXS-C14 Security and trust root

01	**Purpose**	Establish who may do what on and between NEXUS nodes, and keep authority and privacy bounded under compromise of higher layers.
02	**System requirement**	SEC-001 to SEC-004, NEU-001, INV-10, SYS-002.
03	**Inputs**	Boot measurements, attestation requests, capability and policy tokens, keys, update packages.
04	**Outputs**	Attestation results, signed tokens, decrypted streams to authorized consumers, audit entries, revocations.
05	**State**	Key hierarchy, device identity, trust records, revocation list, audit anchor.
06	**Interface**	Secure boot chain; attestation API; key service; Trust AIDL; update service.
07	**Timing requirement**	Authorization checks on the Loop C budget; no cryptographic work in Loop A.
08	**Power budget**	`OPEN` ; cryptographic acceleration in hardware preferred.
09	**Thermal budget**	Negligible `OPEN` .
10	**Bandwidth requirement**	Small control traffic; stream encryption at line rate.
11	**Memory requirement**	`OPEN` ; keys and audit buffers.
12	**Accuracy / quality requirement**	No acceptance of unsigned or rolled-back code; policy tokens unforgeable.
13	**Safety constraint**	Compromise of any application or reservoir service cannot produce an unsafe optical state.
14	**Security boundary**	Hardware root; domain separation; least privilege; signed manifests.
15	**Failure modes**	Key compromise, signer compromise, rollback, token replay, downgrade, supply-chain tamper.
16	**Degradation behavior**	Revoke, re-provision, wipe, boot to a minimal safe image; M0 remains safe.
17	**Calibration requirements**	Device identity and attestation references are calibration-like records.
18	**Manufacturing implications**	Provisioning stations, key injection, tamper-evident packaging.
19	**Verification method**	Attestation tests, penetration tests, fuzzing of interfaces, rollback tests.
20	**Validation experiment**	S0 to S4 conformance; red-team exercise.
21	**Acceptance criterion**	No unsigned artifact runs; attested nodes only for remote execution; audit log complete for authorized actions.
22	**Falsification condition**	A demonstrated path from a compromised application or node to unsafe optical output falsifies the isolation architecture.
23	**Current maturity**	`M1` `V0` ; `NOT STARTED` in implementation.
24	**Next implementation artifact**	Threat-model-derived test list with a first policy-token and manifest-signature demo.

55 Neurodata sovereignty, privacy and bystanders

Data class	Default location	Permission	Retention	Cloud
Raw neural samples	neural node or visor, restricted domain	explicit enable per purpose	minimal, short	never by default
Neural features	visor or reservoir	per-capability grant	policy-governed	privacy filter first
NeuralHypothesis	visor or reservoir	part of intent evidence	episode, then memory policy	derived only
Intent and action records	reservoir	audit class	audit retention	optional, filtered
Camera and audio	visor; features out	explicit; indicator on	minimal	by policy

Neurodata sovereignty means the user owns, inspects, exports and deletes the data and the models adapted to it, and no external party receives neural data without a purpose-bound grant. Bystander protection means capture is visibly indicated, processed locally where possible, and subject to policy, because people in view have not consented. Privacy is the default posture: the system does not record unless a policy permits the write.

56 Human factors

Construct	Why it decides adoption	Measure	Status
Physical comfort	mass, heat, pressure over hours	wear studies; pressure and temperature mapping	OPEN
Visual comfort	near-work eye strain is the top-priority unresolved bottleneck	longitudinal study; validated symptom instruments; objective measures	E5
Cognitive burden	the point of the loop is lower effort	calibrated workload instrument; secondary-task cost	OPEN
Agency and trust	wrong actions break trust faster than correct ones build it	wrong-action and correction rate; trust questionnaire	OPEN
Social acceptability	device worn in public	bystander and wearer studies	OPEN
Accessibility	users who cannot produce a modality	modality-removal study	OPEN

Visual ergonomics include vergence-accommodation consistency, flicker and brightness adaptation, and the possibility that sedentary posture and screen-fixed gaze can be relieved by a see-through spatial interface; that is a hypothesis to test, not a result E4. Comfort claims are made only after longitudinal study (HF-001, SAFE-VIS-001).

PART X

SIMULATION AND VALIDATION

Digital twin · replay · HIL · bench · wearable · humans



Each environment proves one thing and cannot prove the next. A claim states the highest rung it has reached.

57 Digital twin, virtual devices and dataset replay

The digital twin has two halves that share interfaces: an optical twin (geometry, source, wavefront, pupil and eye, exposure, thermal and fault models) and a system twin (sensors, compute nodes, links, human and world models). A virtual device implements the same HAL as hardware, so software above it cannot tell a replayed sensor from a real one (SW-005). The VirtualSensor ABI and the dataset registry are in Appendices H and K; datasets are pinned by version and checked for licence and synchronization fidelity before use.

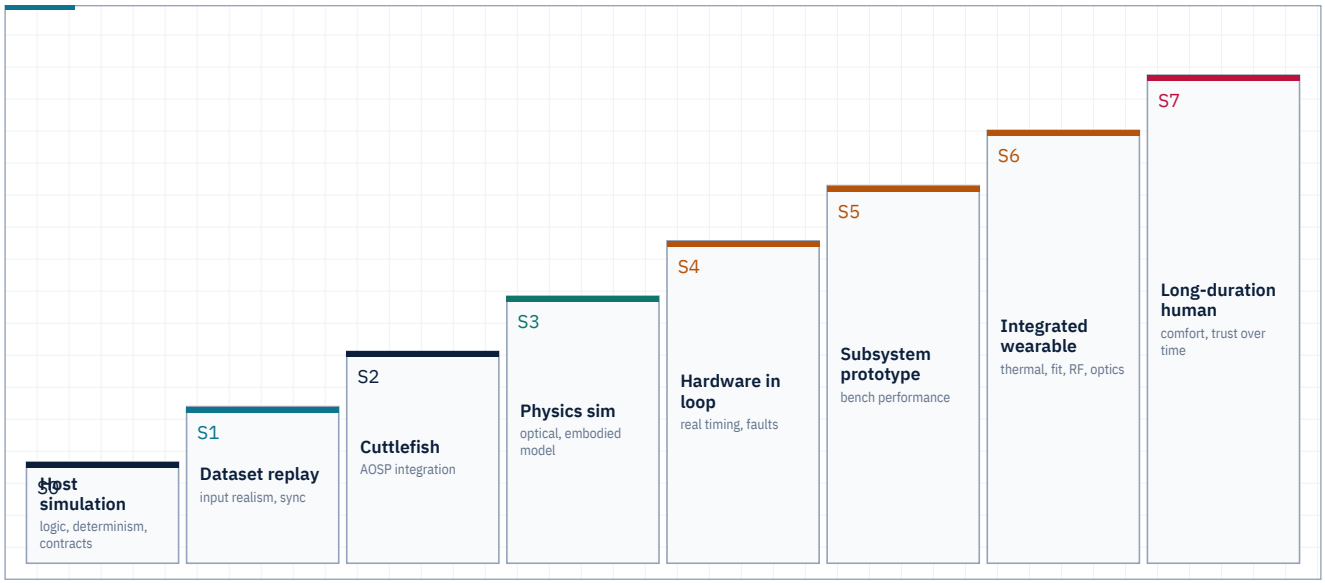


Figure 36. Simulation-to-hardware ladder. Each level proves a different property; a higher level never invalidates a lower one.

ANSWERS What does each execution environment prove, and what can it not?

Level	Environment	Proves	Cannot prove
S0	host simulation	logic, determinism, contracts	real timing, physics
S1	dataset replay	input realism, synchronization handling	closed-loop behaviour
S2	Android and Cuttlefish	platform integration, AIDL, buffers	thermal, optics, real latency
S3	physics simulation (Isaac Lab and Twin)	representation, dynamics, uncertainty	that NEXUS intelligence works for humans
S4	hardware in the loop	real timing, faults, drivers	human comfort, optics yield
S5	subsystem prototype	bench performance of optics, neural, power	integration effects
S6	integrated wearable	thermal, fit, RF, optics together	long-term and population behaviour
S7	long-duration human	comfort, trust, adoption	production yield

Host simulation and replay are available now and carry V0 to V3. Cuttlefish validation is V4. Hardware-in-the-loop (V5) needs boards with hardware timestamps; optical and neural benches (V6) need laboratory instruments; integrated wearable (V7) needs a built device; human validation (V8) needs ethics review and the safety gates; production engineering (V9) needs manufacturing.

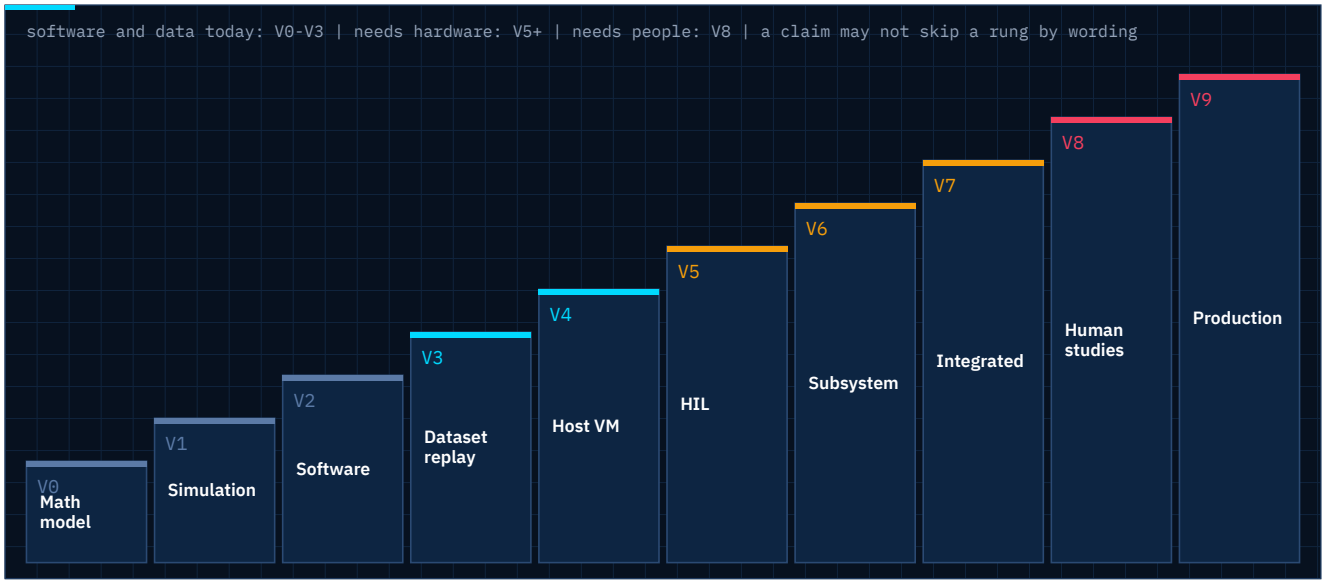


Figure 37. Validation ladder V0 to V9. A claim states the highest rung it has reached. Rung and evidence class are different attributes.

ANSWERS How far has each claim actually travelled?

Reproducibility

Every reported result carries a reproducibility record so a third party can reproduce it, which is what moves an item from REPOSITORY-ASSERTED to VERIFIED:

```
ReproRecord { repo_commit, config_hash, schema_versions, dataset_ids_and_versions, random_seeds,
              host_or_hardware_id, calibration_refs, trace_id, results_hash, test_report_ref, date }
```

PART XI

EXPERIMENTS

Variables · thresholds · falsifiers

HYPOTHESIS · one claim, one variable

MEASUREMENT · instrument, split, baseline

THRESHOLD · fixed before data

DECISION · keep, change, reject

An experiment is defined before it is run: a variable, a success criterion fixed in advance, a failure criterion, and the decision that follows a negative result.

58 Experiment protocol

Acceptance thresholds are set before each experiment from the requirements, not after seeing data, and are recorded in the research log. Every experiment records: experiment_id, hypothesis, baseline, hardware revision, software revision, dataset and version, train and test split, random seed, environment, sensor configuration, model configuration, compute placement, transport, power profile, thermal profile, latency profile, result, confidence intervals, failure cases, interpretation, next decision and artifact links. Rejected designs are kept: a failed optical architecture is valuable if it shows which constraint dominated, and a neural model that fails across sessions is more informative than a high within-session score if it exposes the stability bottleneck.

The experiment identifiers below carry over from v1.0. The optical experiments O1 to O7 map onto the technology ladder O0 to O9 (Section 42) as shown in the last column.

59 Experiment register

OPTICAL, SAFETY AND MANUFACTURING

ID	Name	Measures	Decision it informs	Ladder
O1	source characterization	wavelength, output power, modulation, efficiency, stability	whether a candidate source is usable	O1
O2	field generation	FOV, angular resolution, MTF, brightness, colour, speckle	optical architecture selection	O2, O3
O3	eyebox	$Q(x_p, y_p, z_p)$ and safe accessible output independently	usable eyebox and margin	O4
O4	gaze control	latency, accuracy, prediction horizon, calibration drift	feasibility of eye-locked optics	O5
O5	foveated allocation	uniform, computational, optical and predictive foveation at matched quality	whether optical foveation saves energy	O5
O6	thermal saturation	maximum sustainable operating state	duty and placement limits	O6, O7
O7	fault injection	gaze loss, stale gaze, scan failure, source saturation, monitor failure, clock failure, power fault	safe-state design; Gate 1	O7
O8	human visual evaluation	comfort, task performance, artifacts at matched task	architecture acceptability; Gate 2 to 3	O8
O9	integrated optical subsystem	population yield $P(Q >= Q_{min})$ and safety yield	production feasibility; Gate 4	O9
MX1	tolerance and yield Monte Carlo against measurement	predicted versus measured spread of alignment and Q	tolerance budget; fixture design	O9

NEURAL

ID	Name	Measures	Decision it informs
N1	within-user decoder	accuracy, calibration, latency	decoder feasibility
N2	cross-session decoder	held-out session performance, calibration time	drift handling
N3	cross-user transfer	held-out user performance	generic model viability
N4	adaptation	gain per calibration minute	personalization value
N5	multimodal fusion	incremental information over gaze, EMG, voice, context	whether EEG stays on the critical path
N6	false-activation measurement	false activations per hour, consequence-weighted	usable operating point
N7	intent-commitment study	accuracy together with commitment timing	commitment policy

RESERVOIR AND SEMANTIC OPERATING SYSTEM

ID	Name	Measures	Decision it informs
RX1	transport	throughput, jitter, packet loss, energy per bit, thermal rise	link choice
RX2	sensor fusion and failover	gaze accuracy, EMG classification, EEG artifact rejection, multimodal accuracy, degradation transitions	fusion and degradation design
RX3	reservoir routing	task completion, local versus reservoir compute, energy per task, response distribution, visor temperature	the reservoir thesis
RX4	memory	repeated instructions, corrections, interaction time, effort	memory value
RX5	predictive interaction	tasks per action and unwanted intervention rate	proactivity policy
RX6	closed-loop optical control	state age, gaze prediction error, registration error, temporal stability	control architecture
RX7	wired-wireless equivalence	same logical API, different resource envelope	transport abstraction
H1	interaction effort (includes backend swap)	explicit interactions, time, error on matched tasks; unchanged scenario across providers	semantic OS thesis

INTELLIGENCE AND HUMAN FACTORS

ID	Name	Measures	Decision it informs
H2	wrong-action and correction	wrong-action rate, correction rate	trust policy
H3	externalized working memory	task continuity, effort, understanding; cognitive-burden calibration	context and memory value
H4	proactivity	interruptions, acceptance, suppression	attention policy
H5	focus and visual comfort	fatigue, accommodation and vergence behaviour, longitudinal	optical focus-level choice
H6	social acceptability and bystanders	acceptance, interruption cost, capture indication	form factor and capture policy
EP1	epistemic discrimination	model formation, error detection, experiment selection, transfer	the intelligence-research thread

60 Falsification program

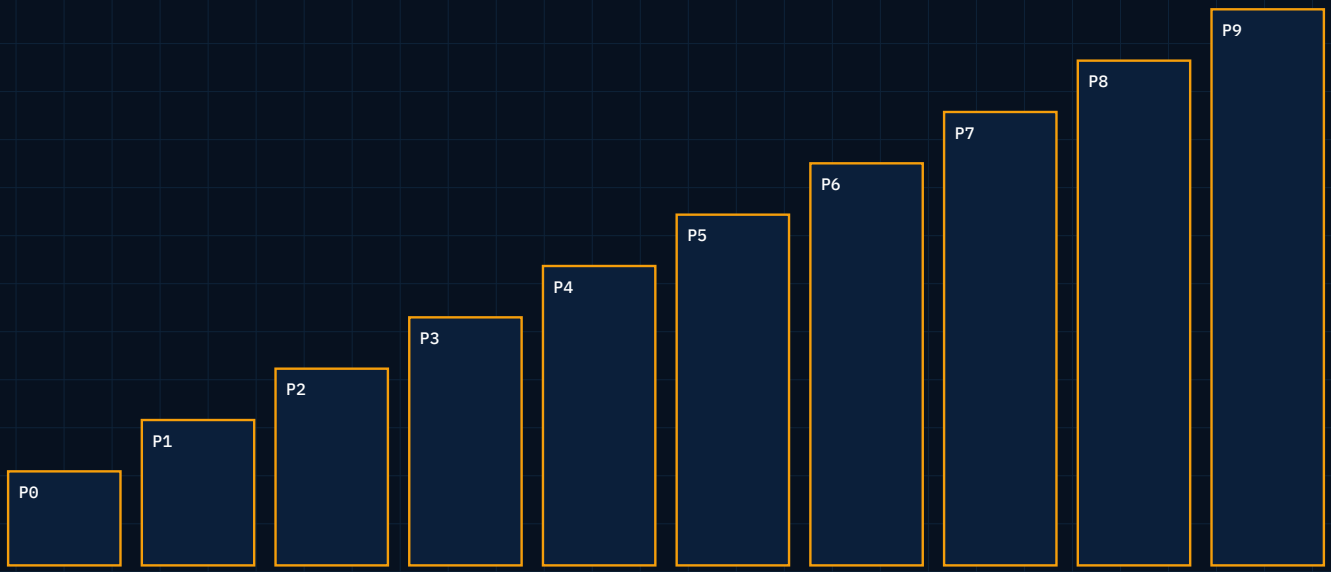
Claim	Measure	Success	Failure	Decision if falsified
Multimodal fusion reduces consequence-weighted false intent	false-action rate at matched sensitivity, held-out users and sessions	lower than strong single and dual-modality baselines	no gain	remove fusion complexity
EEG adds information	incremental information over gaze, EMG, voice, context, memory	positive gain at tolerable power and comfort	none	demote EEG; keep intent-centric architecture
Neural thesis	cross-session drift, calibration cost, false actions	drift bounded by adaptation; calibration short; false actions rare at useful sensitivity	drift overwhelms adaptation; calibration destroys usability; false actions frequent	reconsider neural channels
Gaze-conditioned rendering saves compute	compute against quality frontier under injected latency and noise	dominant frontier	worse than uniform	reject computational foveation
Optical foveation saves optical energy	$P_{optical}$ at matched perceptual and task performance	measurable saving	negligible saving	optical thesis weakened
Optical thesis	joint FOV, eyebox, brightness, efficiency, MTF, latency, safety, mass, thermal	acceptable joint frontier	frontier unreachable, or safety constrains geometry, or yield unrealistic	change source or combiner architecture; the loop does not depend on one source
Reservoir improves the wearable frontier	same task and quality: skin temperature, mass, battery, latency, stability	better joint frontier	latency, thermal or battery penalty dominates, or commodity devices match it	collapse to M0 or M1 plus a commodity host
Semantic OS reduces interaction	task time, interaction count, corrections, errors against strong baselines	net reduction without more errors	existing abstractions suffice	reposition the layer or abandon it
Platform thesis	developer integration effort against existing APIs	measurable leverage	developers need hardware knowledge or complexity rises	narrow the ABI
Personalization beats scaling	fixed model, memory and adapter ablations	long-horizon gain per data	none	stop investing in personalization
Distributed placement	latency, energy and failure behaviour of one task across placements	benefit without added failure	more instability than benefit	collapse placement options

These falsifiers are stated, not hidden. A branch rejected on strong measurement is a scientific result.

PART XII

CURRENT STATE AND FRONTIER

What exists · what is open · what can kill it · what is next



The annex ends in executable proof: what can be run now, what needs hardware, laboratories and people, what can kill the architecture, and which decision cannot be undone.

61 Current implementation

The repository was not inspected for this annex. The statements below are reported by the project and are labelled REPOSITORY-ASSERTED. They become VERIFIED only on a repository snapshot, a passing test report and a recorded trace (Appendix M).

Item	Reported state	Label	Evidence needed
Software baseline	a Python repository with kernel, runtime, SDK and perception baseline (V0)	REPOSITORY-ASSERTED	snapshot, test report
Runtime plan	continuously running runtime fed by real or replayed measurements; a real-time runtime with shared-memory IPC and Python/C++ bridges; neural-input to intent and action policy; world-model loops in a physics simulator	PROPOSED	staged builds with passing tests and state manifests
Ingestion plan	MNE, BrainFlow, LSL and OpenCV adapters; PhysioNet EEG motor imagery, NinaPro DB6 and DB10, emg2pose, GazeCapture replay	PROPOSED	per-dataset licence, access and sync verification (Appendix K)
Semantic objects	Intent IR, Action IR, capability graph, policy envelope, typed memory	REPOSITORY-ASSERTED	schema conformance and replay-equivalence tests
Host UI	Qt6 / PySide6 host interface preferred	PROPOSED	running build
Android integration	AIDL contracts and Cuttlefish validation path	REPOSITORY-ASSERTED for abstractions	Cuttlefish run trace
Optical engine software	beam and field computation, safety-gate logic	REPOSITORY-ASSERTED	bench measurements
Hardware	no integrated wearable; no optical or neural prototype reported	UNRESOLVED	bench artifacts

Status labels for any module in the repository follow one set: IMPLEMENTED, VIRTUAL, REPLAY, HARDWARE-BACKED, EXPERIMENTAL, PARTIAL, STUB, UNRESOLVED; and every build block ends with a state manifest (active module, completed components, next steps). A module with no test run is not counted as done.

62 Verified capabilities, open problems and rejected approaches

Verified capabilities in this annex: none. No NEXUS capability is VERIFIED, because no repository snapshot or test report was available. Externally demonstrated, not NEXUS: generic sEMG decoding across people, flat-panel PIC-plus-LCoS laser display, synthetic-aperture waveguide holography, foveal resolution measurement, retinal scanning displays, predictive latent representations, multi-sensor research eyewear.

Open problem	Why it is open	Class
Near-work eye strain	accommodation, vergence, blink and posture interact; no architecture shown to relieve it	E5
Wearable FOV, eyebox, efficiency frontier	etendue and pupil replication costs; efficiency chain unmeasured	E5
Size-to-functionality ratio for mixed reality	mass, heat and optics conflict inside a frame	E5
Ten-hour to all-day battery	energy arithmetic against skin-contact heat and mass	E5
Wearable EEG usefulness	artifacts, drift, variability; incremental information unmeasured	E5
Gaze prediction at saccade speeds	registration error versus delay and horizon	E4
Accommodation-consistent depth	no candidate shown at wearable scale	E5
Calibrated intent confidence	ECE per verb on held-out users unknown	E4
Manufacturing yield of optics	population and tolerance effects unknown	E5
Long-duration comfort and trust	no longitudinal data	E5

Failed or rejected approaches are listed in Section 1 (rejected positions) and in the Decision register. In addition: three headline metrics from earlier design variants are retired until redefined (Section 8); the earlier practice of treating a pixel count as an optical specification is retired (Section 40); shuffled within-session accuracy as a neural result is retired (Section 33).

63 Critical bottlenecks, next experiments and next decisions

Bottleneck	Decided by	First measurement
Near-work eye strain	optical focus-level choice; H5	comfort protocol design with validated instruments
Size-to-functionality ratio	mass and thermal budget; optical frontier	thermal rig; mass breakdown
Battery runtime	energy per task; reservoir thesis	RX1 power traces
Eye-locked optical registration	delay and prediction	RX6 measurement rig
EEG role	N5	incremental-information test on collected data

Next engineering decisions, in order of irreversibility. (1) Which optical branch to build the first breadboard around; reversible at low cost but it fixes the eye-safety analysis route. (2) The first neural packaging (wrist EMG first); reversible. (3) The semantic ABI major version freeze; costly to reverse once third-party capabilities exist. (4) The safety interface freeze between the safety supervisor and the optical driver; the most irreversible, because it binds hardware and certification. (5) The optical-source class, which determines the standards route and Gate 0.

Long-term frontier

The long-term vision is kept, and made legible by the experiments above: programmable optical information fields, distributed personal compute, a semantic operating environment, multimodal human state, neural and physiological interaction, persistent world and context models, adaptive optical presentation, a human-centred compute fabric and personalized intelligence. Each is a research hypothesis or engineering target with a stated falsifier, not a claim of what exists today.

64 Executable proof

Question	Answer
What can be executed now?	host simulation, schema validation, deterministic runtime tests, multimodal replay, placement simulation, calibration and fusion analysis on public data
What can be measured now?	ECE and false-action arithmetic on replayed data; timing jitter in the host runtime; placement energy in a model
What requires hardware?	Loop B state-age histograms, radio energy per bit, rail power, skin-adjacent thermal behaviour, failover (RX1, RX2, RX6)
What requires laboratory optics?	source, field, combiner, eyebox and FOV, efficiency chain, fault injection (O1 to O7)
What requires human subjects?	N2 to N7 on real wear, H1 to H6, O8; each behind the safety gates and ethics review
What requires manufacturing?	population yield, fixtures, factory calibration time, field monitoring
What can kill the architecture?	a credible fault combination leaving optical output above the limit; Loop B bound unreachable; reservoir thesis false for the Loop C mix; fused intent no better than gaze plus confirmation; thermal limit incompatible with optical presentation
Next irreversible decision	the safety interface freeze between supervisor and optical driver

Proof stack

Proof	Statement proven	Artifact	Requires
P0	schema validity	schemas, conformance suite	software
P1	deterministic runtime	seeded trace reproducible bit-for-bit	software
P2	multimodal replay	Observation-equivalent replay with preserved sync and calibration	public datasets
P3	distributed compute	placement benchmark with failover and migration trace	two or more hosts
P4	intent and action closed loop	canonical trace (Appendix M) with rejection demo	software, simulator
P5	hardware in the loop	Loop B state-age histogram and power trace	boards, timestamps
P6	optical bench	K_{opt} measured; fault-injection record	laboratory optics, Gate 0 to 1
P7	integrated wearable	worst-case thermal map; M0 with link removed	built device, Gate 2
P8	human validation	H1 to H6, O8 reports	ethics review, Gate 3
P9	production engineering	yield, fixtures, field monitoring	manufacturing, Gate 4 to 5

Appendices

Schemas are carried from v1.0 and shown as normative sketches. Types follow the rules in Section 17. Registers are the authoritative record; the main text reads without them.

Appendix A Intent IR

```
IntentIR {
  intent_id: uuid;  verb: enum{LOOK,SELECT,OPEN,MOVE,PLACE,COMPARE,EXPLAIN,REMEMBER,SEARCH,ACT,CONFIRM,CANCEL};
  targets: list<EntityRef>;  parameters: map;  context_refs: list<Ref>;
  confidence: float [0,1]  // calibrated posterior; calibration_ref required
  urgency: float [0,1];  reversibility: enum{high,medium,low,none};
  commitment: enum{emerging,speculative,supported,confirmed,committed,executing,verified,rejected,cancelled};
  evidence: list<{source: enum, weight: float}>;  timestamp: Time(ns, domain, sigma_ns);
  provenance: ModelRef;  calibration_ref: CalibrationRef;  schema_version: Version }
```

Appendix B Action IR

```
ActionIR {
  action_id: uuid;  capability: CapabilityName;  arguments: map;
  authorization: PolicyToken(nonce, expiry);  risk: enum{low,medium,high,physical};
  reversibility: enum;  resource_hint: enum{visor,R0,R1,R2,remote,auto};
  presentation: {mode, priority};
  optical_request?: {eye, pose, depth, field, brightness, foveation, temporal_profile};
  trace_id: uuid;  schema_version: Version }
// optical_request passes the safety envelope; it is never application-controlled
```

Appendix C Observation

```
Observation {
  id; source; modality;  timestamp: Time(ns, clock_domain, sigma_ns);  sequence: int64;
  payload_handle: BufferHandle | inline(small);  pose: Transform(frame, covariance);
  coordinate_frame;  calibration_ref;  confidence;  quality: QualityState;
  sampling_rate;  compression_state;  staleness;  schema_version }
```

Appendix D Capability manifest

```
Capability { name; version; provider; inputs; outputs; required_context; modalities;
             latency_class; privacy_class; compute_requirement; trust_requirement; side_effects;
             reversibility; cost_model; signer; signature }
```

Appendix E ComputeNode ABI

```
ComputeNode { id; tier: enum{visor,R0,R1,R2,remote}; accelerators; memory; thermal;
              power; network; locality; security_domain; health;
  admit(workload) -> Admit | Refuse(reason);
  run;  checkpoint -> Handle;  restore(Handle);  cancel;  report_health }
```

Appendix F Optical parameterization

Group	Parameters exposed to the twin	Sensitivities reported
Source	wavelength, spectrum, beam divergence, output power, modulation	dQ/d(parameter)
Field	phase map, FOV, resolution, brightness, speckle	dQ/d(parameter)
Combiner	waveguide index and thickness, grating period and depth, coupling efficiency, surface roughness	dQ/d(parameter)
Eye	eyebox, eye relief, pupil diameter, eye position, head pose	dQ/d(parameter)
Exposure and thermal	accessible output model, thermal model, fault model	dE_{acc}/d(parameter)
Manufacturing	tolerances, material variation	$sigma_{sys}$, $P(Q < Q_{min})$, $P(E_{acc} > E_{limit})$

Appendix G NeuralDevice

```
NeuralDevice HAL { configure(params); start(); stop();
  get_stream(class: raw | feature | hypothesis);  get_quality();  get_calibration();
  get_artifact_state();  get_health() }       // raw, feature and hypothesis streams have separate permissions
```

Appendix H VirtualSensor

```
VirtualSensor { id; kind; dataset_ref; adapter_version; clock_domain_map; calibration_ref;
                replay(rate, seek); emits Observation identical in schema to hardware; reports sync_fidelity }
```

Appendix I Safety case

Claim	Argument	Evidence (slot)	Gate
Accessible optical output is bounded	passive-first barrier stack plus independent active path	radiometry; fault-injection records	1
Loss of tracking drives a verified safe state	confidence gate and independent monitor	dropout injection	1
No single software fault creates uncontrolled output	hardware path independent of software	watchdog and fault injection	1
Skin-contact heat inside the validated envelope	throttle ladder; redundant sensing	worst-case thermal maps	2
Battery faults contained	independent protection; containment	abuse tests	2
Visual comfort acceptable over long use	staged human studies	longitudinal data	3
Units screened in production	end-of-line fixtures	per-unit records	4

Appendix J Safety requirements registry

ID	Requirement	Verification record	Gate
NXS-SAFE-OPT-001	Accessible optical output stays within the verified exposure envelope.	VR-OPT-001: radiometric classification, fault tests	1
NXS-SAFE-OPT-002	Loss of valid eye-state confidence forces a verified safe optical state.	VR-OPT-002: tracking-loss injection	1
NXS-SAFE-OPT-003	No single software fault creates an uncontrolled optical output state.	VR-OPT-003: watchdog and fault injection	1
NXS-SAFE-VIS-001	Visual-system testing includes long-duration exposure.	VR-VIS-001: H5	3
NXS-SAFE-THM-001	Skin-contact thermal conditions are measured under worst-case compute and optical load.	VR-THM-001: O6, O7	2
NXS-SAFE-PWR-001	Optical-source power has an independently enforceable upper bound.	VR-PWR-001: hardware limiter test	1
NXS-SAFE-HF-001	Safety-critical state is communicated predictably and consistently.	VR-HF-001: usability study	2
NXS-SAFE-MFG-001	Production units receive objective safety verification before shipment.	VR-MFG-001: end-of-line fixture	4

Appendix K Dataset registry

Registry fields: name, version, license, citation, modalities, subjects, sessions, sampling, labels, coordinate systems, timestamps, calibration, synchronization, access method, adapter, purpose, limitations, and verification status of license and access. Label sets and coordinate conventions are never merged without a documented mapping.

Resource	Modalities	Purpose	Status
Project Aria recordings (VRS) with open tooling	multi-camera, IMU, audio, eye-tracking images, gaze	egocentric replay; virtual camera, IMU, gaze	access and licence UNRESOLVED
PhysioNet EEG Motor Movement/Imagery v1.0.0	multichannel EEG, motor tasks	EEG virtual device; pipeline tests	licence UNRESOLVED
BCI Competition IV	EEG with labelled tasks	decoder benchmarks; cross-session design	licence UNRESOLVED
emg2pose (Meta Research)	sEMG with hand pose, many participants	EMG decoding; held-out-user evaluation	licence UNRESOLVED
NinaPro DB6 and DB10; GazeCapture	sEMG and hand data; gaze images	named in the software build plan	licence and sync UNRESOLVED
BrainFlow, LSL playback and synthetic boards	generic biosignal streaming	virtual neural device plumbing	to be checked

Appendix L Reservoir model sheet

```
P(t)  = P_compute + P_RF + P_memory + P_sensor + P_conversion
C_th dT/dt = P(t) - hA (T - T_a)
instrument: CPU package, skin-adjacent, enclosure, optical source, battery,
            radio duty cycle, workload occupancy;  optimize task utility per joule
runtime  t = E eta / P_avg
offload break-even:  E_local > E_tx(bits) + E_wait   (plus the visor thermal term)
```

Appendix M System trace schema

```
TraceRecord { trace_id; observation_ids[with clock_domain]; model_versions; intent_id;
              policy_decision; capability; compute_node; transport; render_plan; execution_result;
              power_state[per rail]; thermal_state[skin-adjacent, junction, source, battery];
              latency{ T_sensor, T_sync, T_fusion, T_intent, T_plan, T_transport,
                       T_render, T_optical }[p50, p95, p99, jitter, drops];
              uncertainty{ sigma_t, pose_covariance, posterior_calibration, quality_state };
              failure_code: F0..F8; failure_reason }
```

Appendix N Neural benchmark protocol

Separate training users from held-out users, training sessions from test sessions, development devices from evaluation devices, calibration time from steady-state performance, and clean data from realistic motion. Report accuracy, FPR, FNR, precision, recall, consequence-weighted false-action rate, latency, calibration time, energy, robustness and retention. Report calibration (reliability diagram, ECE) per verb. Within-session shuffled accuracy alone is not a result.

Appendix O Requirements traceability matrix

Requirement	Architecture element	Where defined	Verification	Rung / gate
SYS-002, VIS-002 reflex independent of network	Loop A and B on visor; Class A terminates on visor	Sections 9, 14, 36	RX2, O7	V5
OPT-003, SAFE-OPT-001..003 bounded output	independent safety path; hardware inhibit	Section 45; NXS-C10	O7; radiometry	Gate 1
NEU-001, INV-2 neural evidence never authorizes	trust chain; gating table	Sections 22, 31	N6, N7, policy tests	V2 to V8
SYS-003 modality and provider independence	Intent IR; capability graph	Sections 17, 20, 21	H1 backend swap	V3
CMP-003 placement transparency	ComputeNode ABI; task identity	Sections 34, 35	RX3, RX7	V4
INV-10 raw neural data local by default	data-class policy	Section 55	access-control tests	V2
SW-004 calibrated confidence	calibration requirement; ECE per verb	Sections 20, 33	N5, replay	V3

Requirement	Architecture element	Where defined	Verification	Rung / gate
SW-001, SW-002 no serialized tensors; declared buffers	data plane; ownership table	Section 18	static buffer audit	V2
OPT-004 state-age bound	visor-local Loop B; predictor	Sections 13, 44	O5, RX6	V5
THM-001..004 thermal control	sensors; throttle ladder; separate envelopes	Section 49; NXS-C12	O6, O7, wear test	V7
PWR-001, PWR-004 source ceiling, brownout	independent rail and limiter	Section 48	hardware limiter test	Gate 1
SEC-001..004 trust root	secure boot, keys, signing, audit	Section 54; NXS-C14	attestation, penetration tests	V6
MFG-001..003 yield and traceability	tolerance stack; calibration records	Sections 46, 51, 52	MX1, population study	V6
HF-001..003 comfort, bystanders, cognitive load	studies H3, H5, H6	Section 56	H3, H5, H6	V8
CMP-001 admission control	ComputeNode ABI	Section 34; NXS-C07	RX3	V4
RF-002 link loss no effect on Loop A	transport class A	Section 36	RX2	V5

Appendix P Decision register

Decision	Alternatives	Current choice	Why and evidence	Cost and risk	Revisit when
Android/AOSP substrate	custom Linux; RTOS plus Linux	semantic layer on Android and Linux PREFERRED	inherit drivers, HAL, tooling E1	integration effort; partition and compositor access unknown	target configuration fixed
Semantic kernel	event bus; application-centric	typed semantic ABI PREFERRED	authority and lifetime differ by object E4	schema discipline; ABI not independent of providers	H1 fails
Local reflex loop	cloud-assisted loop	visor-local Loop A and B FROZEN	delay to angular error E0	visor compute and heat; thermal limit	never, unless physics changes
Reservoir architecture	phone-only; cloud; all on head	R0 to R2 plus remote FROZEN (packaging open)	thermal and memory frontier E4	extra device; link and body RF	RX1, RX3 negative
Neural node separation	integrated in visor	logical node, packaging open FROZEN	analog isolation and placement E1	extra link; comfort and acceptability	N1 to N5
Optical architecture branch	families A to F	none INVESTIGATING	no measurement selects one E2 E4	parallel breadboards; premature commitment	O2 to O4 results
Data-plane transport	IPC serialization	handles and shared buffers FROZEN	bandwidth and latency E1	ownership discipline; platform buffer limits	RX1
Control-plane transport	raw sockets	stable AIDL or typed IDL PREFERRED	versioning, security E1	interface freeze; release differences	target release
Memory architecture	record all; none	typed, governed, write-gated PREFERRED	privacy and utility E4	write policy design; under- or over-remember	RX4
Safety architecture	software-only safety	independent hardware path FROZEN	single-fault safety E0	extra hardware; interface freeze	never relaxed
Wired and wireless split	wireless only; wired only	same logical API, both PREFERRED	power and latency trade E4	two transports; complexity	RX1, RX7
See-through optics	passthrough	optical see-through PREFERRED	real photons as substrate product choice	optical complexity; depth cues, brightness	user study
Role of EEG	central; absent	research channel; demotable INVESTIGATING	unproven information gain E5	channel cost; wasted effort	N5
GUI versus optical engine	one pipeline	separate FROZEN	safety envelope placement E0	extra layer; latency	never

Appendix Q Equation index

Equation	Section	Design variable it constrains	Falsifier
dtheta = omega dt	13	where Loop B runs; state age	O5, RX6
$t_{ref} = (1 + delta)\, t_s + b +/- sigma_t$	14	fusion pair validity	clock residual
$T_{tot} = T_{sen} + \ldots + T_{opt}$	14	latency allocation	trace sum
$O(I\|Y) = O(I)\ prod\ LR_m$	32	fusion calibration	N5
$ECE = sum\ (n_b/n)\ \|acc - conf\|$	20	confidence meaning	N5
$EU_{spec} = P\ B - (1-P)\ C$	22	speculation	H4
J(i,w) = sum lambda terms	35	placement	RX3
$L_{tot} = L_{queue} + \ldots + L_{jitter}$	35	offload latency	trace
$Delta\ P_{local} > Delta\ P_{RF}$	34	reservoir thesis	RX1

Equation	Section	Design variable it constrains	Falsifier
$P_{loss} = P^2 R / V^2$	34, 48	tether voltage	bench
$G = n^2 A \Omega$	41	FOV, eyebox, aperture	O4
$E_{FOV} = N \lambda$	41	SLM pixels	bench
$d < \lambda/(1 + \sin\theta_{max})$	42	OPA pitch	array test
$C = 1/\sqrt{N}$	42	speckle	bench
$S = [\pi r^2 + \rho (A - \pi r^2)] / A$	43	foveation saving	O5
$\eta_{sys} = \prod \eta$	43	efficiency and heat	O1 to O3
$\tau_{tot} = \tau_{det} + \ldots + \tau_{opt}$	45	safety removal time	O7
$\sigma_{sys}^2 = \sum (df/dx)^2 \sigma^2$	46	tolerance	MX1
$Q_{pop} = P(Q \geq Q_{min})$	46	yield	O9
$P_{sys} = \sum P$	48	power budget	rail traces
$C_{th} dT/dt = P - (T - T_a)/R_{th}$	49	thermal envelope	thermal rig
$SNR = P_{sig} / (e_n^2 B + \ldots)$	29	neural front end	neural bench

Appendix R Evidence and revision register

EVIDENCE REGISTER

Claim	Evidence	Status	Artifact	Verification	Next proof
Foveal resolution limit about 94 ppd (achromatic gratings)	Ashraf et al. 2025 E2	EXTERNALLY DEMONSTRATED	DOI 10.1038/s41467-025-64679-2	publisher abstract checked	none needed
Generic sEMG decoding across people at modest rates	Kaifosh et al. 2025 E2	EXTERNALLY DEMONSTRATED	Nature 645, 702 to 711	publisher abstract checked	N1 to N5 on NEXUS data
Flat-panel PIC plus LCoS laser display, over 80 % volume reduction	Shi et al. 2025 E2	EXTERNALLY DEMONSTRATED	Nature 644, 652 to 659	abstract checked	O1 to O3
Synthetic-aperture waveguide holography, eyebox and FOV relations	Choi et al. 2025 E2	EXTERNALLY DEMONSTRATED ; 9 x 8 mm eyebox and 38 deg FOV unverified	Nature Photonics	abstract and supplement checked	O3, O4
Etendue conserved in lossless passive optics	physics E0	established	-	-	O4 frontier
Delay maps to angular error	physics E0	established; omega working value	-	-	O5, RX6
Reservoir reduces visor power and heat	design hypothesis E4	PROPOSED	rule table, model	none	RX1, RX3
Fused intent beats gaze plus confirmation	design hypothesis E4	PROPOSED	fusion code	none	N5, N6, H2
Optical foveation saves optical energy	design hypothesis E4	PROPOSED	twin	none	O5
Wearable EEG adds useful information	unresolved E5	UNRESOLVED	-	none	N5
Semantic ABI independent of providers	design hypothesis E4	REPOSITORY-ASSERTED schemas	repository	not inspected	H1 backend swap
Cuttlefish and AOSP integration works	path E4	REPOSITORY-ASSERTED	repository	not inspected	Cuttlefish trace

REVISION REGISTER: V1.0 TO V2.0

Change	Detail
Restructure	twelve Parts; requirements hierarchy, budgets, loop budgets, subsystem contracts, canonical trace, normative ABI, state-machine tables, security architecture, calibration architecture, decision register, failure matrix, proof stack added
Visual system	new identity, flagship figure set on one grammar, equation cards, status badges, chapter openers, document-control page
Terminology	earlier design variants carried conflicting terminology and requirement scales; canonical definitions used here are the ones in Sections 1 to 8; internal editorial machinery moved to this register
Correction: efficiency chain	six stages at 0.5 is a 64-fold loss; v1.0 wrote "about 62 times"
Correction: availability notation	two redundant paths of availability 0.9: $1 - (0.1)^2 = 0.99$; v1.0 printed "1 - 0.12"
Metric correction	three headline metrics retired until a measured attention proxy and a paired wrong-action rate are defined (Section 8)
Experiment mapping	v1.0 optical experiments O1 to O7 mapped onto ladder O0 to O9; neural, reservoir, human IDs unchanged
Status relabelling	asserted implementation state relabelled REPOSITORY-ASSERTED; no item raised to VERIFIED
Not re-verified	standards, platform documentation and reference items marked [C] in v1.0 remain not re-verified

Appendix S Reference register

V: checked against a publisher or preprint record during preparation of v1.0 and carried forward. C: carried from project reference lists, not re-verified. P: placeholder, not used as evidence.

Status	Reference	Establishes (and does not)
V	Ashraf M., Chapiro A., Mantiuk R. K., "Resolution limit of the eye: how many pixels can we see?", Nature Communications (2025), DOI 10.1038/s41467-025-64679-2	foveal achromatic, red-green and yellow-violet limits in ppd under stated conditions; not a display specification
V	Shi et al., "Flat-panel laser displays through large-scale photonic integrated circuits", Nature 644, 652 to 659 (2025), DOI 10.1038/s41586-025-09107-7	thin PIC plus LCoS laser display, over 80 % volume reduction; not a direct retinal emitter; not a wearable thermal or eye-safety result
V	Choi S. et al., "Synthetic aperture waveguide holography for compact mixed-reality displays with large etendue", Nature Photonics (2025), DOI 10.1038/s41566-025-01718-w	holographic waveguide with synthetic aperture; eyebox and FOV relations; reported eyebox and FOV figures unverified
V	Kaifosh P., Reardon T. R. et al., "A generic non-invasive neuromotor interface for human-computer interaction", Nature 645, 702 to 711 (2025), DOI 10.1038/s41586-025-09255-w	generic sEMG decoding across people, modest closed-loop rates; not multimodal fusion, not EEG
C	IEC TS 60825-20:2025; IEC 60825-1:2014; IEC 62471:2006; ISO 15004-2:2024; IEC 62368-1:2023; IEC 62133-2:2017+A1:2021; IEC 62311:2019; IEEE C95.1-2019; IEC 61000-4-2:2025; CISPR 32:2015+A1:2019; ISO 9241-392:2015; ISO 9241-210:2019; ISO 26800:2011; ISO 10993-1:2025; ANSI Z136.1-2022	candidate standards; applicability per selected architecture; not re-verified
C	emg2pose (Meta Research); PhysioNet EEG Motor Movement/Imagery v1.0.0; BCI Competition IV; Project Aria documentation and tools; TI ADS1299 documentation	public resources and example hardware; licence and access to be checked per use
C	I-JEPA; V-JEPA 2 technical materials (2025)	predictive latent representation methods; not evidence for NEXUS world-state use
C	Android AOSP, Binder, stable AIDL, SurfaceFlinger, Hardware Composer; OpenXR 1.1; Vulkan; Android XR documentation; USB4 specification family; IEEE 802.11be-2024; Qualcomm Snapdragon AR2 Gen 1	platform documentation and vendor materials; to be re-verified against the target release
C	de Wit G. C., retinal scanning displays (1999); single-layer achromatic metagrating waveguides, Nature Nanotechnology (2025); visible OPA, Optics Letters (2025); freeform holographic curved-waveguide retinal projection, Optics Express (2026)	named in project references; some citations incomplete
P	unsourced USB4 and lifelong-learning statements; incomplete tool-citation tokens in earlier source papers	to be resolved before external publication; not used as evidence

Appendix T Failure and degradation matrix

Failure	Detection	Immediate response	Degraded mode	User-visible effect	Recovery
Gaze loss	eye-state confidence drop	confidence gate; widen foveal region	uniform presentation; reduced energy	slightly dimmer, no foveation	recovered confidence; recalibrate
Camera failure	frame timeout; sequence gap	mark sensor failed; drop dependent estimates	IMU-only pose; reduced scene awareness	world anchoring weaker	restart sensor service
IMU drift	bias residual; disagreement with vision	inflate pose covariance	vision-led pose; reduce world-lock	content drifts slowly	recalibrate IMU
Neural signal loss	contact loss; quality collapse	drop neural evidence; widen posterior	gaze, voice, gesture, manual input	less implicit input	reseat; re-check impedance
Neural decoder drift	ECE drift; held-out check	tighten thresholds; reduce proactivity	neural demoted to low-gate use	more confirmations	adapter rollback or recalibration
Radio loss	heartbeat; link-quality drop	Class B and C to local; Class A unaffected	M0 or wired	reduced assistant	resync on return
Reservoir loss	node health timeout	M2 to M0	local small models; read-only memory if cached	basic operation only	restore node; resync
Cloud loss	connection failure	local and reservoir only	no scale features	some features unavailable	queued uploads resume
Clock failure	sync residual; exchange loss	hold estimate with growing $sigma_t$	local timestamps; fusion down-weighted	less precise fusion	re-estimate; hardware sync
Thermal overload	sensor threshold; derivative	throttle ladder T1 to T4	lighter models; dimmer optics	notice; slower features	cool-down with hysteresis
Battery brownout	rail monitor	optical source to safe-off first; save state	M0 minimal	device may power down	charge; restart
Optical monitor failure	self-test; sensor disagreement	inhibit source	static safe presentation or off	display off or minimal	service; Gate 1 re-check
Policy failure	misconfiguration check; rejection demo fail	deny by default	manual confirmation for all actions	more confirmations	restore signed policy
World-model inconsistency	coherence checks; residuals	mark stale; rebuild from observations	reactive operation	assistant less proactive	rebuild; re-anchor
Memory corruption	integrity check	read-only mode	working memory only	forgets history	restore signed snapshot

Failure	Detection	Immediate response	Degraded mode	User-visible effect	Recovery
Model corruption	signature or checksum failure	reject model; load previous	rule-based baseline	degraded intelligence	redeploy signed model

Appendix U Engineering budget tables

LATENCY STAGE ALLOCATION (OWNER AND MEASUREMENT)

Term	Owner	Runs on	Measured by	Budget
T_{sensor}	sensor service	visor	hardware timestamp to descriptor publish	OPEN from loop budget
T_{sync}	time service	visor	timestamp residual	OPEN
T_{fusion}	perception, fusion	visor or reservoir	trace stage time	OPEN
T_{intent}	intent runtime	visor or reservoir	trace	OPEN
T_{plan}	planner, scheduler	reservoir	trace	OPEN
$T_{transport}$	transport	link	RX1 distribution	OPEN
T_{render}	compositor	visor	frame timing	OPEN
$T_{optical}$	optical engine	visor	high-speed measurement	DERIVED Loop B bound

POWER AND THERMAL SOURCE ALLOCATION

Source	Power term	Thermal path	Instrumentation
SoC and memory	$P_{comp} + P_{mem}$	spreader to frame; skin-adjacent	junction, skin-adjacent
Optical source and driver	P_{opt}	isolated path; near eye	source, near-eye structure
RF	P_{RF}	antenna and PA location	radio duty cycle; local sensor
Sensors and AFE	P_{sense}	distributed; low	rail current
Conversion and loss	$P_{conv} + P_{loss}$	PMIC location	efficiency curves; rail pairs
Battery	charge and discharge heat	cell to enclosure	cell temperature

Appendix V Reference implementation map

The repository was not available. No path below is verified. Language entries reflect the reported state (Python baseline, REPOSITORY-ASSERTED) and the planned real-time runtime (C++), and interfaces are proposals. A subsystem moves from this table to the Current implementation section only with a verified path, build target, test and trace.

Subsystem	Repository path	Language	Process and interface (proposed)	Test	Trace
Semantic kernel and runtime	IMPLEMENTATION PATH NOT VERIFIED	Python today; C++ real-time runtime planned	NexusRuntime; AIDL	schema and replay-equivalence tests	trace_id
Intent runtime	IMPLEMENTATION PATH NOT VERIFIED	Python; C++ for visor-resident models	NexusIntentService; AIDL	calibration report, replay	trace_id
Capability and policy	IMPLEMENTATION PATH NOT VERIFIED	Python	NexusCapabilityService, policy engine	rejection demo	policy token id
Observation and data plane	IMPLEMENTATION PATH NOT VERIFIED	C++ with Python bridge; shared-memory rings	buffer pools; typed descriptors	buffer audit, gap injection	sequence ids
Virtual devices and replay	IMPLEMENTATION PATH NOT VERIFIED	Python adapters (MNE, BrainFlow, LSL, OpenCV)	VirtualSensor HAL	adapter conformance	dataset_ref
Neural pipeline	IMPLEMENTATION PATH NOT VERIFIED	Python; C++ filters	NeuralDevice HAL; quality and decoder services	N1 to N5 harness	window ids
World model and memory	IMPLEMENTATION PATH NOT VERIFIED	Python	NexusWorldService, NexusMemoryService	RX4, coherence tests	snapshot_id
Compute placement	IMPLEMENTATION PATH NOT VERIFIED	Python	NexusComputeService; ComputeNode ABI	RX3 harness	placement log
Transport	IMPLEMENTATION PATH NOT VERIFIED	C++	NexusTransport; class channels	RX1, RX7	session id
Optical engine software	IMPLEMENTATION PATH NOT VERIFIED	Python twin; C++ control	Optical HAL	O1 to O6 tooling	frame id
Optical safety path	IMPLEMENTATION PATH NOT VERIFIED	hardware plus minimal firmware	hardware lines; frozen interface	O7 harness	fault record
Time and calibration	IMPLEMENTATION PATH NOT VERIFIED	C++ estimator; Python tools	NexusClockService, calibration service	drift injection	calibration_ref
Telemetry and trace	IMPLEMENTATION PATH NOT VERIFIED	Python; C++ sink	NexusTelemetry	trace completeness	trace_id

Subsystem	Repository path	Language	Process and interface (proposed)	Test	Trace
Android integration	`IMPLEMENTATION PATH NOT VERIFIED`	Kotlin and C++ (AOSP)	AIDL services; HAL implementations	Cuttlefish run	trace_id

Appendix W Glossary

Term	Meaning
ABI	application binary interface; here the typed semantic objects and their rules
AFE	analog front end of a biopotential channel
Capability	an authorized operation a provider can perform
Class A, B, C	transport timing contracts: reflex, interaction, intelligence
ECE	expected calibration error
Etendue	$G = n^2 A \Omega$, a conserved optical throughput
Eyebox	set of pupil positions where the image meets quality and exposure criteria
FOV	field of view
Intent IR	canonical representation of what the human intends
Action IR	authorized commitment lowered to resources
Loop A, B, C, D	safety, reflex, intelligence, background loops
M0 to M3	operating modes by node presence
NeuralHypothesis	calibrated evidence from a neural decoder, never authority
ppd	pixels per degree
R0, R1, R2	reservoir tiers: pocket, portable, stationary
Reservoir	off-head persistent and sustained compute
SLM	spatial light modulator
State age	time between measurement and use of a state
Trace	record of one interaction across all stages
PIC	photonic integrated circuit
OPA	optical phased array
LCoS	liquid crystal on silicon
AOSP	Android Open Source Project
HIL	hardware in the loop
MTF	modulation transfer function